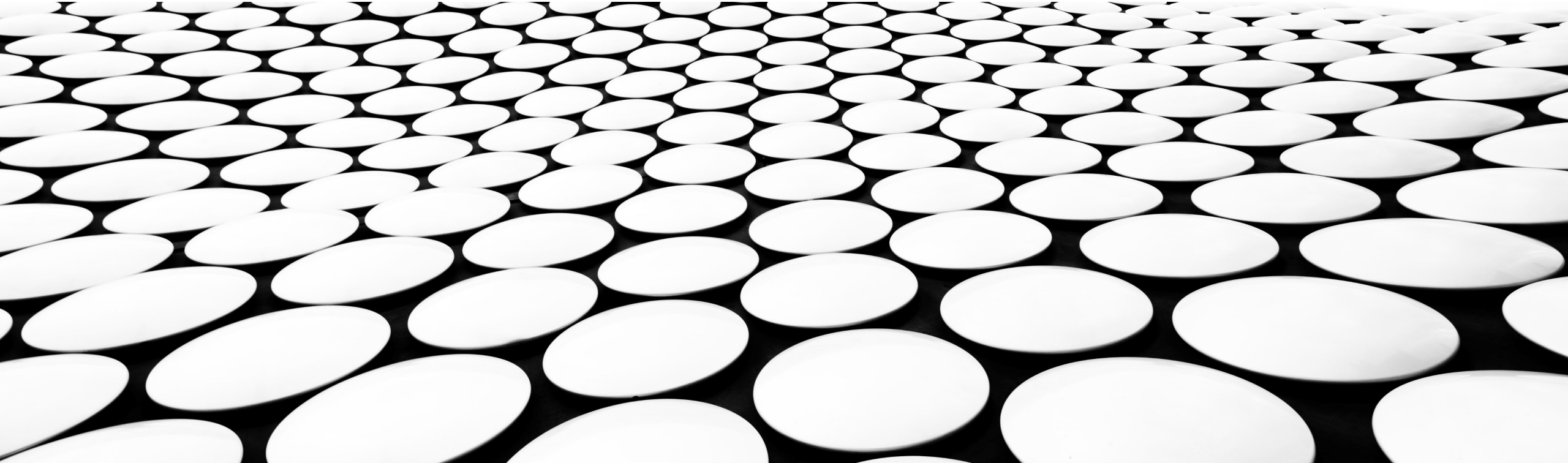
Introduction to Machine Learning

part 1



References

Parts of this lecture are based on:

1. Slides from the Harvard Course **CS109A Introduction to Data Science** by Pavlos Protopapas, Kevin Rader and Chris Tanner, available at https://github.com/Harvard-IACS/2019-CS109A

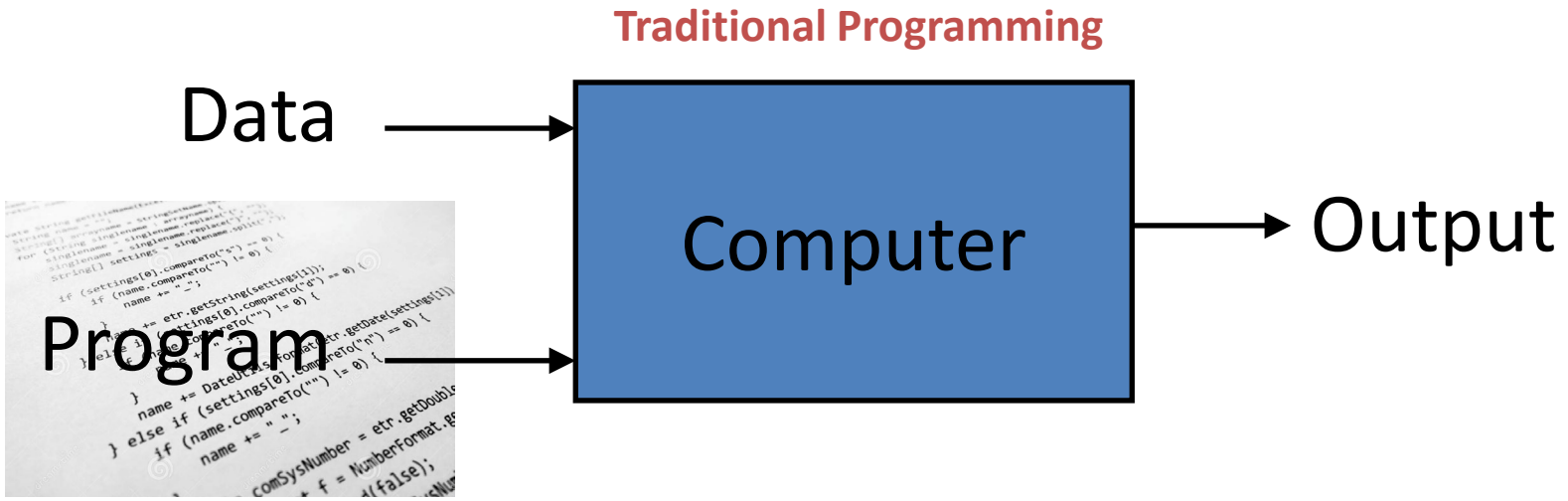
2. James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. **An introduction to statistical learning**. Vol. 112. New York: springer, 2013.

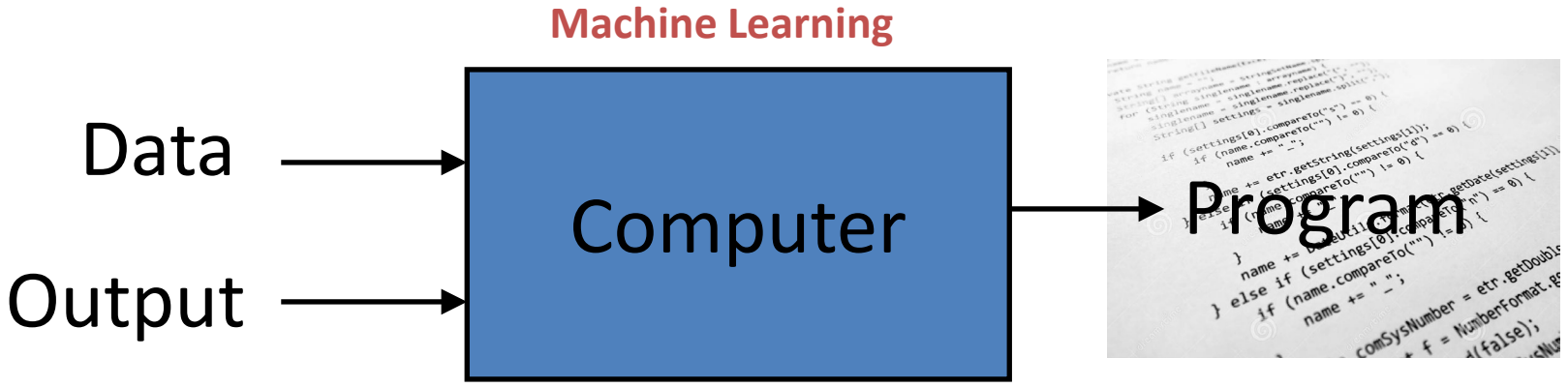
3. Cielen, Davy, Arno Meysman, and Mohamed Ali. **Introducing data science: big data, machine learning, and more, using Python tools**. Manning Publications Co., 2016.

Agenda

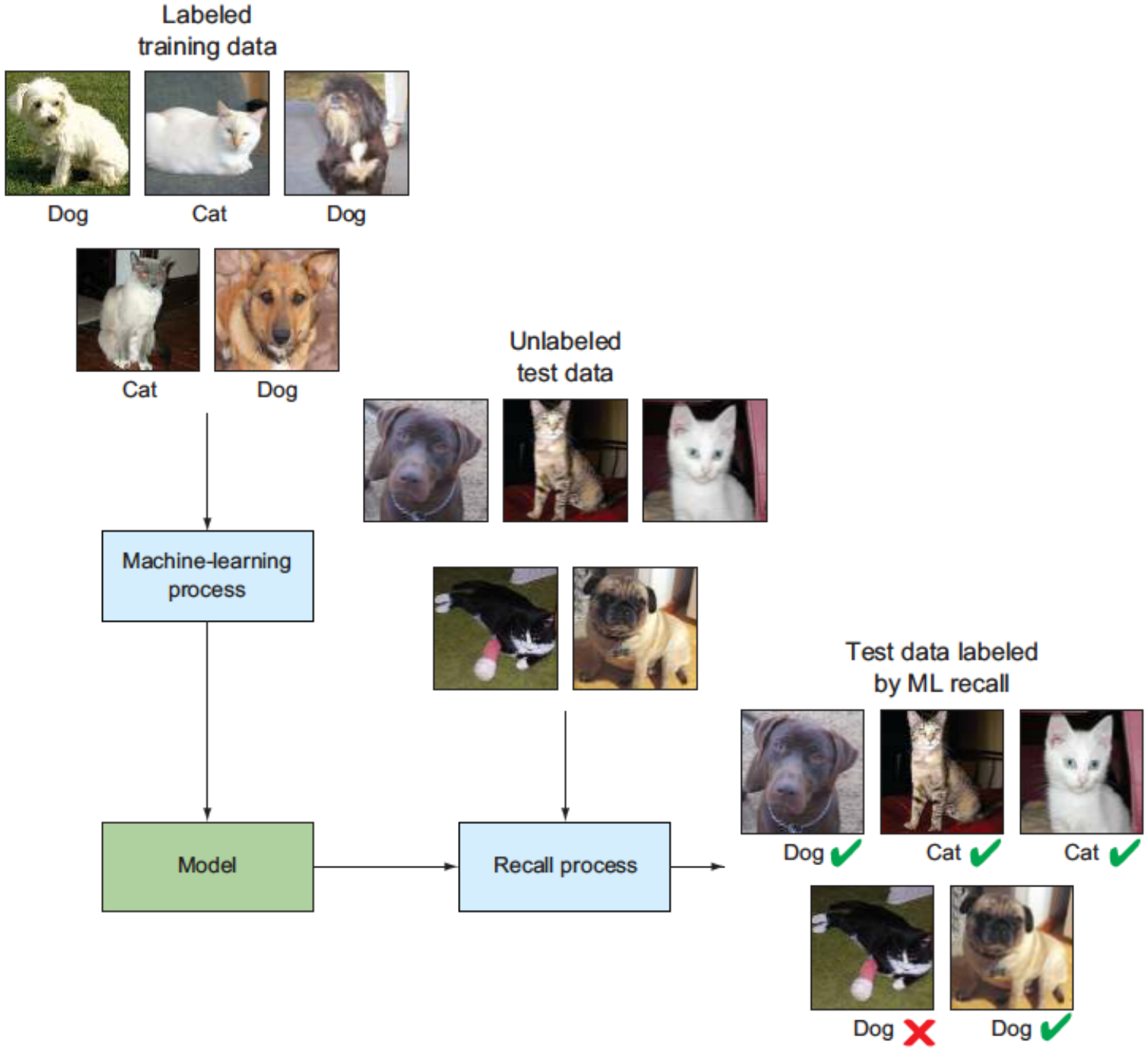
- Introduction to ML
  - Types of ML
  - Using ML to make predictions
- Regression
  - Defining the problem
  - Linear Regressing Models
  - Regressing model evaluation
- Classification
  - Logistic regression
  - Classification metrics

Paradigm shift





Understanding how machines learn

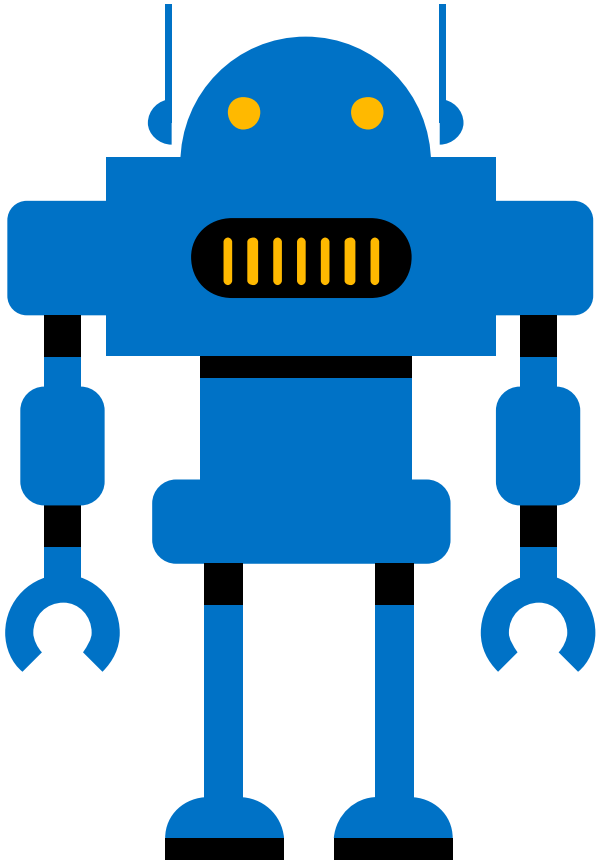


What is machine learning?

- A branch of artificial intelligence concerned with designing and developing algorithms that allow computers to **evolve behaviors based on empirical data**.
- As intelligence requires **knowledge**, it is necessary for computers to acquire knowledge
- Automating automation
- Getting computers to program themselves
  - Let the data do the work instead!

Machine Learning Definition

- The Formal one:
  - “A computer program is said to learn from experience E concerning some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.”
- A Practical Example:
  - Look at data. Try something. Get the correct answer? No? Look at the data. Do something different. Better? Yes? Do that again. (Repeat)
- Human learning is still far more sophisticated than even the most advanced machine-learning algorithms, but computers have the advantage of greater capacity to memorize, recall, and process data.



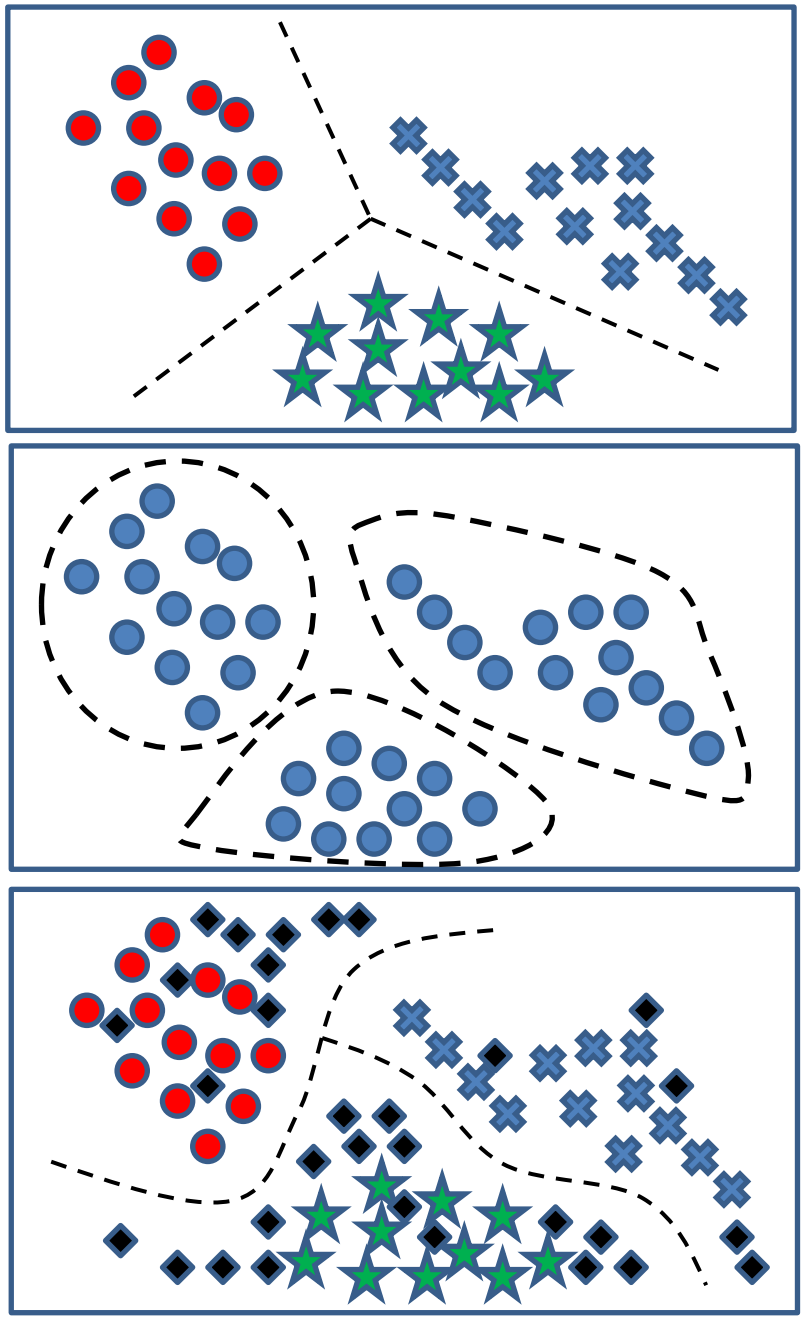
Types of Learning

- **Supervised (inductive) learning**
  - Labels provided
  - Training data includes desired outputs
  - Predicting the future.
  - Learn from known past examples to predict future

- **Unsupervised learning**
  - Labels not provided
  - Training data does not include desired outputs
  - Making sense of data
  - Understanding the past
  - Learning the structure of data

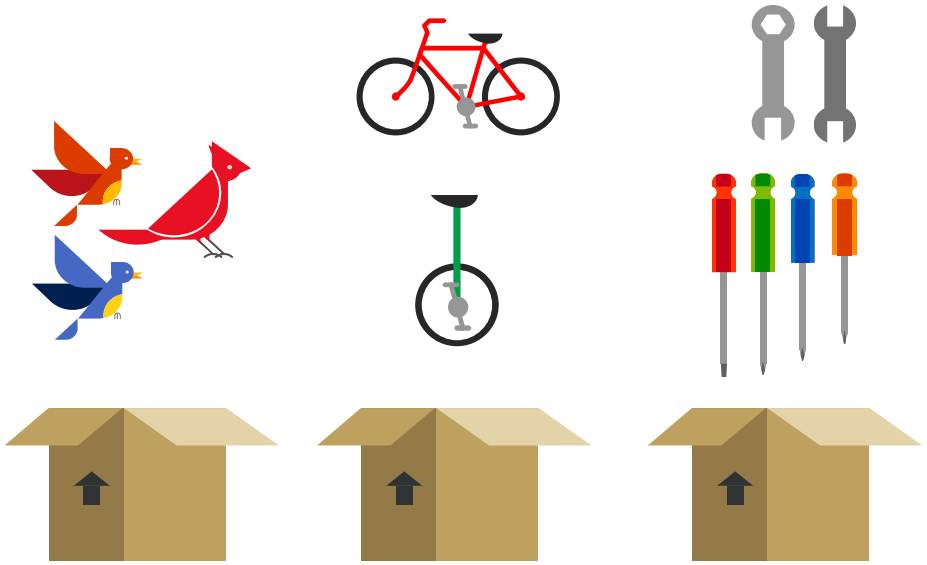
- **Semi-supervised learning**
  - Training data includes a few desired outputs

- **Reinforcement learning**
  - Rewards from sequence of actions

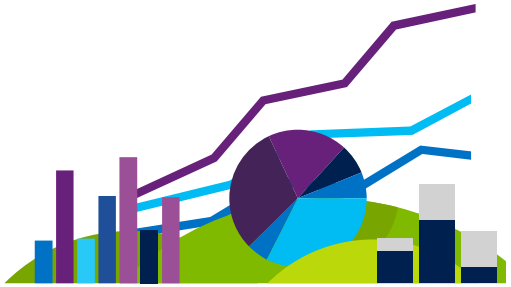


Machine Learning Capabilities

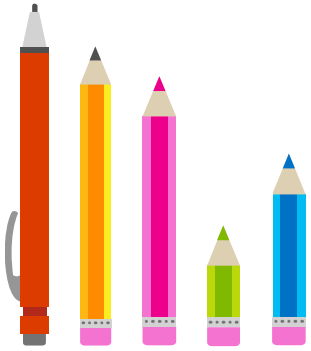
Which category
(Classification)



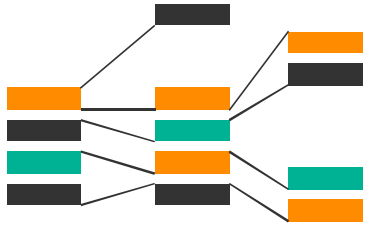
How much/many
(Regression)



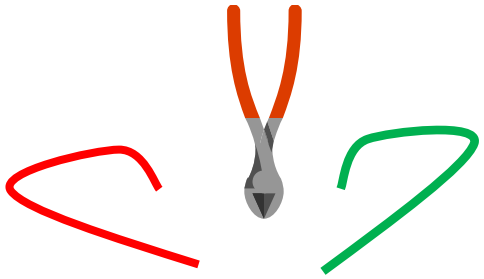
Is it odd
(Anomaly)



Which group
(Clustering, Recommender)



Which action
(Reinforcement Learning)



ML in a Nutshell

- Tens of thousands of machine learning algorithms
- Hundreds of new every year
- Every machine learning algorithm has three components:
  - **Representation**
  - **Evaluation**
  - **Optimization**

Representation

- Decision trees
- Sets of rules / Logic programs
- Instances
- Graphical models (Bayes/Markov nets)
- Neural networks
- Support vector machines
- Model ensembles
- Etc.

Evaluation

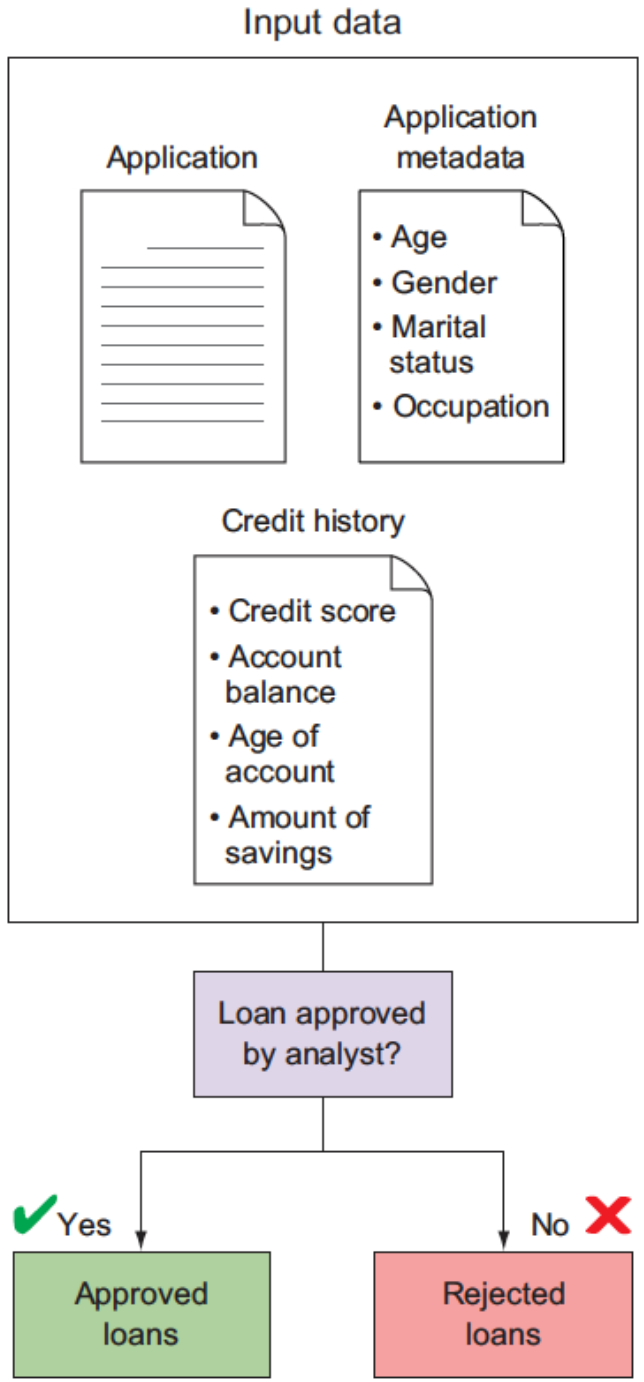
- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- Etc.

Optimization

- Combinatorial optimization
  - E.g.: Greedy search
- Convex optimization
  - E.g.: Gradient descent
- Constrained optimization
  - E.g.: Linear programming

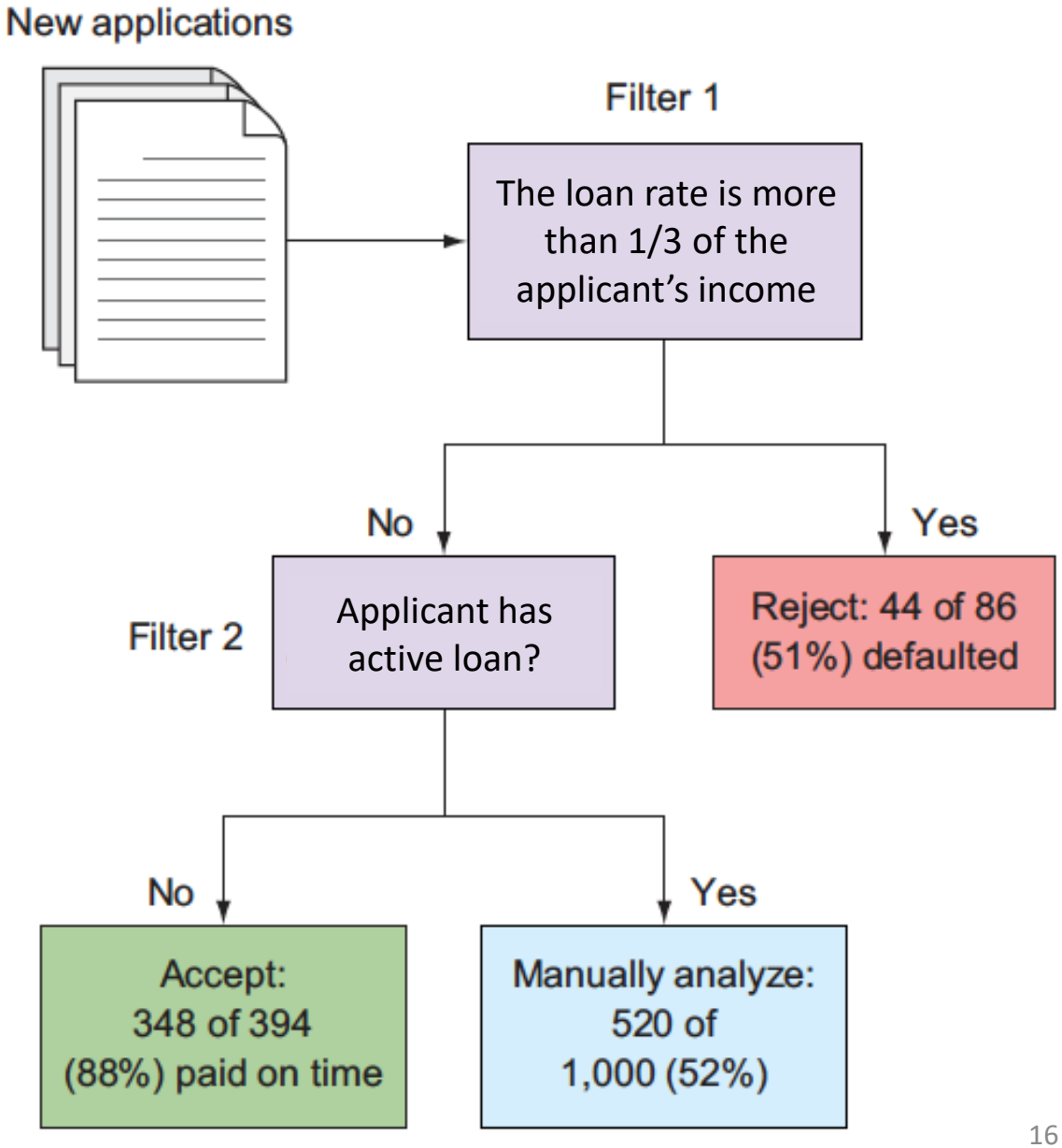
Using data to make decisions

- Traditional approaches for the loan-approval process for the microlending
- Human Analyst look at the documents
- Human Analyst makes the decision (YES or NO)



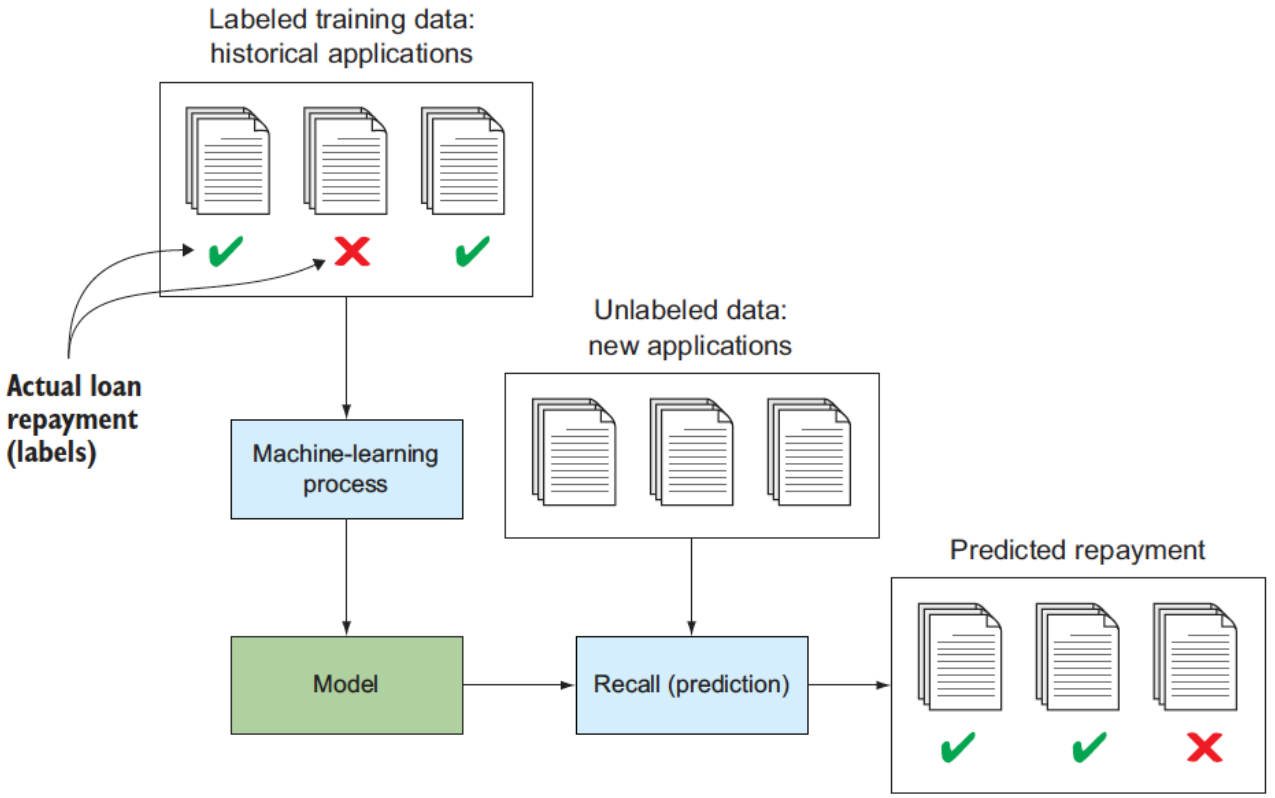
Using data to make decisions

- You need to manually analyze incoming applications
- Apply filtering rules and intuition to approve or disapprove the application



The machine-learning approach

- ML learns the optimal decisions directly from the data without arbitrarily hardcoding decision rules.
- This graduation from rules-based to ML-based decision-making means that your decisions will be more accurate and improve over time as more loans are made.
- Data provides the foundation for deriving insights about the problem at hand
- The input data consists of a set of features and numerical or categorical metrics that capture the relevant aspects of each application, such as the applicant's credit score, gender, and occupation.



The machine-learning approach

- In **logistic regression**, the logarithm of the odds that loan will be accepted is modeled as a linear function of the input features:
  - the applicant's credit line,
  - education level,
  - age

- The optimal values of each coefficient of the equation are learned from the training data examples.

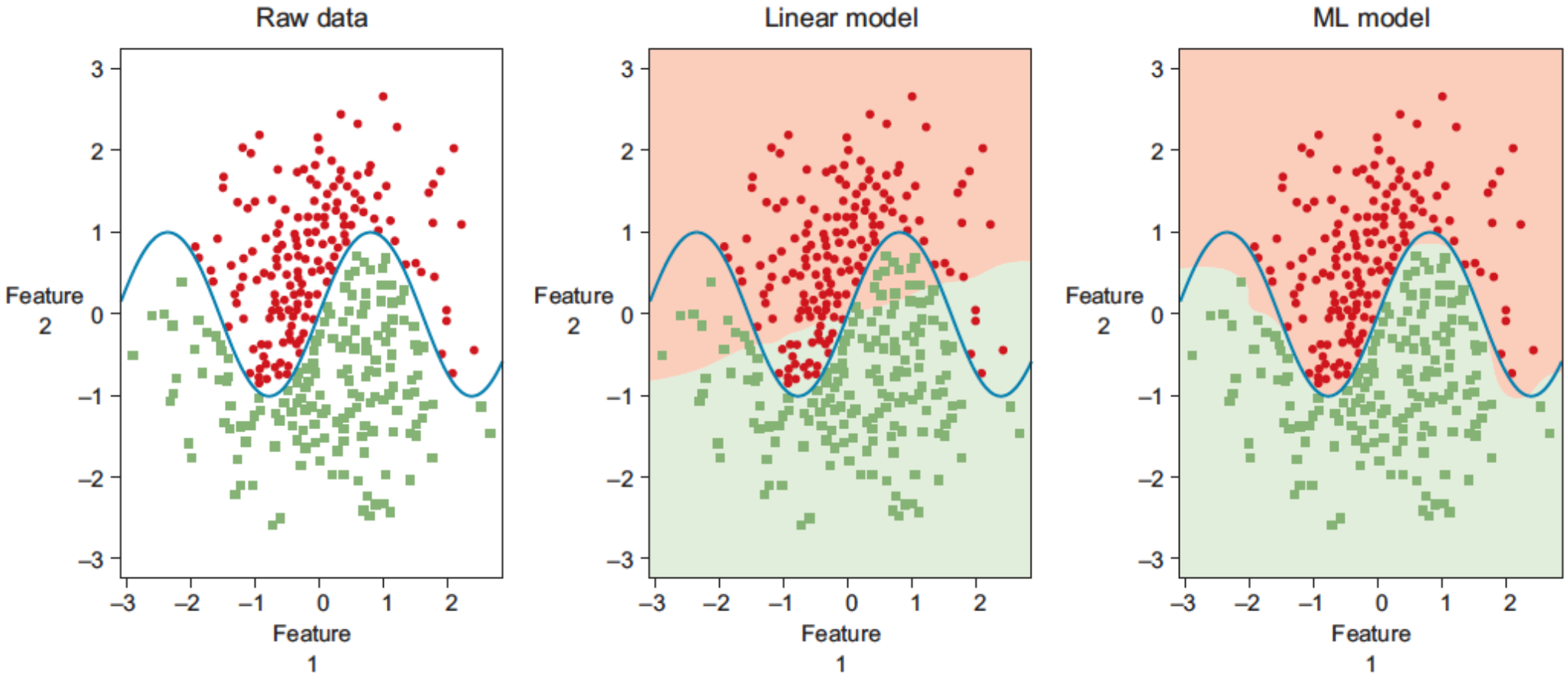
Log odds that applicant will get the loan

$$y = \beta_0 + \beta_1 * \text{Credit_Line} + \beta_2 * \text{Education_Level} + \beta_3 * \text{Age}$$

coeficients

Linear vs Non-Linear ML Algorithms

- When the relationship between the inputs and the response is complicated, models such as logistic regression can be limited. We need to use more complicated models.



Predicting a Variable (Regression)

- Let’s imagine a scenario where we'd like to predict one variable using another (or a set of other) variables.
- Thus, we'd like to define two categories of variables:
  - variables whose value we want to predict
  - variables whose values we use to make our prediction

- **Examples:**
  - Predict the number of views a YouTube video will get next week based on video length, the date it was posted, the previous number of views, etc.
  - Predict which movies a Netflix user will rate highly based on their previous movie ratings, demographic data, etc.

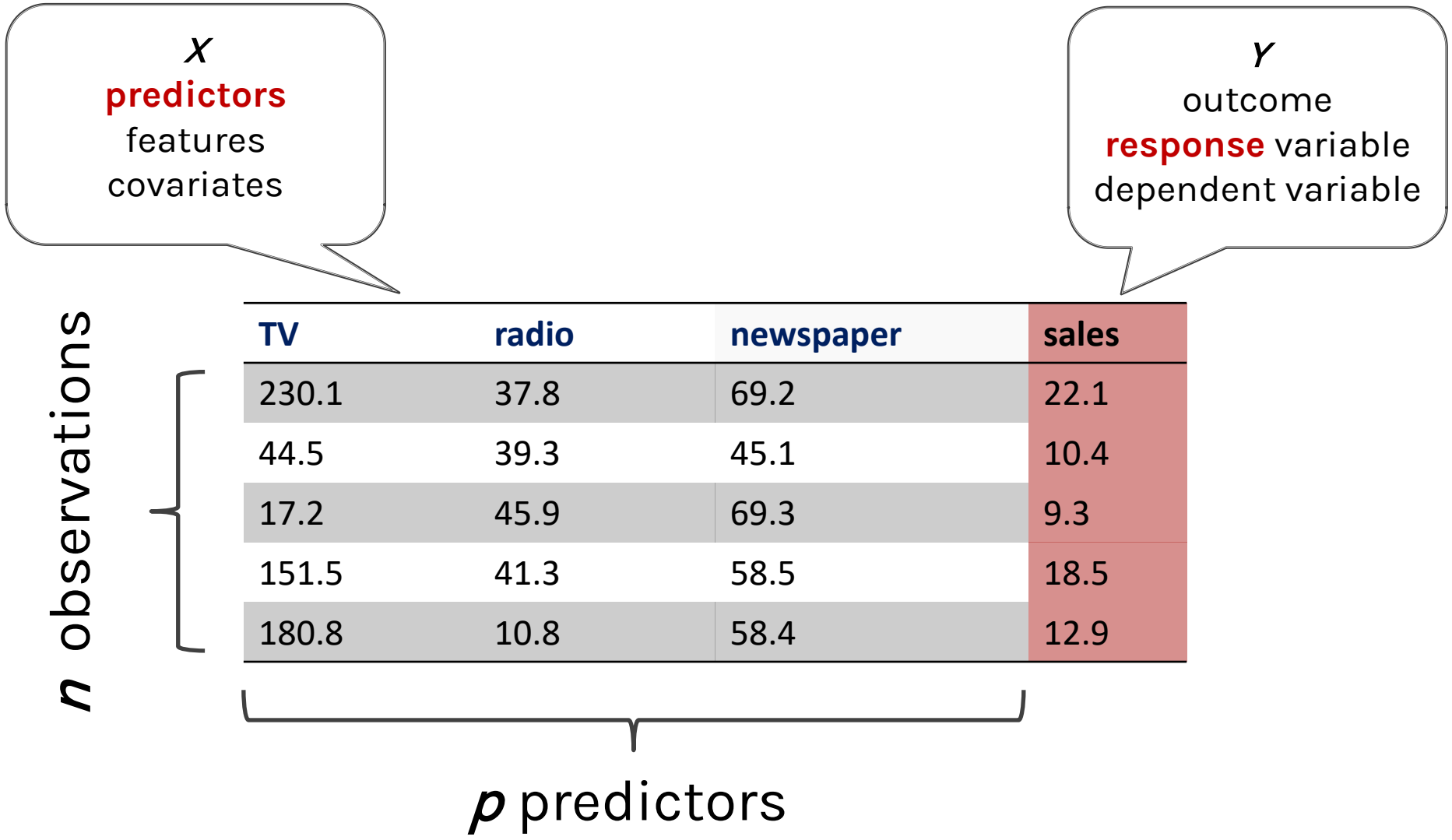
Data

The **Advertising data set** consists of the sales of that product in 200 different markets, along with advertising budgets for the product in each market for three different media: TV, radio, and newspaper. Everything is given in units of $1000.

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

Some of the figures in this presentation are taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013)

Response vs. Predictor Variables



TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

Response vs. Predictor Variables

$X = X_1, \ldots, X_n$

$X_j = x_{1j}, \ldots, x_{ij}, \ldots, x_{pj}$

predictors

features

covariates

$Y = y_1, \ldots, y_n$

outcome

response variable

dependent variable

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

n observations

p predictors

True vs. Statistical Model

- We will assume that the response variable, Y, relates to the predictors, X, through some unknown function expressed generally as:

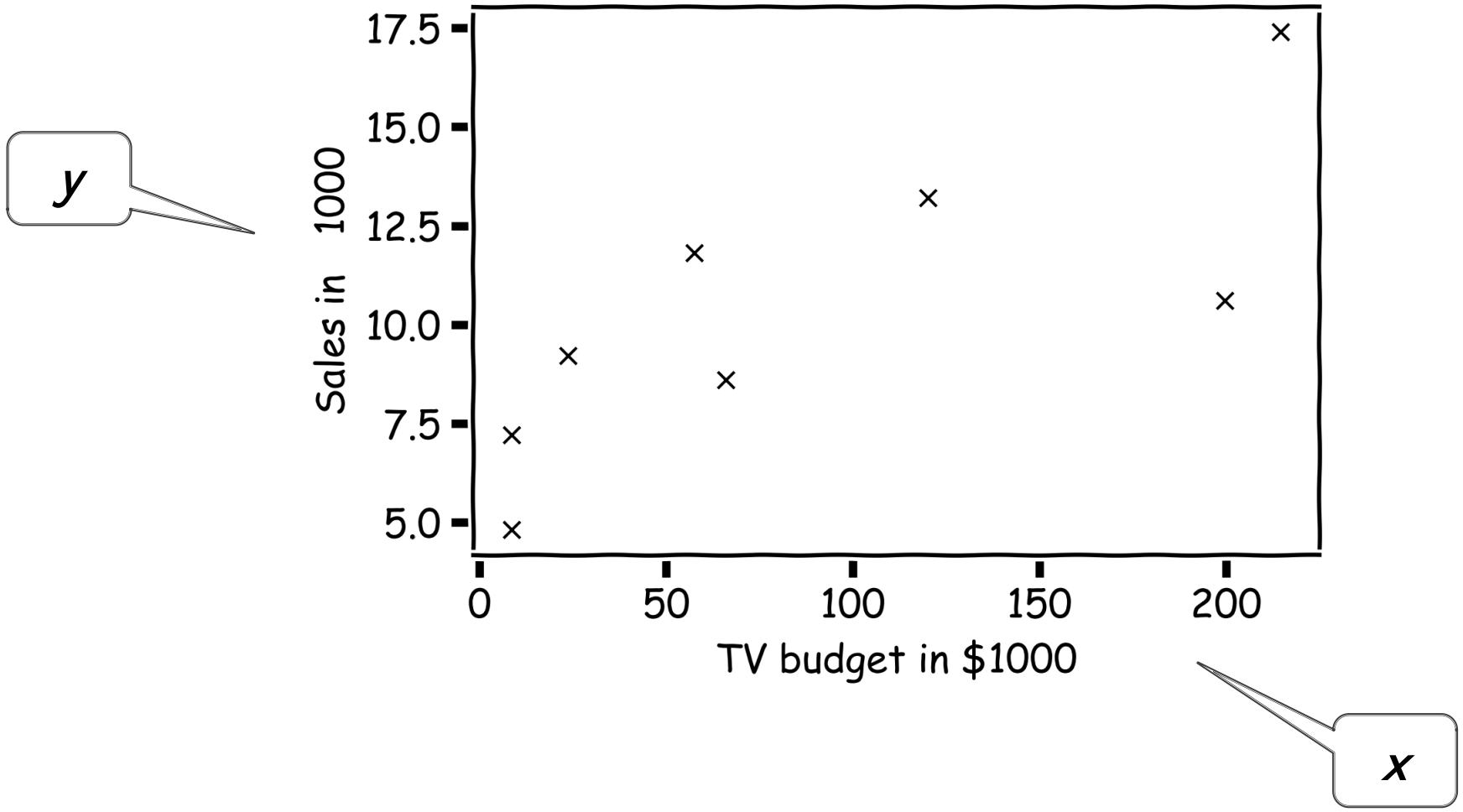
$$Y = f(X) + \varepsilon$$

- Here, f is the unknown function expressing an underlying rule for relating Y to X, ε is the random amount (unrelated to X) that Y differs from the rule $f(X)$.

- A ***statistical model*** is any algorithm that estimates f.
  We denote the estimated function as $\widehat{f}$.

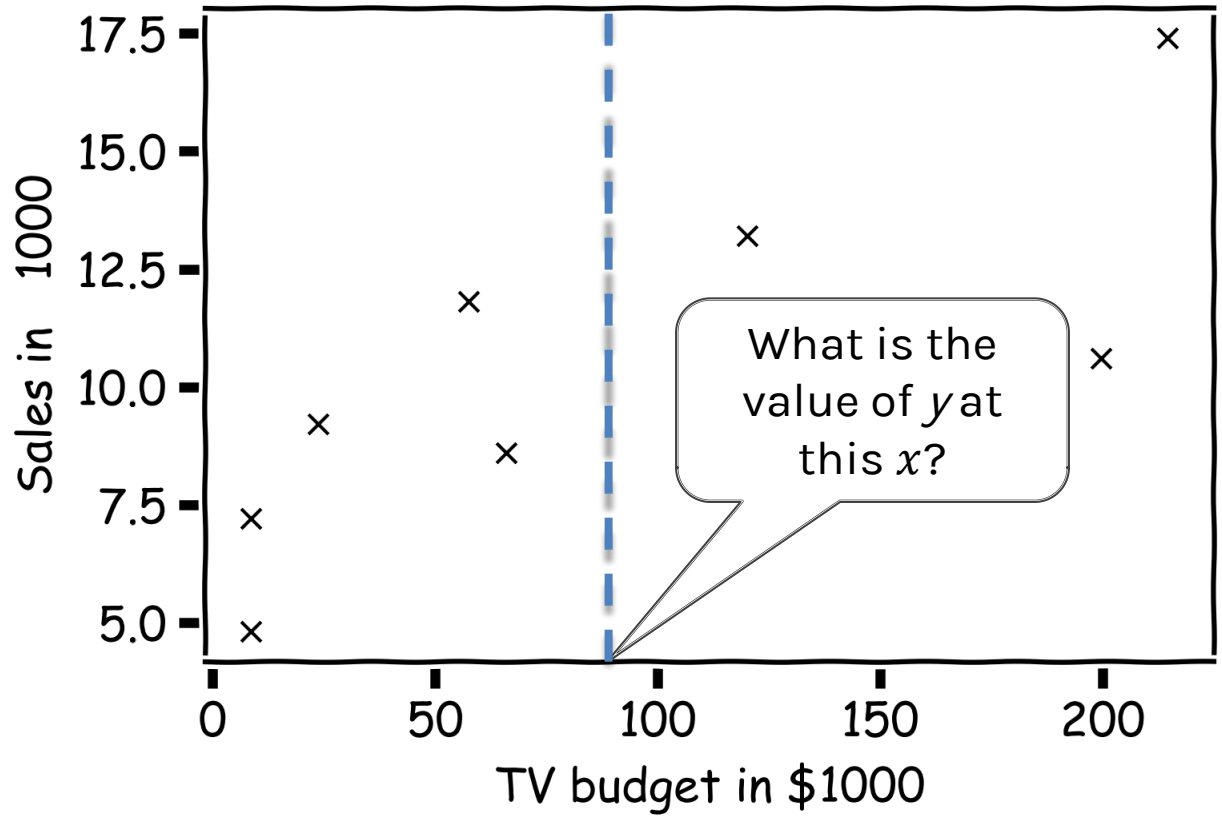
Statistical Model

TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35



Statistical Model

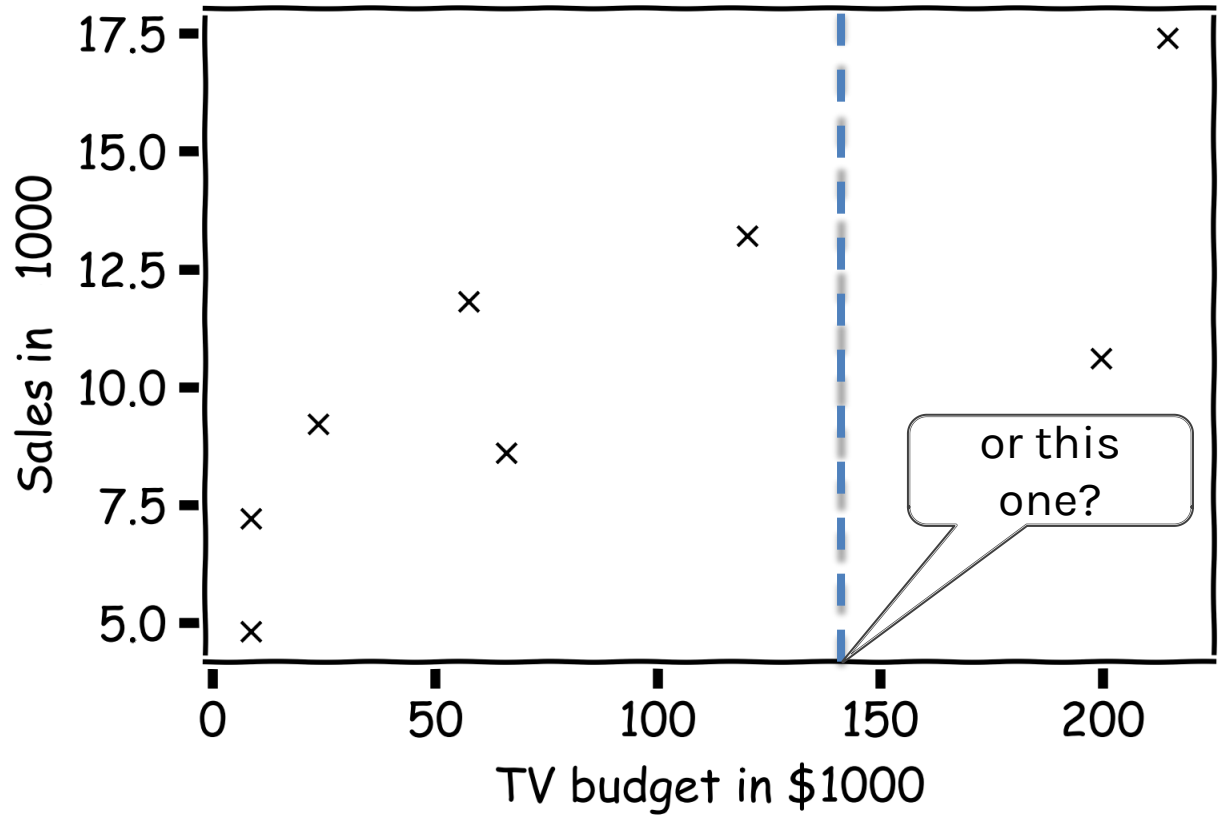
How do we find $\hat{f}(x)$?



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

Statistical Model

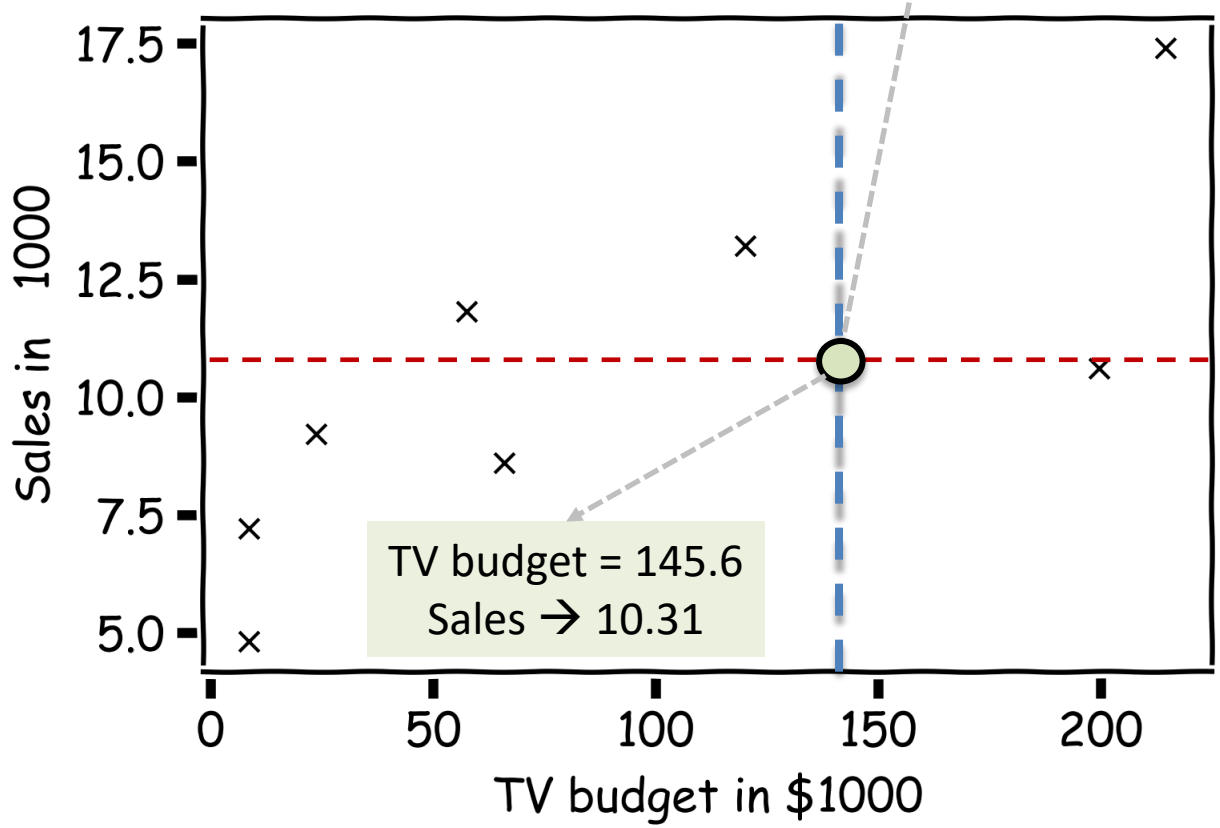
How do we find $\hat{f}(x)$?



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

Statistical Model

Simple idea is to take the mean of all y's, $\hat{f}(x) = \frac{1}{n}\sum_1^n y_i$



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

avg(Sales) = 10.31

Prediction vs. Estimation

- For some problems, what's important is obtaining $\hat{f}$, our estimate of f. These are called ***inference*** problems.

- When we use a set of measurements, $(x_{i,1}, \dots, x_{i,p})$ to predict a value for the response variable, we denote the ***predicted*** value by:

$$\hat{y}_i = \hat{f}(x_{i,1}, \dots, x_{i,p}).$$

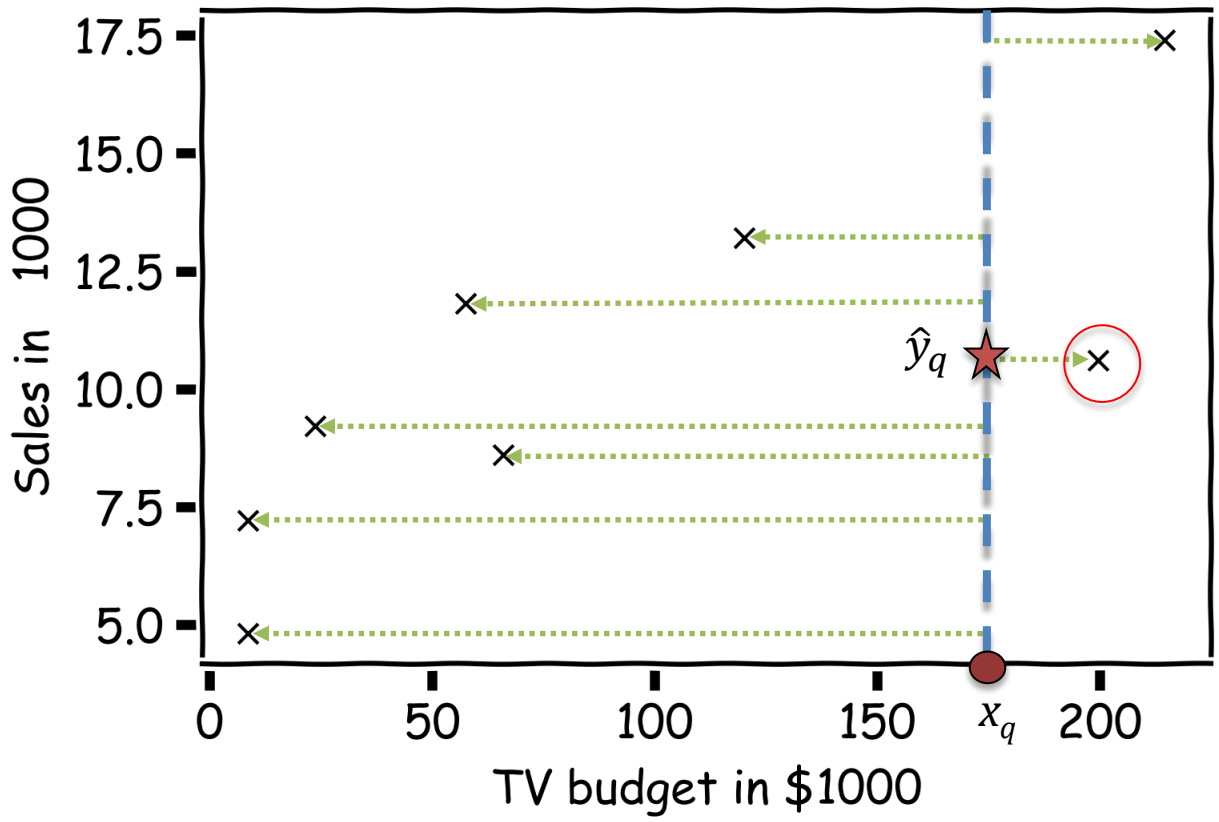
- For some problems, we don't care about the specific form of $\hat{f}$, we just want to make our predictions $\hat{y}$'s as close to the observed values y' as possible. These are called ***prediction*** problems.

Simple Prediction Model

What is $\hat{y}_q$ at some x_q ? For example x_q = 174

Find distances to all other points $D(x_q, x_i)$

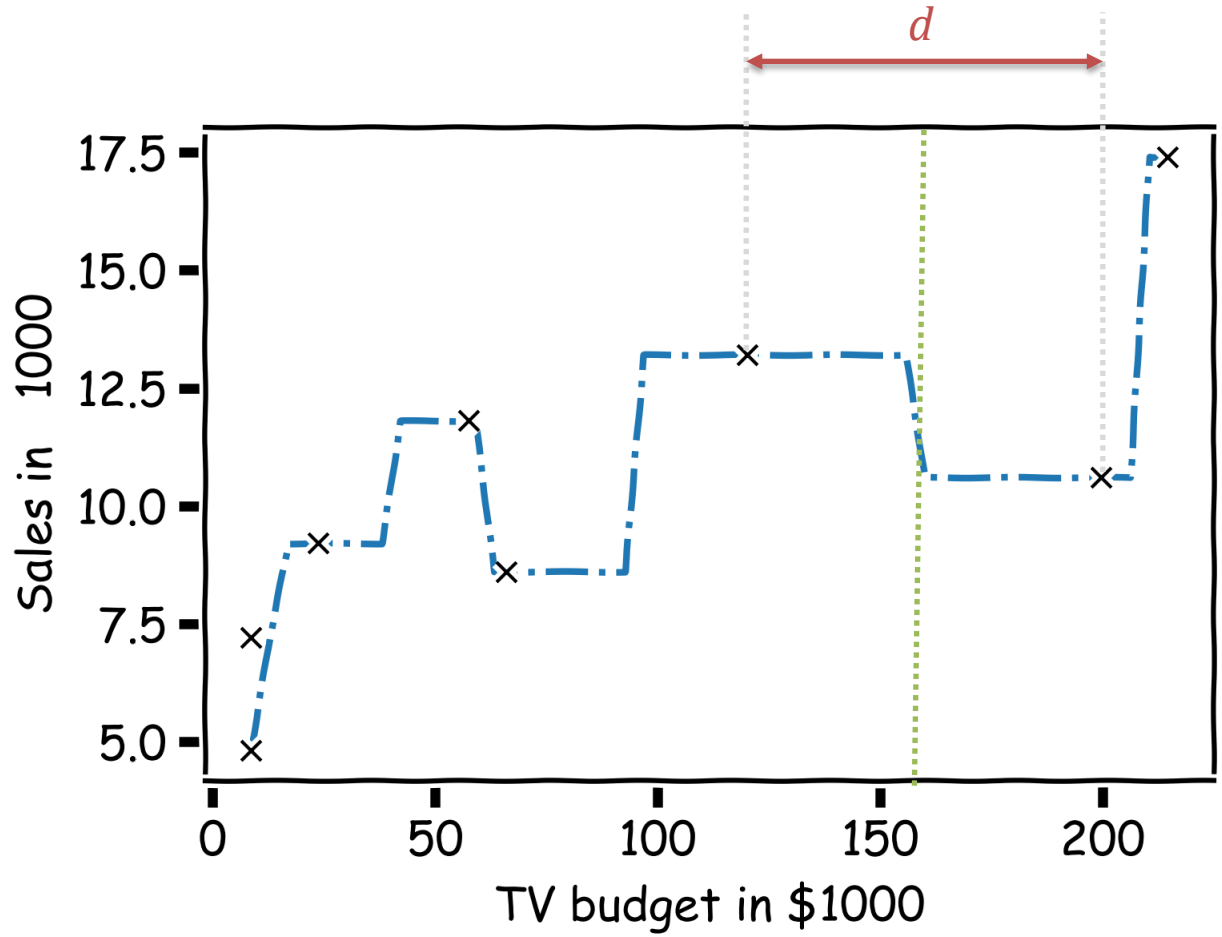
TV budget	|Xq-TV bud|
9.7	164.3
9.9	164.1
23.8	150.2
58.7	115.3
67.1	106.9
120	54
202	28
223	49



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

Simple Prediction Model

Do the same for "all" $x's$



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

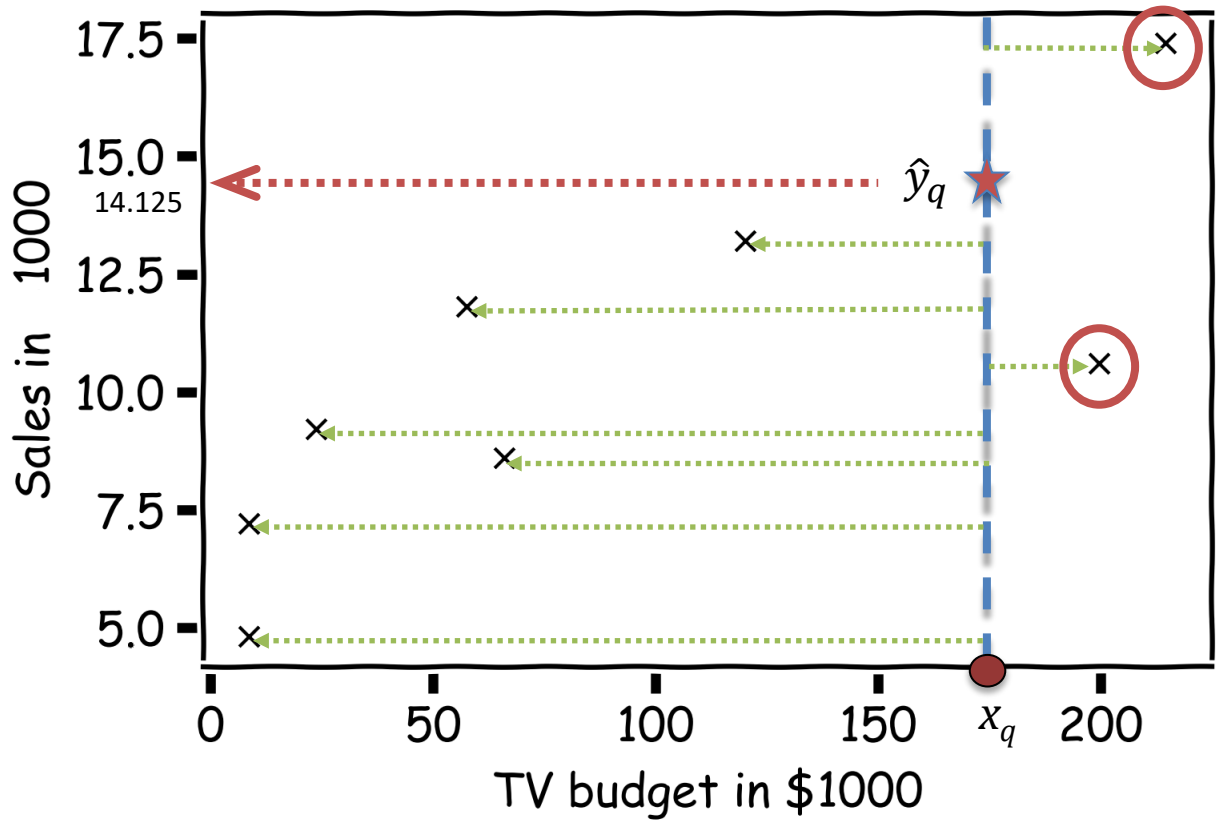
Extend the Prediction Model

What is $\hat{y}_q$ at some x_q ?

Find distances to all other points $D(x_q, x_i)$

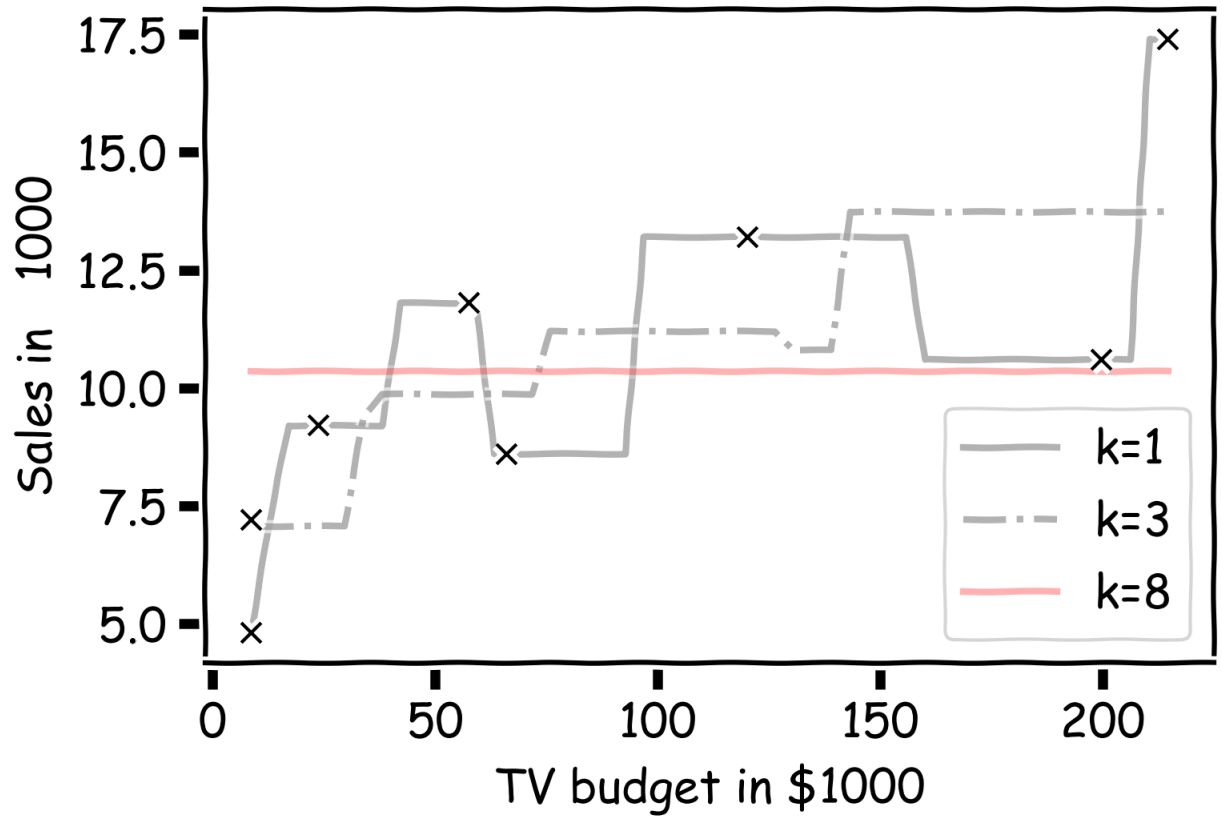
Find the k-nearest neighbors, $x_{q_1}, \ldots, x_{q_k}$

Predict $\widehat{y_q} = \frac{1}{k}\sum_i^k y_{q_i}$



TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35

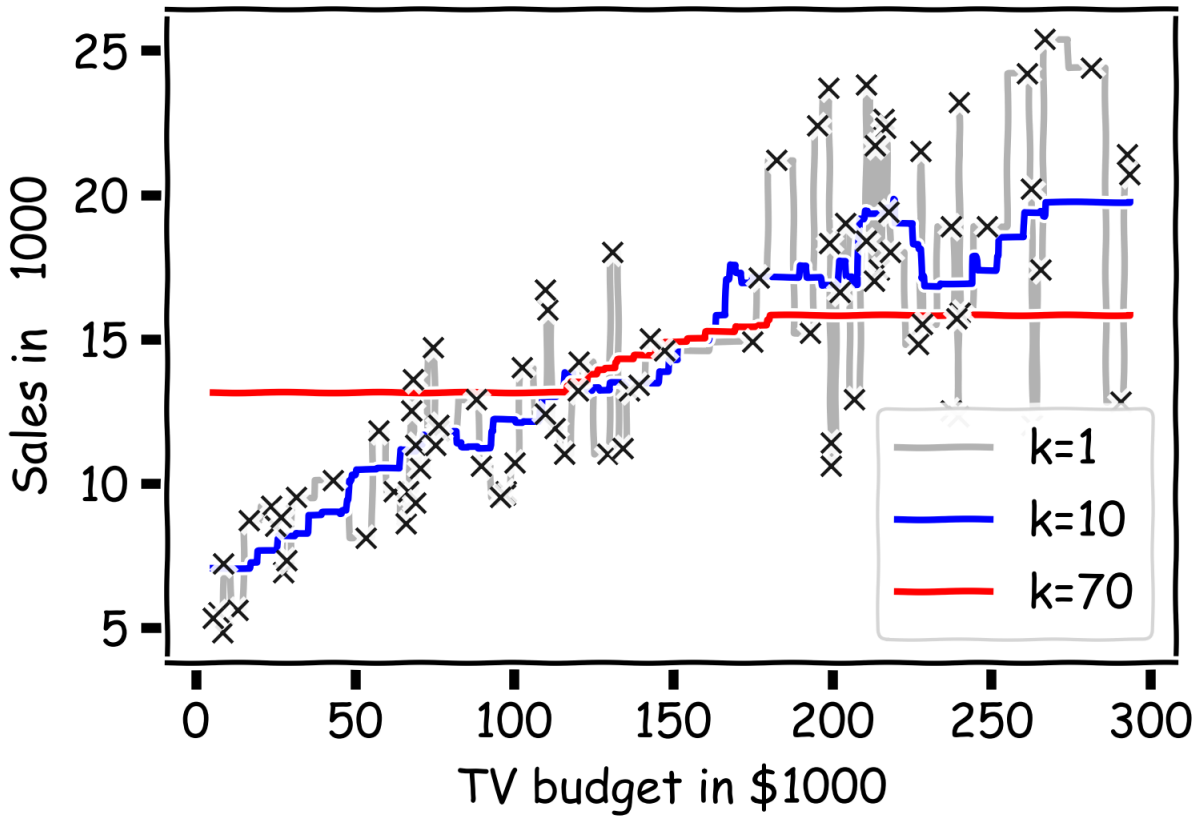
Simple Prediction Models



Which model is the best for our data?

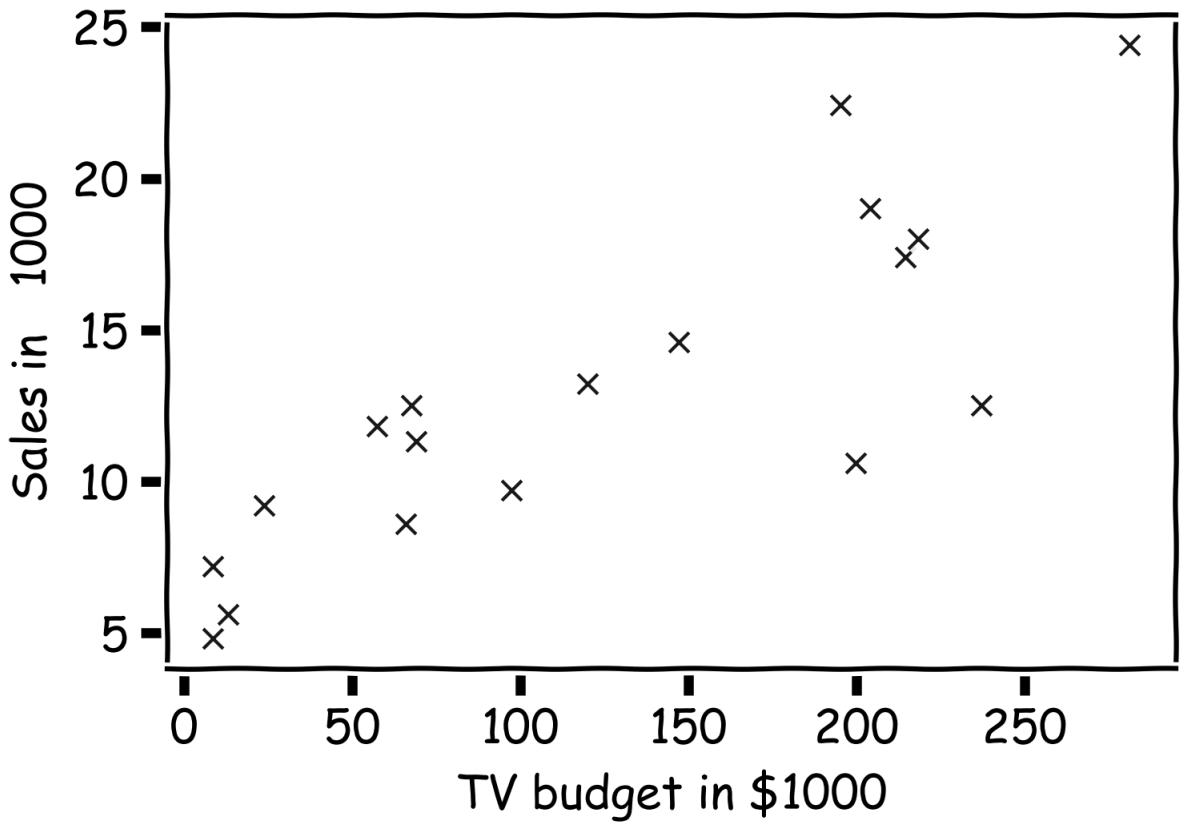
Simple Prediction Models

We can try different k-models on more data



Error Evaluation

Start with some data (18 observations).

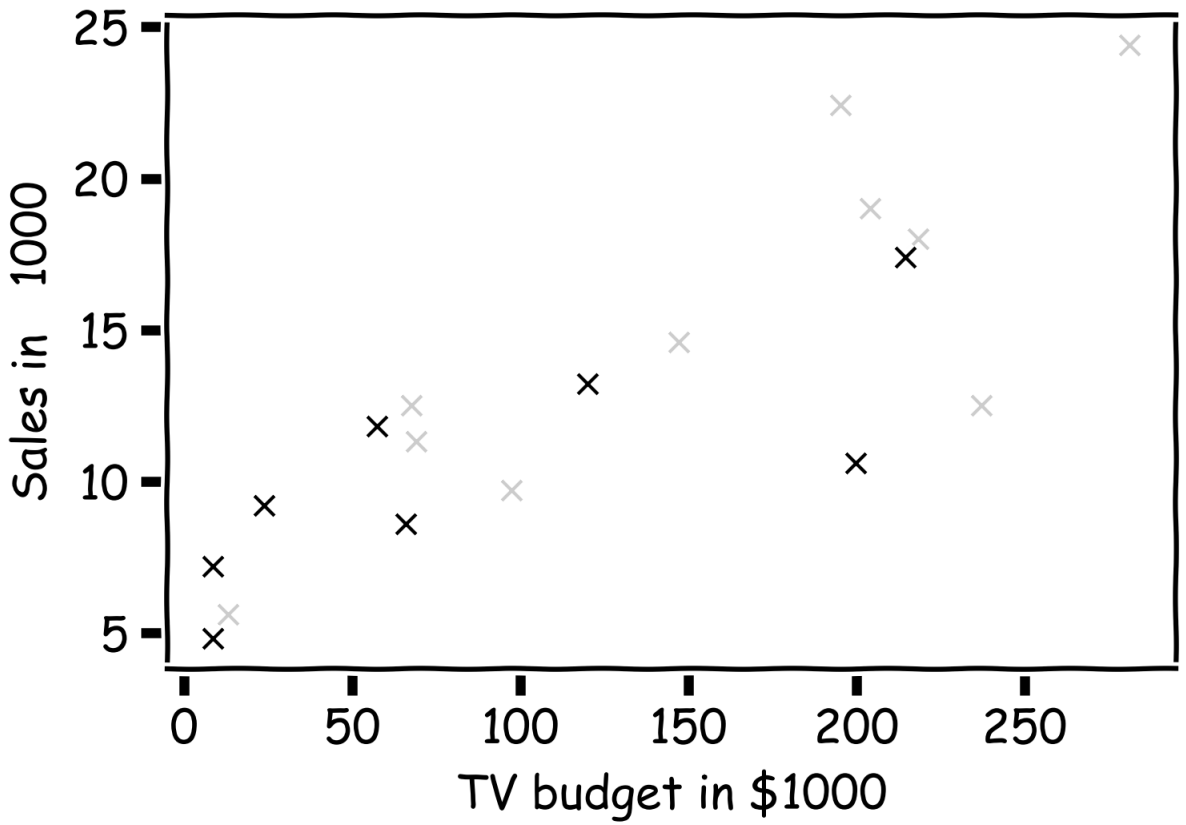


TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35
10.1	5.2
67.2	12.8
67.6	10.8
99.6	8.78
149.1	14.6
200	23.1
209	19.25
225	17.86
242	11.3
293.4	24.69

Error Evaluation

Hide some of the data from the model.
This is called **train-test** split.

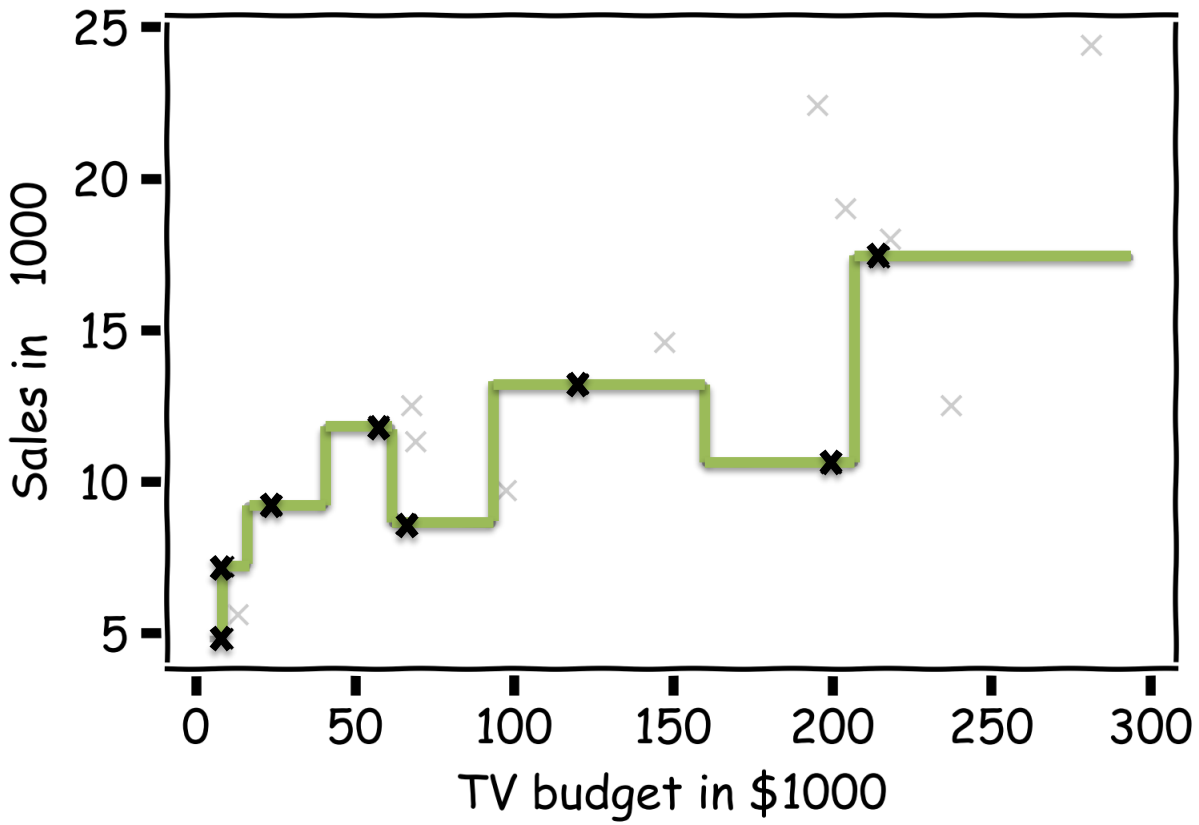
We use the train set (8 observations) to estimate $\hat{y}$, and the test set to evaluate the model.



	TV budget	Sales
TRAIN	9.7	4.75
	9.9	7.41
	23.8	9.1
	58.7	12.1
	67.1	8.12
	120	12.75
	202	10.9
	223	17.35
TEST	10.1	5.2
	67.2	12.8
	67.6	10.8
	99.6	8.78
	149.1	14.6
	200	23.1
	209	19.25
	225	17.86
	242	11.3
	293.4	24.69

Error Evaluation

Estimate $\hat{y}$ for *k=1*, by using only train data.



TRAIN

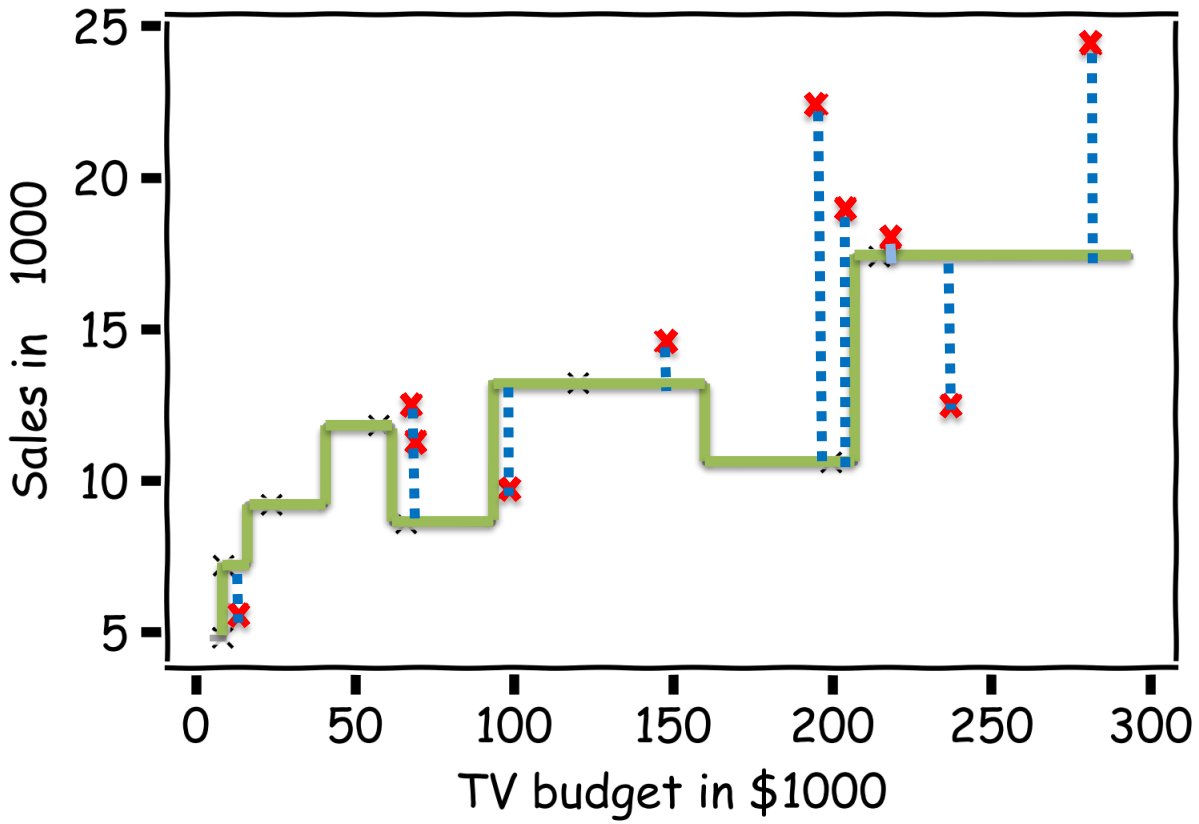
TV budget	Sales
9.7	4.75
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58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35
10.1	5.2
67.2	12.8
67.6	10.8
99.6	8.78
149.1	14.6
200	23.1
209	19.25
225	17.86
242	11.3
293.4	24.69

Error Evaluation

Now, we look at the data we have not used, the **test data** (red crosses).

Calculate the **residuals** $(y_i - \hat{y}_i)$.

TV budget	Sales	$\widehat{Sale}$	Δ
10.1	5.2	7.41	-2.21
67.2	12.8	8.12	4.68
67.6	10.8	8.12	2.68
99.6	8.78	12.75	-3.97
149.1	14.6	12.75	1.85
200	23.1	10.9	12.2
209	19.25	10.9	8.35
225	17.86	17.35	0.51
242	11.3	17.35	-6.05
293.4	24.69	17.35	7.35



TEST

TV budget	Sales
9.7	4.75
9.9	7.41
23.8	9.1
58.7	12.1
67.1	8.12
120	12.75
202	10.9
223	17.35
10.1	5.2
67.2	12.8
67.6	10.8
99.6	8.78
149.1	14.6
200	23.1
209	19.25
225	17.86
242	11.3
293.4	24.69

Error Evaluation

- In order to quantify how well a model performs, we define a ***loss*** or ***error function.***
- A common loss function for quantitative outcomes is the **Mean Squared Error (MSE):**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

- The quantity $y_i - \hat{y}_i$ is called a ***residual*** and measures the error at the *i*-th prediction.
- ***Note:*** The **R**oot of the **M**ean of the **S**quared **E**rrors (RMSE) is also commonly used.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}$$

Error Evaluation

TV budget	Sales	$\widehat{Sale}$	Δ	Δ²
10.1	5.2	7.41	-2.21	4.88
67.2	12.8	8.12	4.68	21.9
67.6	10.8	8.12	2.68	7.18
99.6	8.78	12.75	-3.97	15.76
149.1	14.6	12.75	1.85	3.42
200	23.1	10.9	12.2	148.84
209	19.25	10.9	8.35	69.72
225	17.86	17.35	0.51	0.26
242	11.3	17.35	-6.05	36.6
293.4	24.69	17.35	7.35	54.02

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

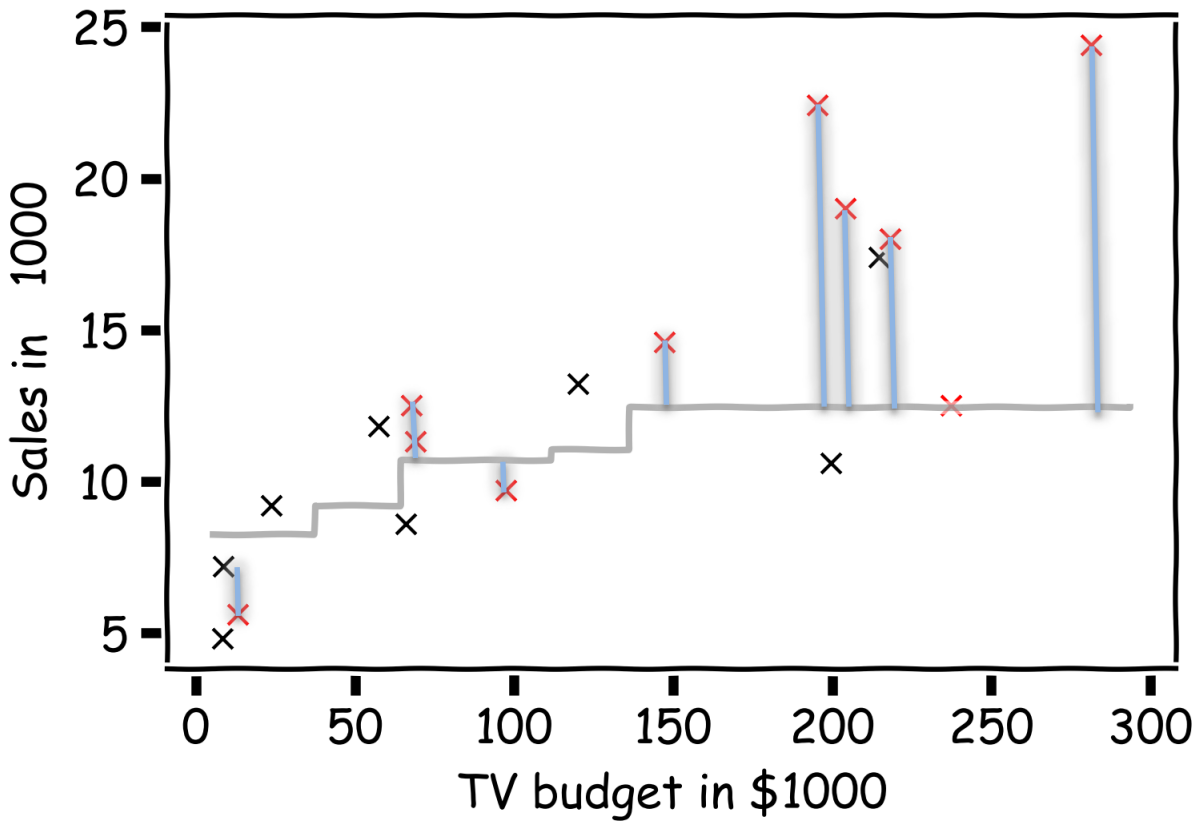
MSE = 36.2453

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}$$

RMSE = 6.0204

Error Evaluation

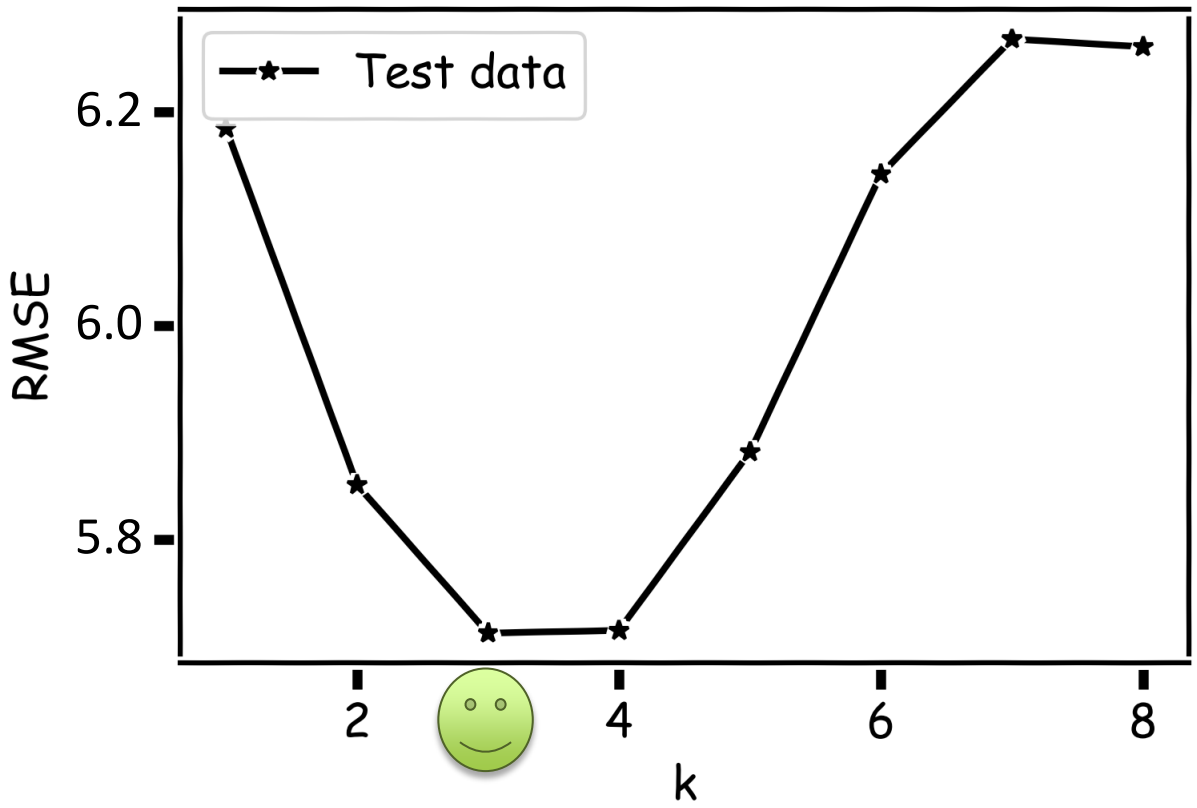
Do the same for *k=3.*



Model Comparison

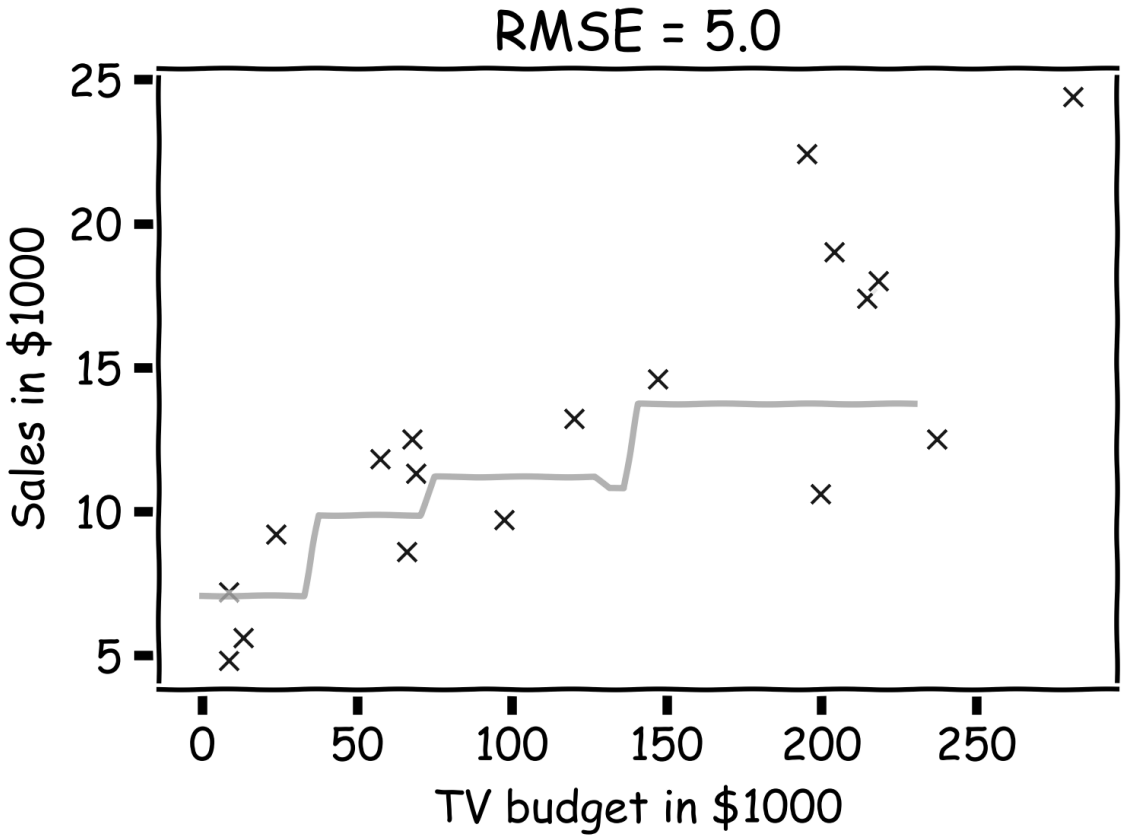
Do the same for all *k*'s and compare the RMSEs.

For *k=3* seems to be the best KNN model.



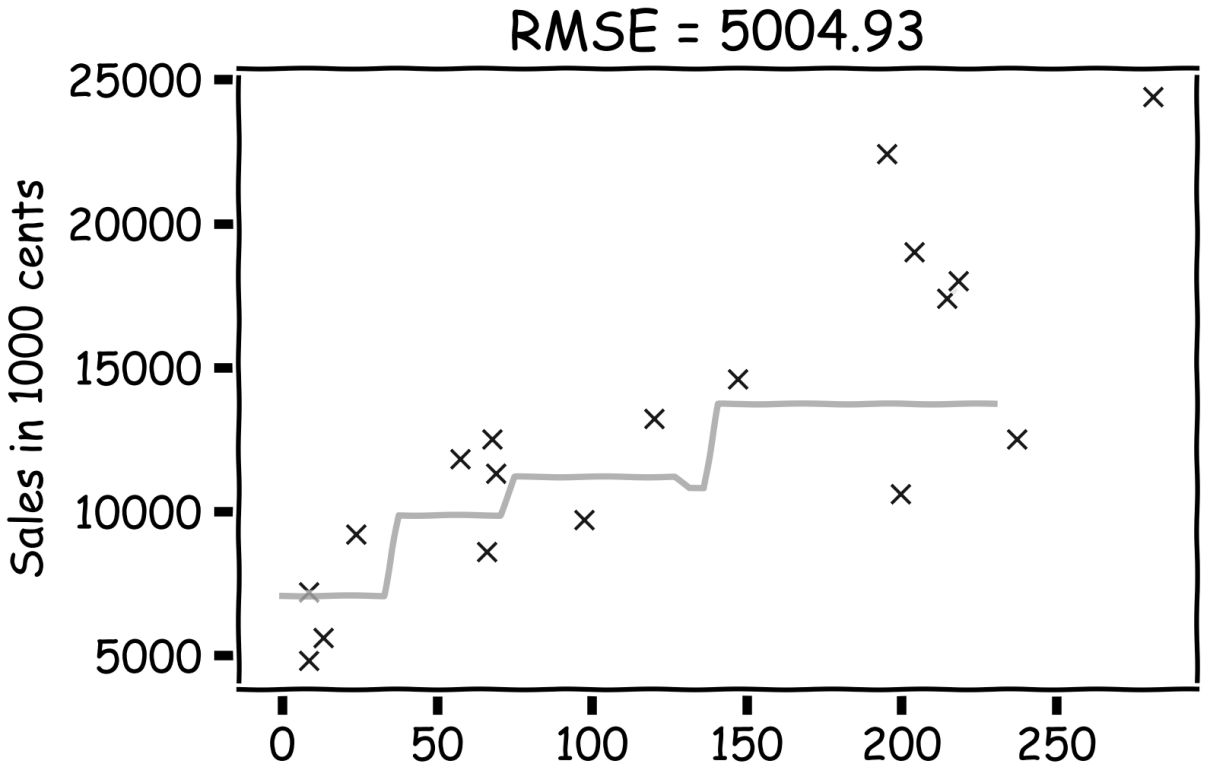
Model fitness

For a subset of the data, calculate the RMSE for *k=3.*



Is RMSE=5.0 good enough?

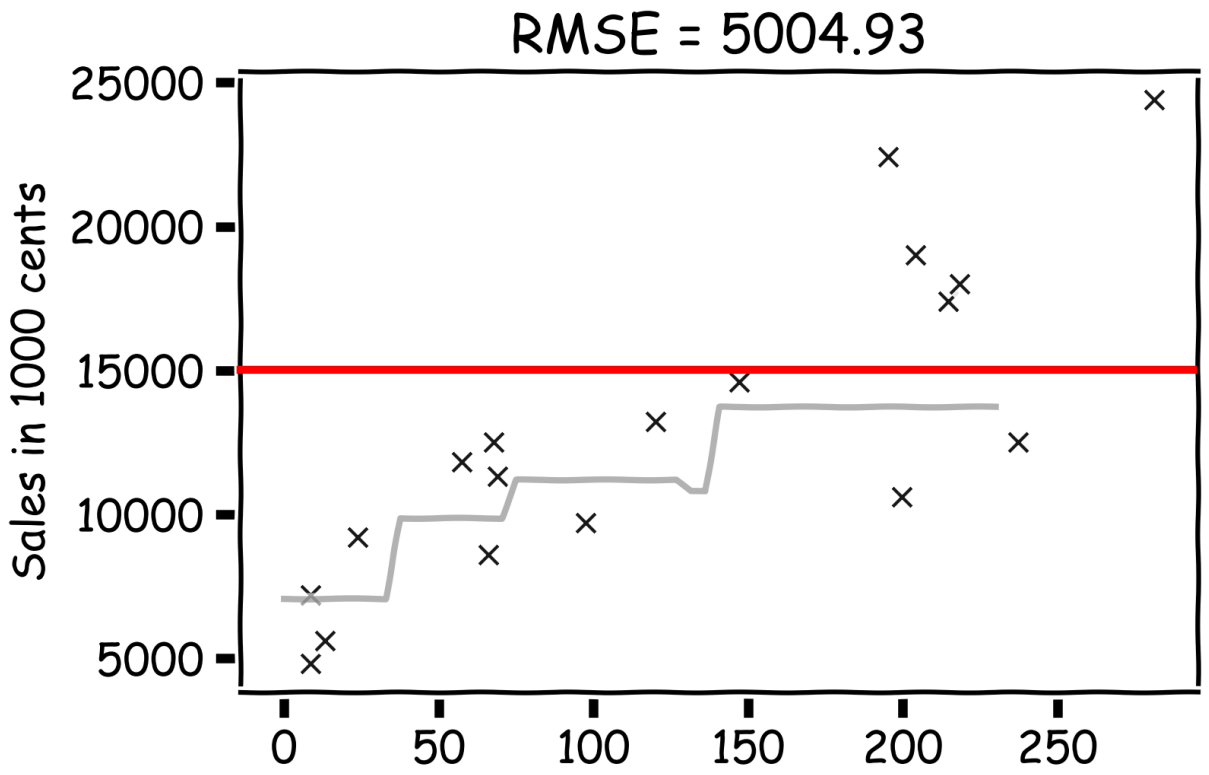
What if we measure the Sales in cents instead of dollars?



Is RMSE=5004.93 that good?

Model fitness

It is better if we compare it to something.



We will use the simplest model:

$$\hat{y} = \frac{1}{n}\sum_{i}^{n} y_i$$

R-squared

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2}$$

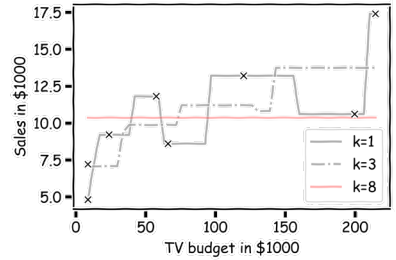
- If our model is as good as the mean value, $\bar{y}$, then $R^2 = 0$
- If our model is perfect then $R^2 = 1$
- R^2 can be negative if the model is worst than the average. This can happen when we evaluate the model in the test set.

$\bar{y}$ = 10.31 (calculated from the train data)

TV budget	Sales	$\widehat{Sale}$	Δ	Δ²	$\bar{\Delta}^2$
10.1	5.2	7.41	-2.21	4.88	26.11
67.2	12.8	8.12	4.68	21.9	6.2
67.6	10.8	8.12	2.68	7.18	0.24
99.6	8.78	12.75	-3.97	15.76	2.34
149.1	14.6	12.75	1.85	3.42	18.4
200	23.1	10.9	12.2	148.84	163.58
209	19.25	10.9	8.35	69.72	79.92
225	17.86	17.35	0.51	0.26	57
242	11.3	17.35	-6.05	36.6	0.98
293.4	24.69	17.35	7.35	54.02	206.78
				362.6	561.57

R^2 = 0.354

Linear Models (Inference)



- Note that in building our kNN model for prediction, we did not compute a closed form for $\hat{f}$.

- What if we ask the question: **"For particular market, how much sales do we expect if we double the TV advertising budget?"**

- **Alternatively**, we can build a model by first assuming a simple form of f:

$$Y = f(X) + \epsilon = \beta_1 X + \beta_0 + \epsilon.$$

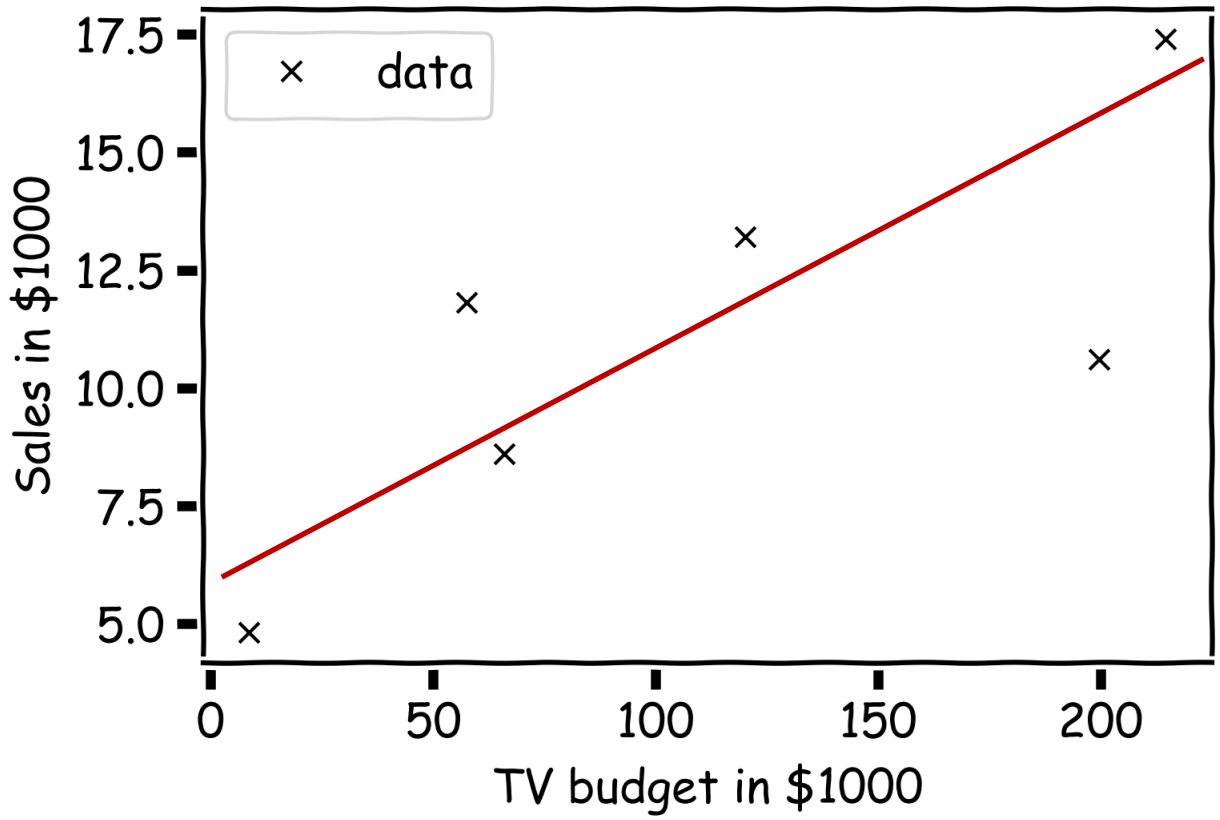
- … then it follows that our estimate is:

$$\widehat{Y} = \widehat{f}(X) = \widehat{\beta_1} X + \widehat{\beta_0}$$

- where $\hat{\beta}_1$ and $\hat{\beta}_0$ are **estimates** of β_1 and β_0, respectively, that we compute by using the observations.

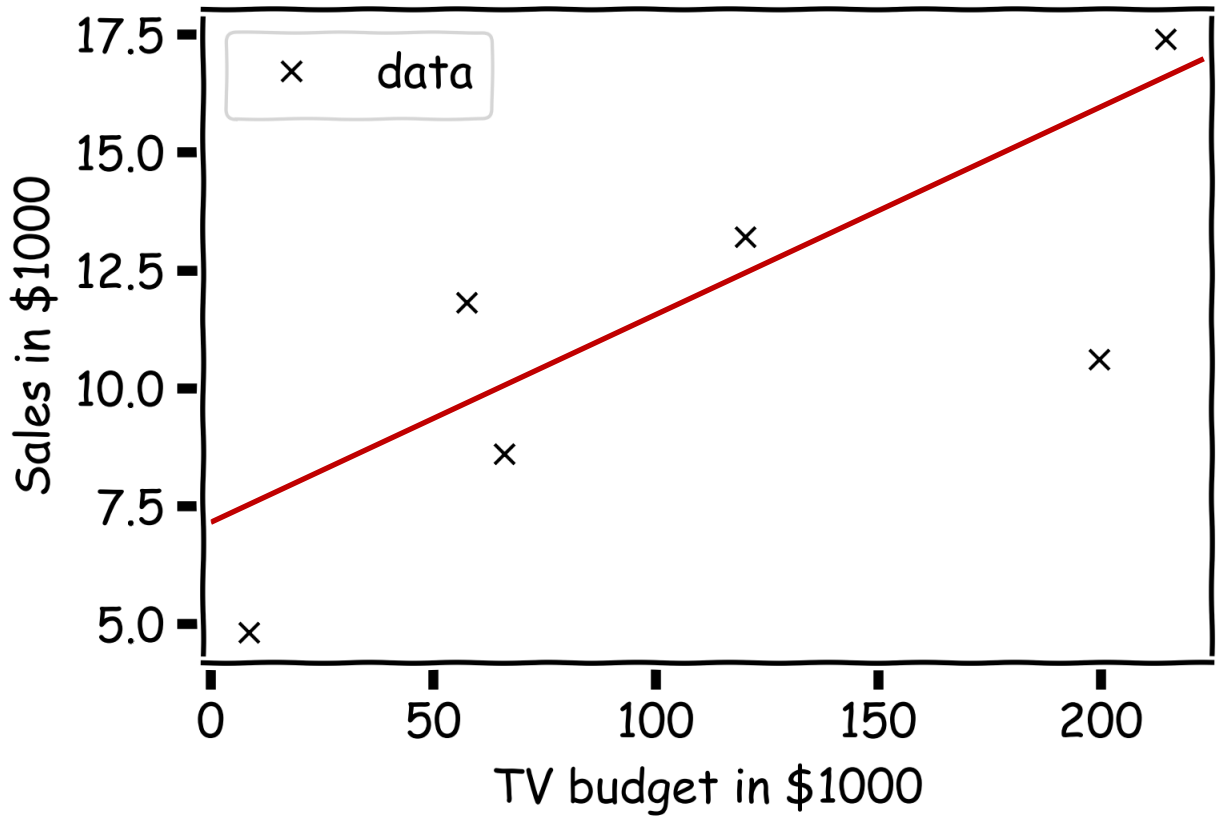
Estimate of the regression coefficients

Is this line good?



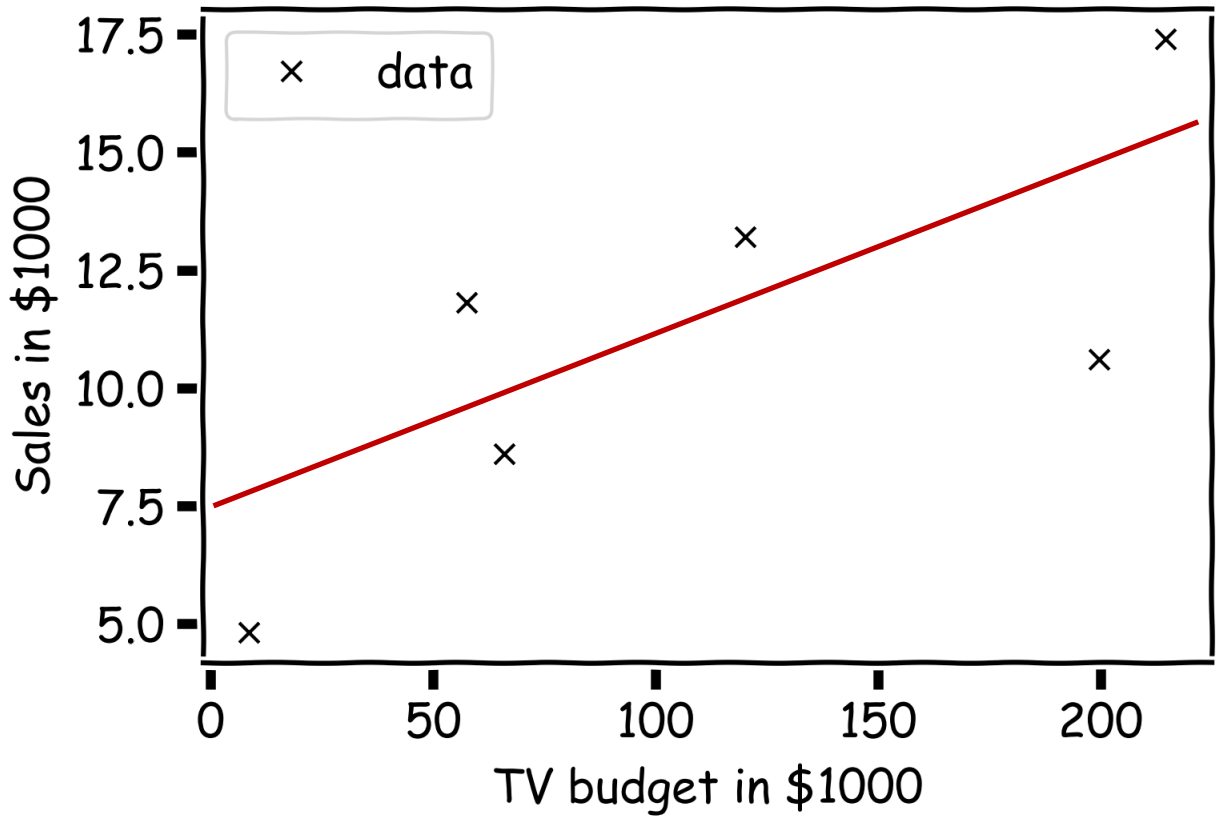
Estimate of the regression coefficients (cont)

Maybe this one?



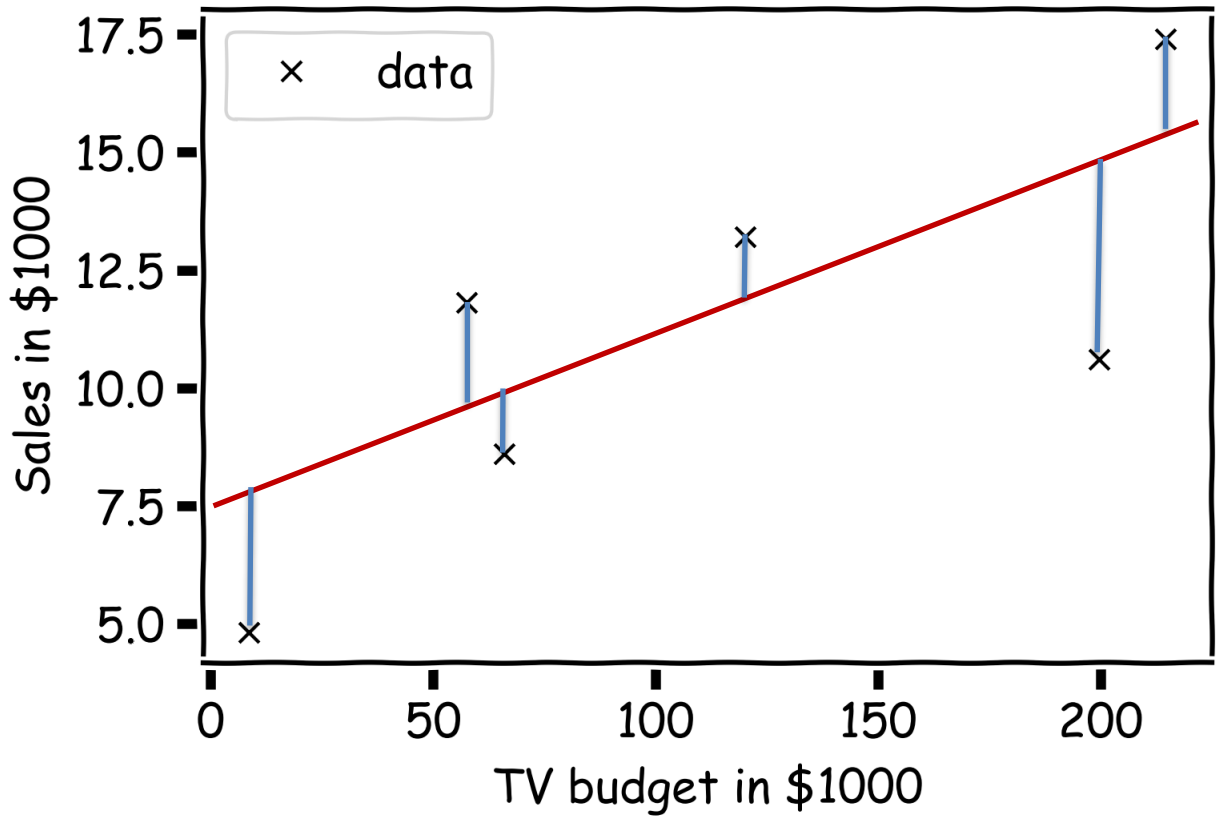
Estimate of the regression coefficients (cont)

Or this one?



Estimate of the regression coefficients (cont)

Question: Which line is the best?
First calculate the residuals



Estimate of the regression coefficients (cont.)

- Again, we use MSE as our **loss function**,

$$L(\beta_0, \beta_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - (\beta_1 X + \beta_0)\right]^2.$$

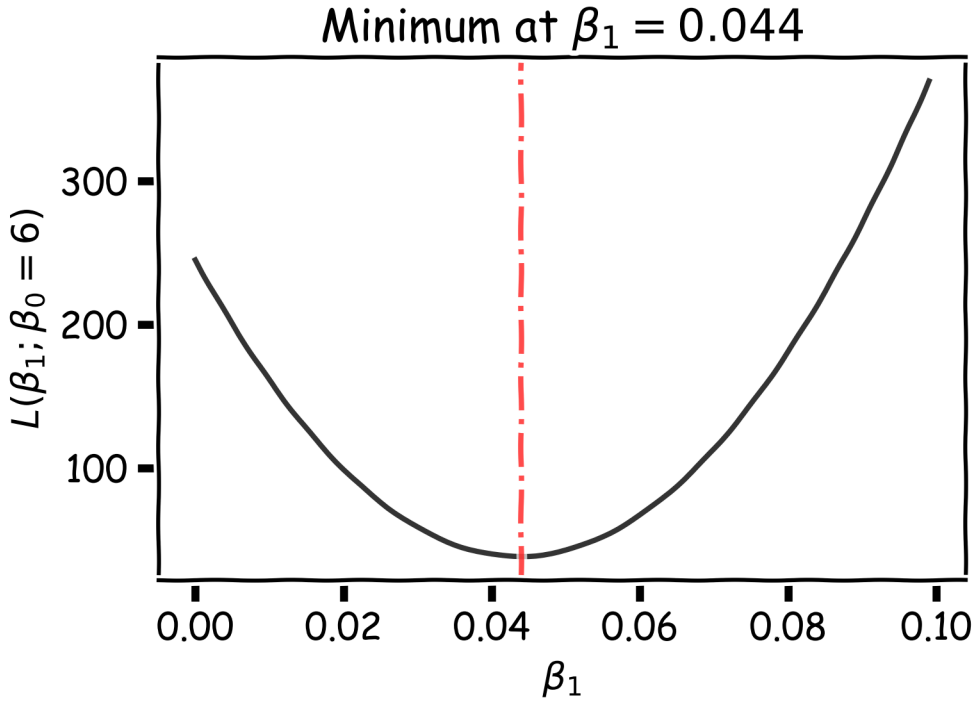
- We choose $\hat{\beta}_1$ and $\hat{\beta}_0$ in order to minimize the predictive errors made by our model, i.e., minimize our loss function.
- Then the optimal values for $\hat{\beta}_0$ and $\hat{\beta}_1$ should be:

$$\widehat{\beta}_0, \widehat{\beta}_1 = \underset{\beta_0, \beta_1}{\operatorname{argmin}} L(\beta_0, \beta_1).$$

Estimate of the regression coefficients: brute force

- A way to estimate $\text{argmin}_{\beta_0,\beta_1} L$ is to calculate the loss function for every possible β_0 and β_1. Then select the β_0 and β_1 where the loss function is minimum.

- E.g., the loss function for different β_1 when β_0 is fixed to be 6:



Estimate of the regression coefficients: exact method

Take the partial derivatives of L with respect to β_0 and β_1, set to zero, and find the solution to that equation. This procedure will give us explicit formulae for $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sum_i (x_i - \overline{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y}$ and $\bar{x}$ are sample means.

Regression line: $\widehat{Y} = \widehat{\beta}_1 X + \widehat{\beta}_0$

$$L(\beta_0, \beta_1) = \frac{1}{n} \sum_i [y_i - (\beta_0 - \beta_1 x_i)]^2$$

$$\frac{dL(\beta_0, \beta_1)}{d\beta_0} = 0$$

$$\Rightarrow \frac{2}{n} \sum_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\Rightarrow \frac{1}{n} \sum_i y_i - \beta_0 - \beta_1 \frac{1}{n} \sum_i x_i = 0$$

$$\Rightarrow \beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\frac{dL(\beta_0, \beta_1)}{d\beta_1} = 0$$

$$\Rightarrow \frac{2}{n} \sum_i (y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0$$

$$\Rightarrow -\sum_i x_i y_i + \beta_0 \sum_i x_i + \beta_1 \sum_i x_i^2 = 0$$

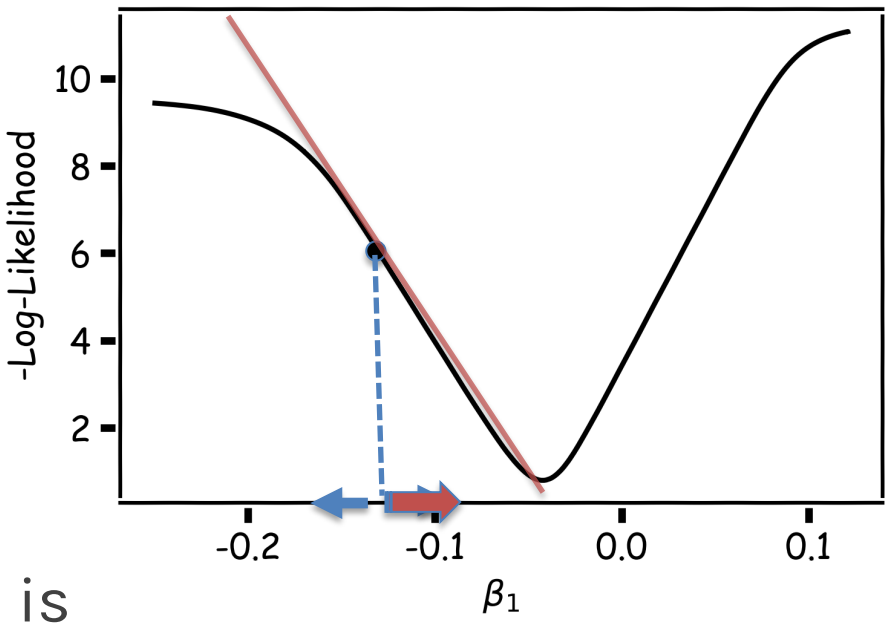
$$\Rightarrow -\sum_i x_i y_i + (\bar{y} - \beta_1 \bar{x}) \sum_i x_i + \beta_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \beta_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right) = \sum_i x_i y_i - n\bar{x}\bar{y}$$

$$\Rightarrow \beta_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2}$$

$$\Rightarrow \beta_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

Estimate of the regression coefficients: gradient descent



A more flexible method is

- Start from a random point
  1. Determine which direction to go to reduce the loss (left or right)
  2. Compute the slope of the function at this point and step to the right if the slope is negative or step to the left if the slope is positive
  3. Goto to #1

Estimate of the regression coefficients: gradient descent

- We know that we want to go in the opposite direction of the derivative, and we know we want to be making a step proportionally to the derivative.

Notation: $w = [\beta_0, \beta_1]$

- Making a step means:

$$w^{new} = w^{old} + step$$

Opposite direction of the derivative and proportional to the derivative means:

$$w^{new} = w^{old} - \lambda \frac{d\mathcal{L}}{dw}$$

Learning Rate

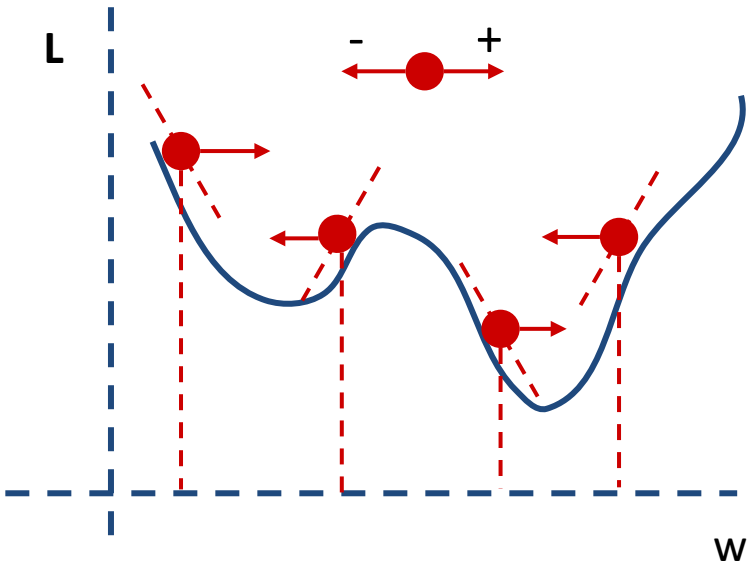
Change to more conventional notation:

$$w^{(i+1)} = w^{(i)} - \lambda \frac{d\mathcal{L}}{dw}$$

Estimate of the regression coefficients: gradient descent

- **Summary of Gradient Descent**
  - Algorithm for first-order optimization to find a minimum of a function.
  - It is an iterative method.
  - L is decreasing in the direction of the negative derivative.
  - The learning rate is controlled by the magnitude of λ.

$$w^{(i+1)} = w^{(i)} - \lambda \frac{d\mathcal{L}}{dw}$$



Interpretation of Predictors

Question: What do you think a predictor coefficient means?

$$Sales = 7.5 + 0.04\,TV$$

- What does 7.5 mean and what does 0.04 mean?
- If we increase the TV by \$1000, what would you expect the increase in sales to be?
- What if? $Sales = 7.5 + 1.01\,TV$

The interpretation of the predictors depends on the values.
Decisions depend on how much we trust these values.

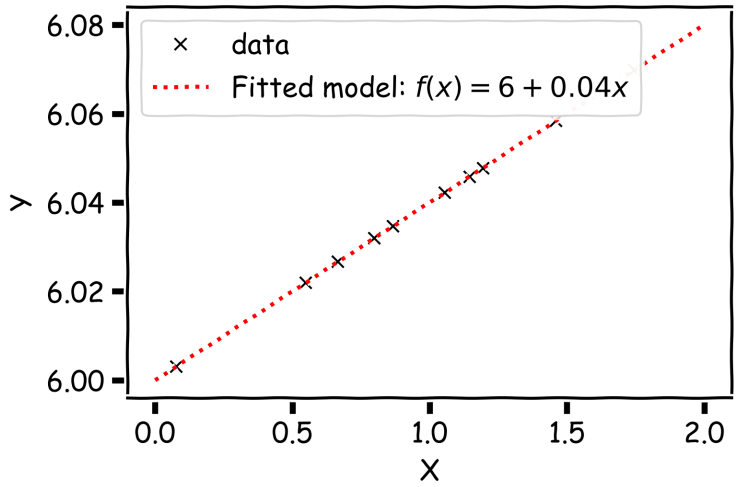
Confidence intervals for the predictors estimates

- We interpret the ε term in our observation

$$y = f(x) + \epsilon$$

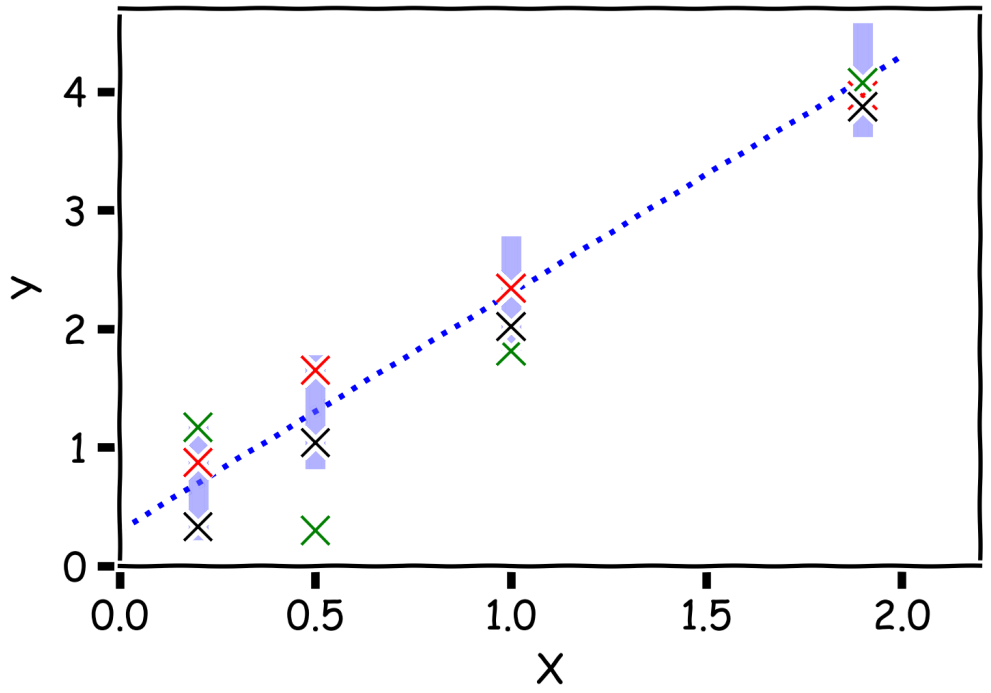
to be the noise introduced by random variations in natural systems or imprecisions of our scientific instruments.

- If we knew the exact form of $f(x)$, for example, $f(x) = \beta_0 + \beta_1 x$, and there was no ε , then estimating the $\hat{\beta}'s$ would have been exact.
- **However**, three things happen, which result in mistrust of the values of $\hat{\beta}'s$:
  - $\boldsymbol{\varepsilon}$ is always there
  - we do not know the exact form of $f(x)$
  - limited sample size



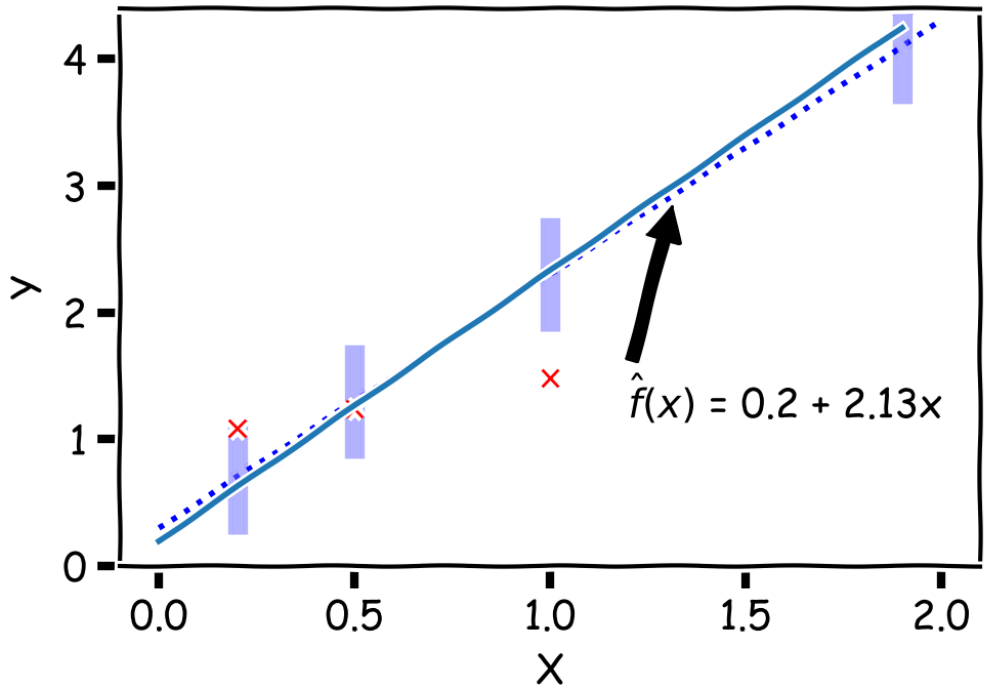
Confidence intervals for the predictors estimates (cont)

- But due to error, every time we measure the response Y for a fixed value of X, we will obtain a different observation.
- We have 3 “realizations” on the picture



Confidence intervals for the predictors estimates (cont)

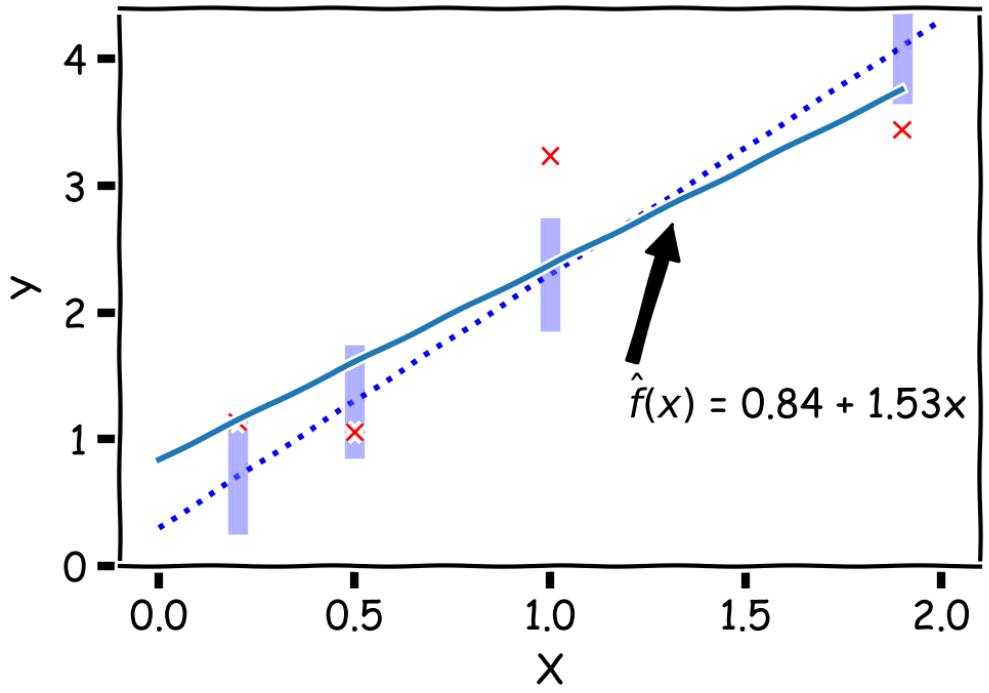
- For each one of those “realizations”, we could fit a model and estimate $\hat{\beta}_0$ and $\hat{\beta}_1$.



$\beta_0 = 0.2$
$\beta_1 = 2.13$

Confidence intervals for the predictors estimates (cont)

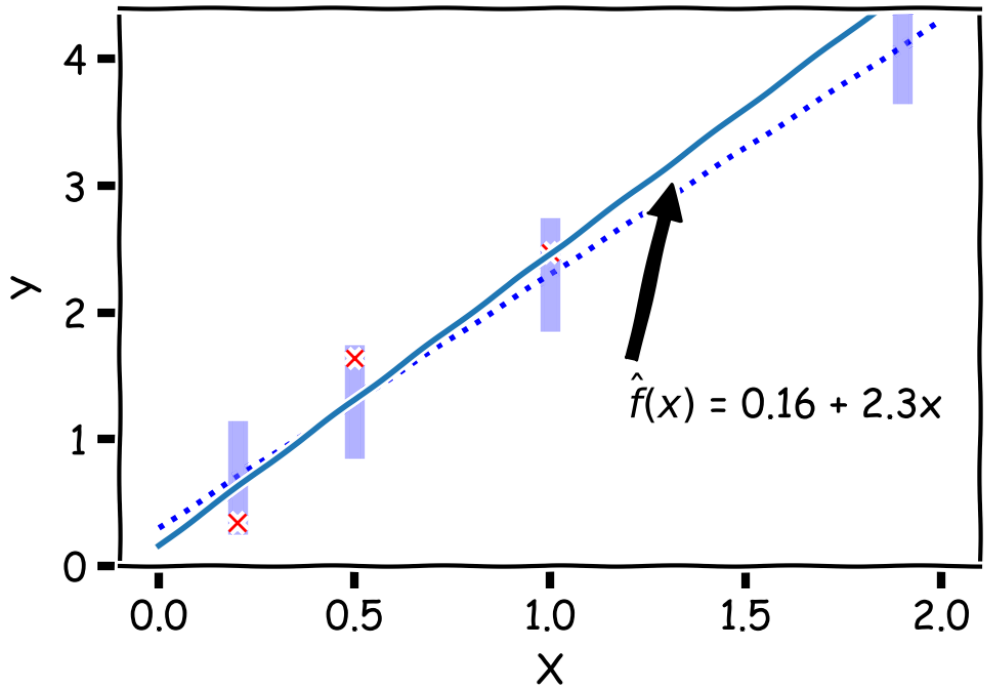
- For each one of those “realizations”, we could fit a model and estimate, $\hat{\beta}_0$ and $\hat{\beta}_1$.



$\boldsymbol{\beta}_0 = 0.84$
$\boldsymbol{\beta}_1 = 1.53$

Confidence intervals for the predictors estimates (cont)

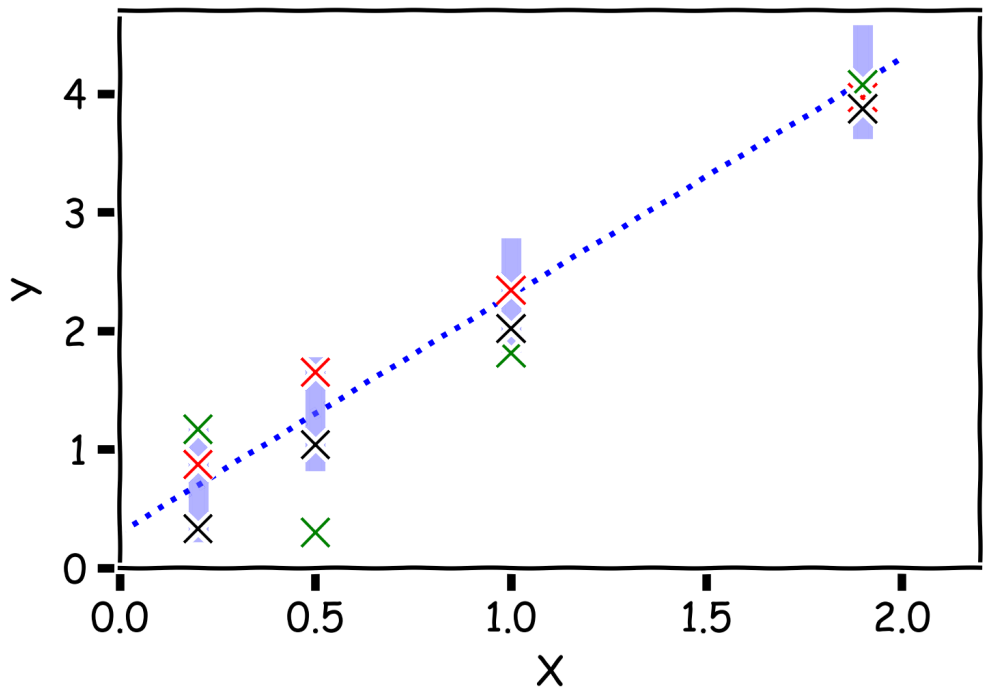
- For each one of those “realizations”, we could fit a model and estimate, $\hat{\beta}_0$ and $\hat{\beta}_1$.



$\boldsymbol{\beta}_0 = 0.16$
$\boldsymbol{\beta}_1 = 2.3$

Confidence intervals for the predictors estimates (cont)

- For the three realizations of the experiment we obtain different regression lines.



I) $\beta_0 = 0.2$, $\beta_1 = 2.13$

II) $\beta_0 = 0.84$, $\beta_1 = 1.53$

III) $\beta_0 = 0.16$, $\beta_1 = 2.3$

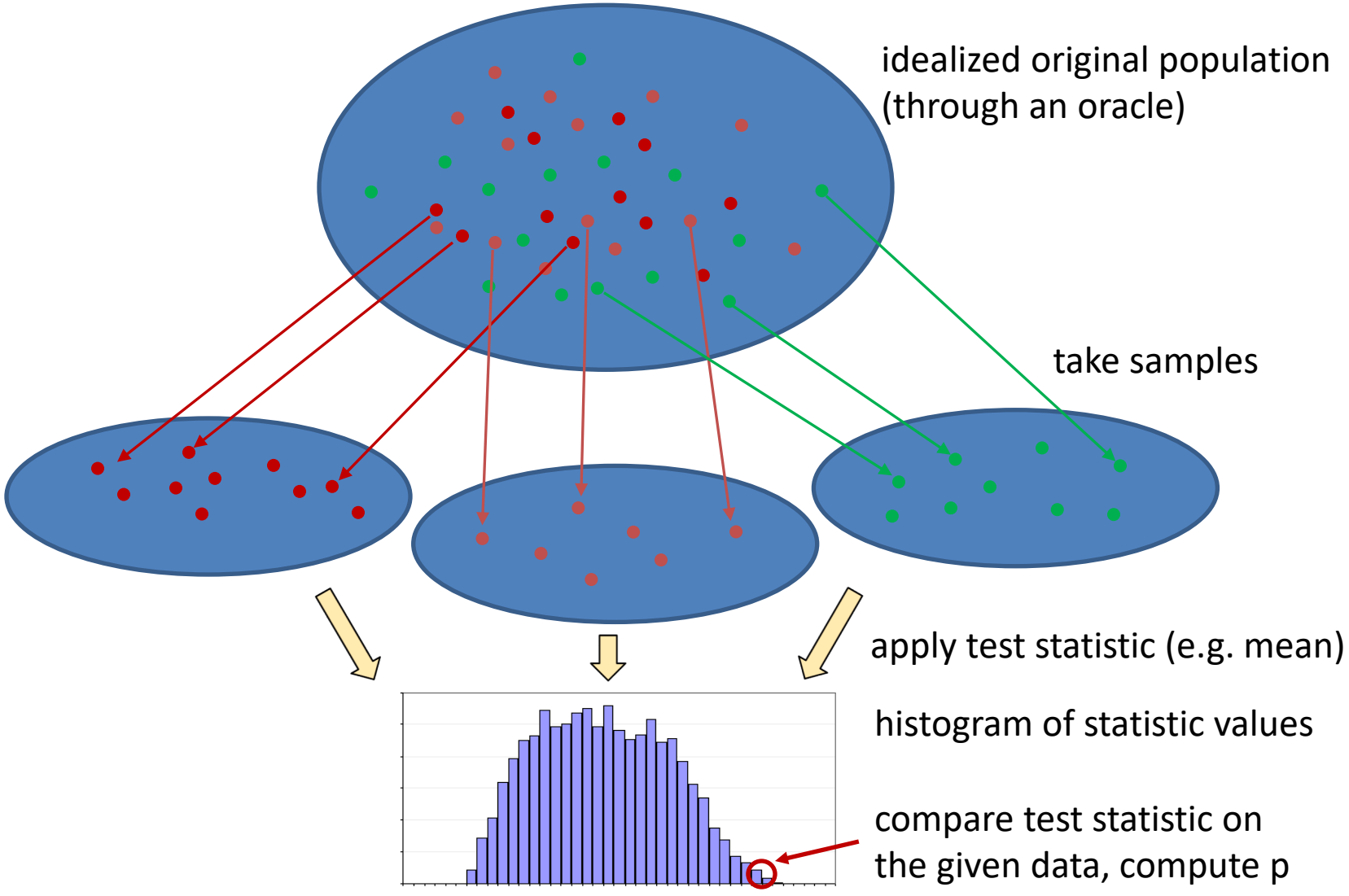
Bootstrapping for Estimating Sampling Error

Definition

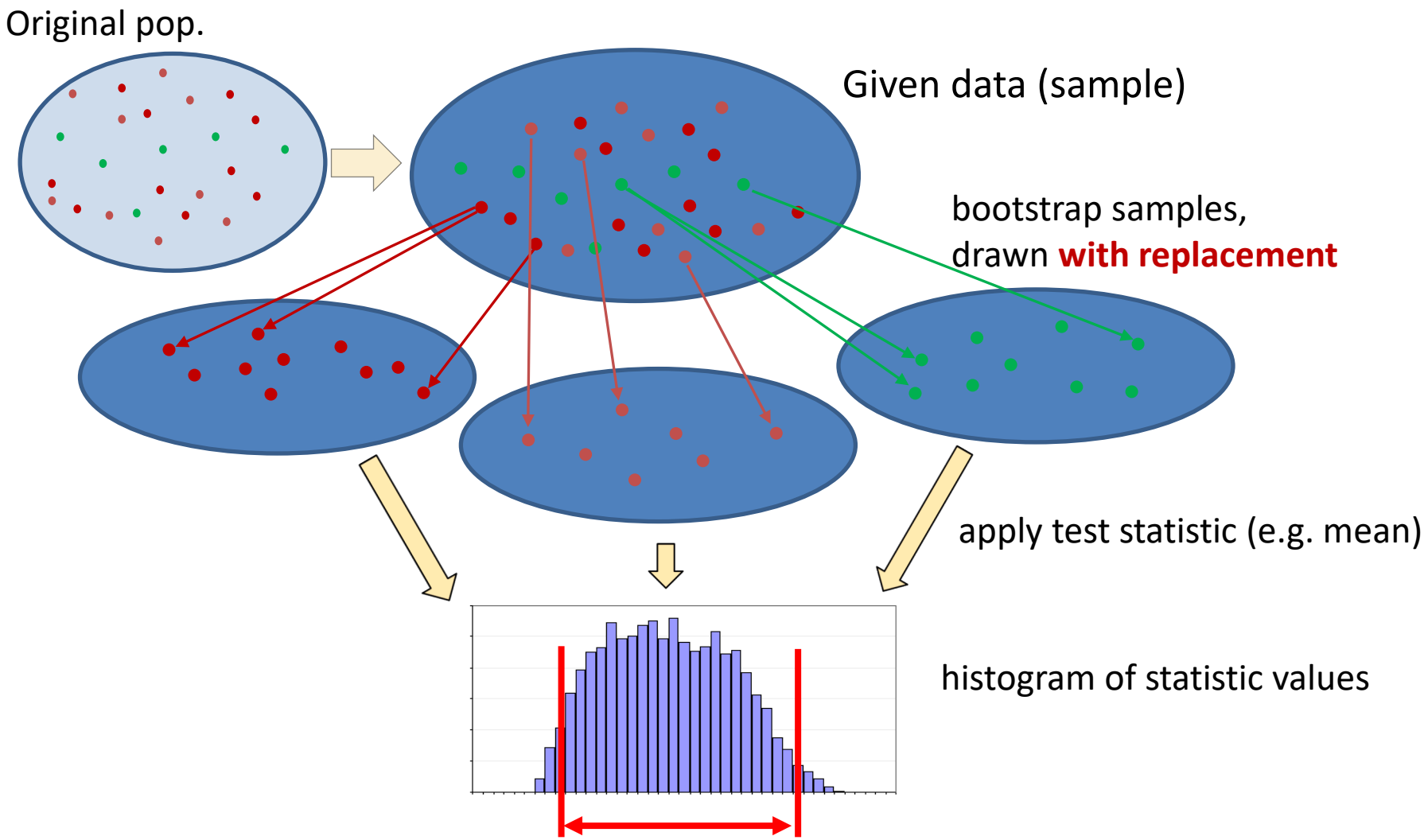
- Bootstrapping is the practice of estimating properties of an estimator by measuring those properties by, for example, sampling from the observed data.

- For example, we can compute $\hat{\beta}_0$ and $\hat{\beta}_1$ multiple times by randomly sampling from our data set. We then use the variance of our multiple estimates to approximate the true variance of $\hat{\beta}_0$ and $\hat{\beta}_1$.

Idealized Sampling



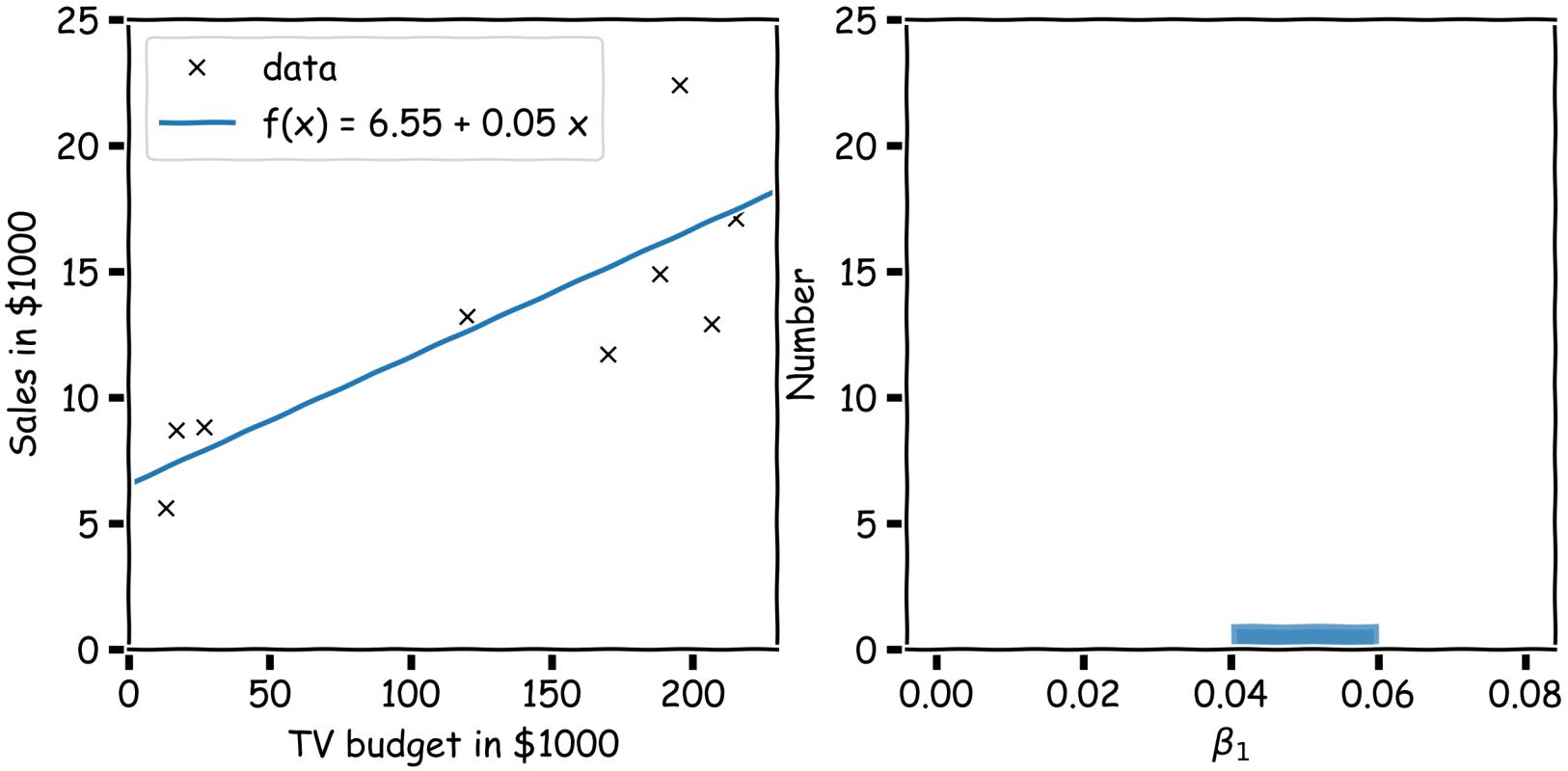
Bootstrap Sampling



The region containing 95% of the samples is a 95% confidence interval (CI)

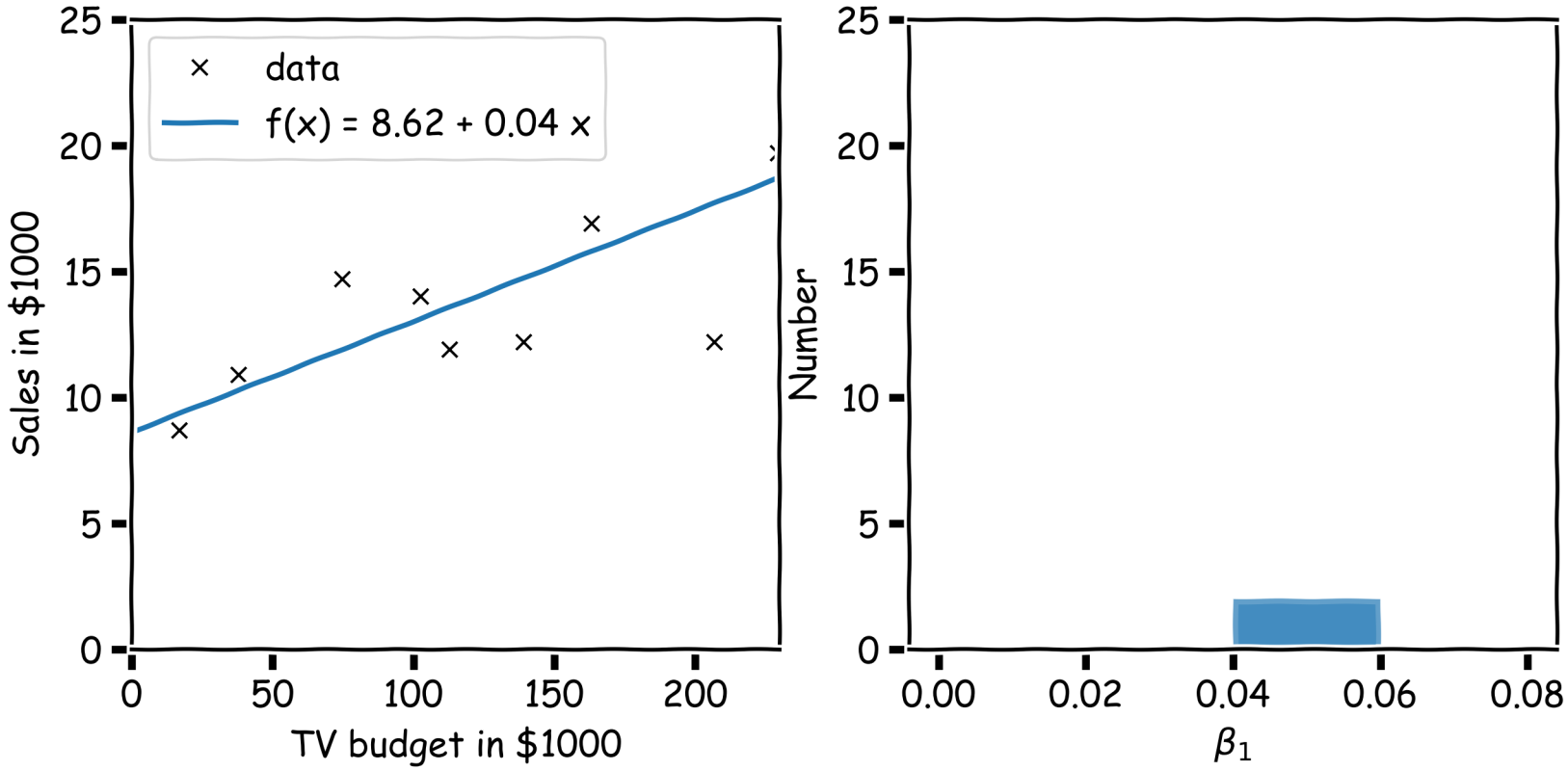
Confidence intervals for the predictors estimates (cont)

We sample multiple times and calculate $\hat{\beta}_0$ and $\hat{\beta}_1$



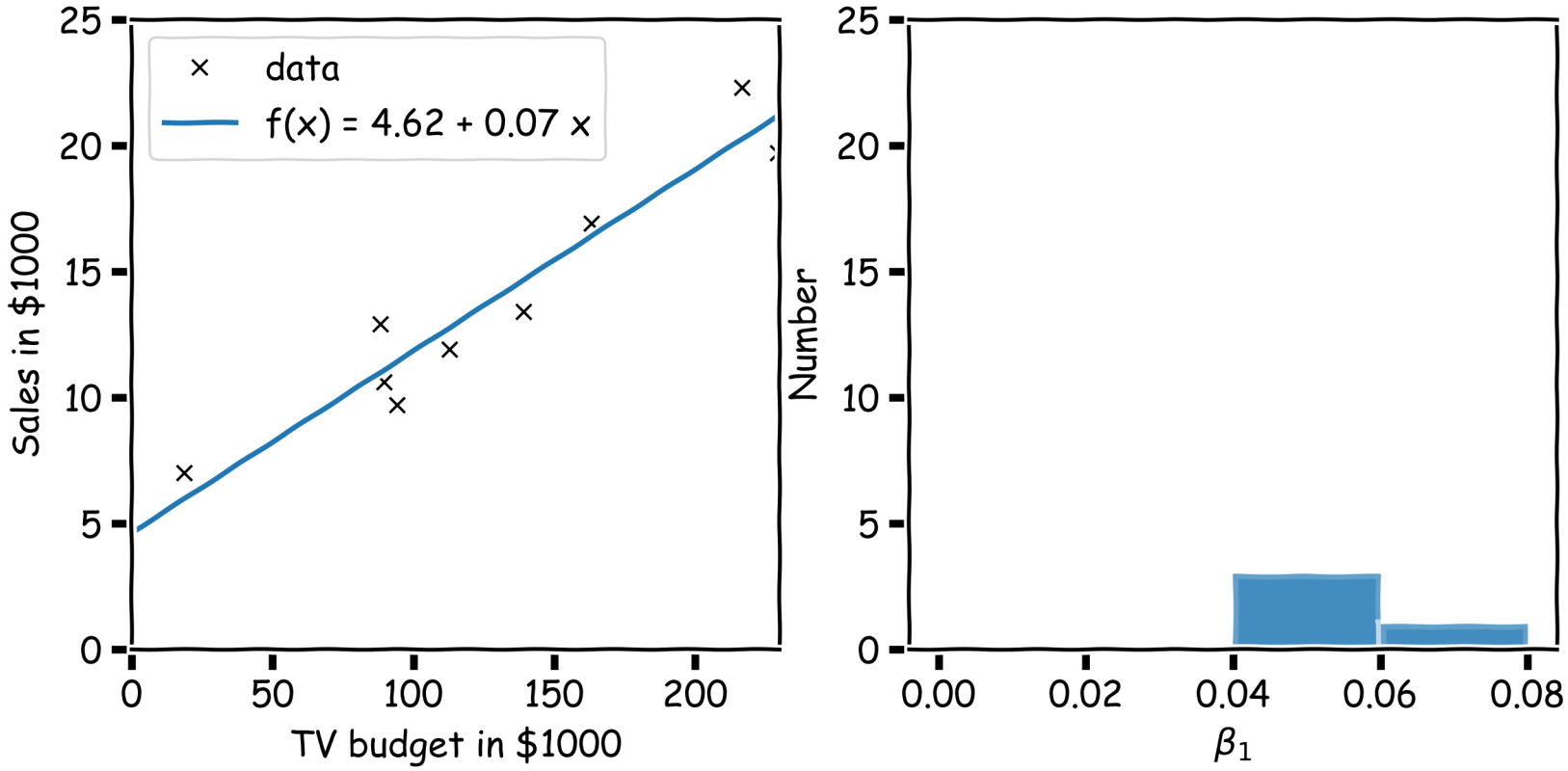
Confidence intervals for the predictors estimates (cont)

Another sample



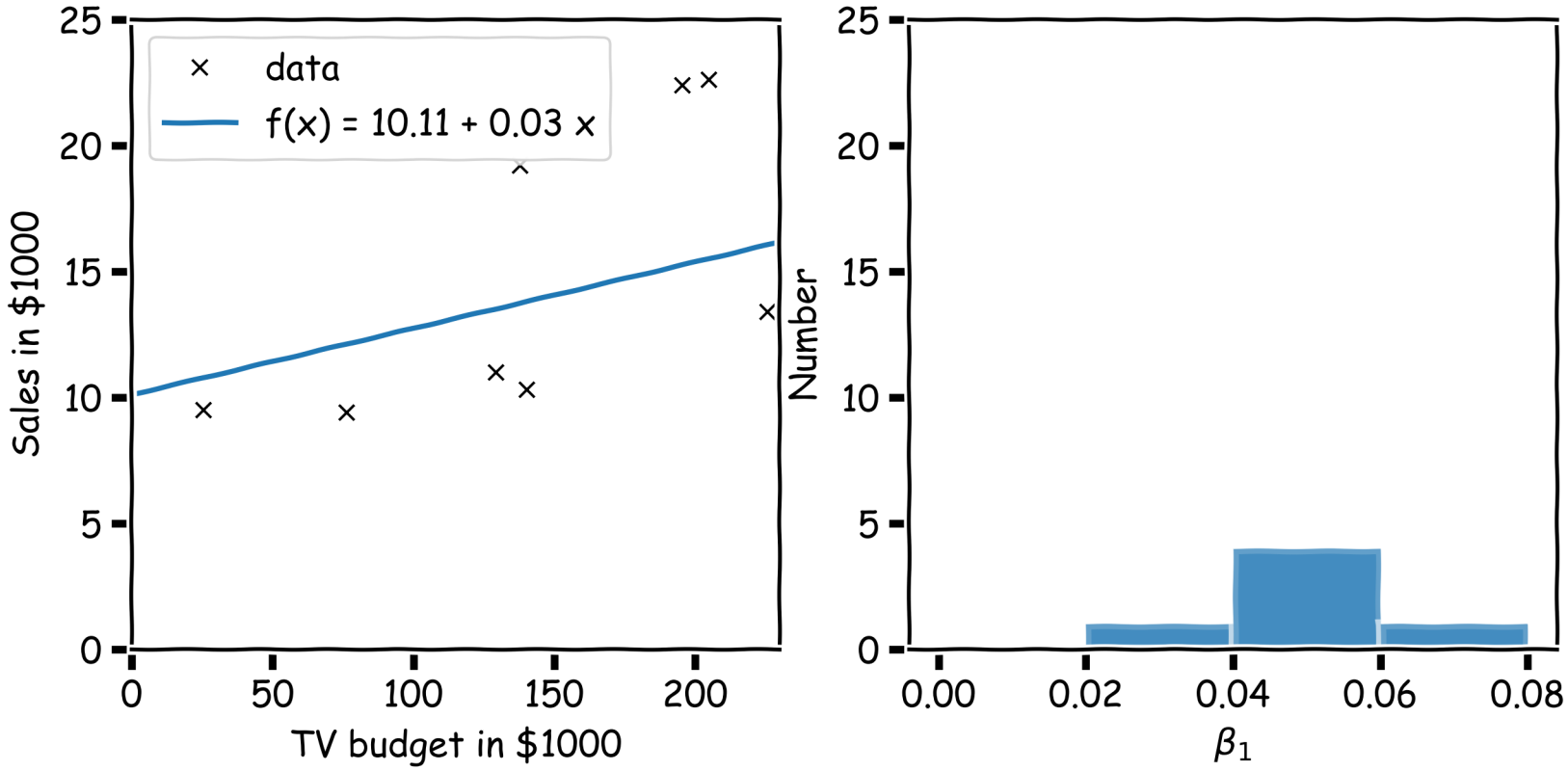
Confidence intervals for the predictors estimates (cont)

Another sample



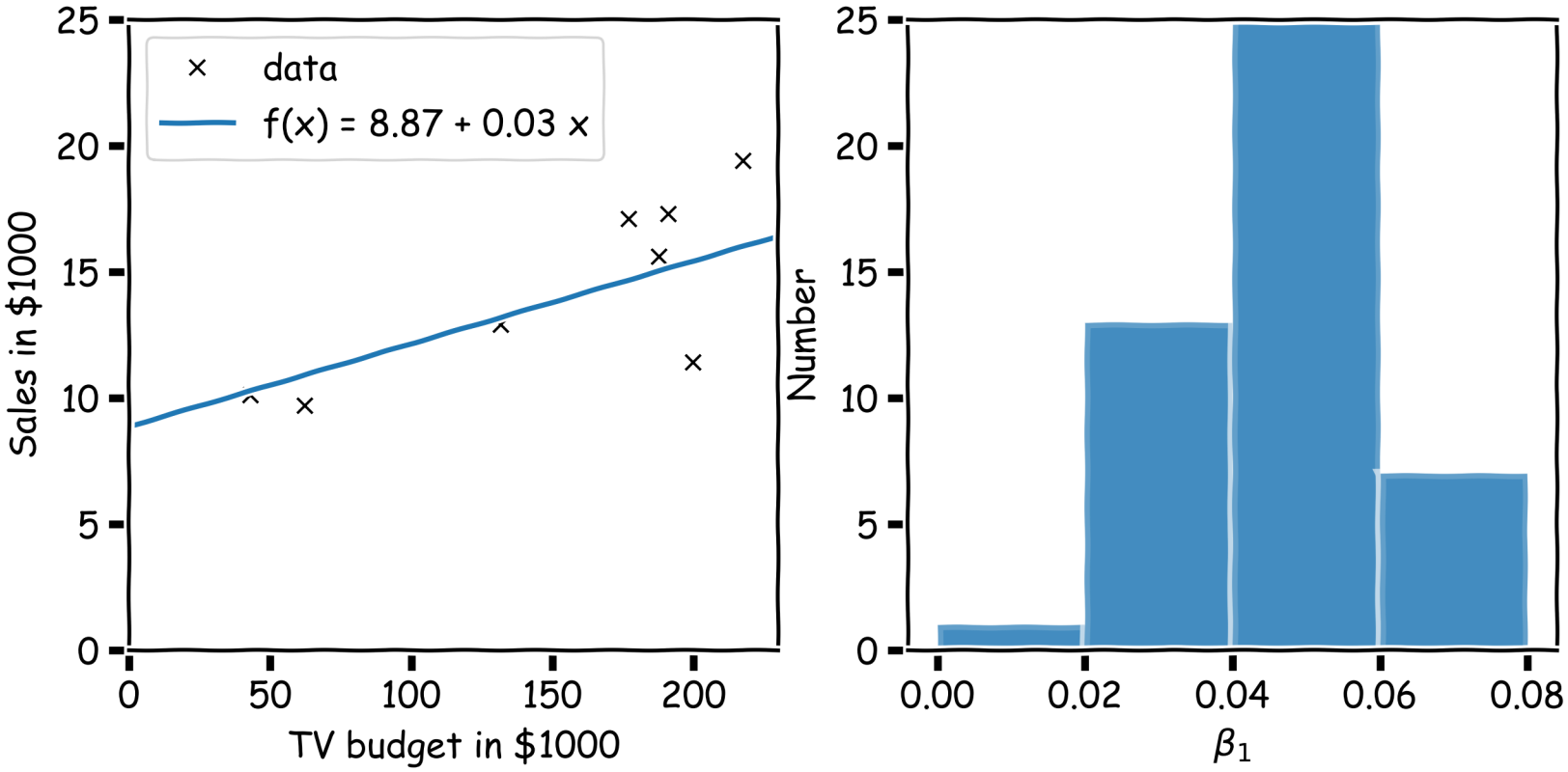
Confidence intervals for the predictors estimates (cont)

And another sample



Confidence intervals for the predictors estimates (cont)

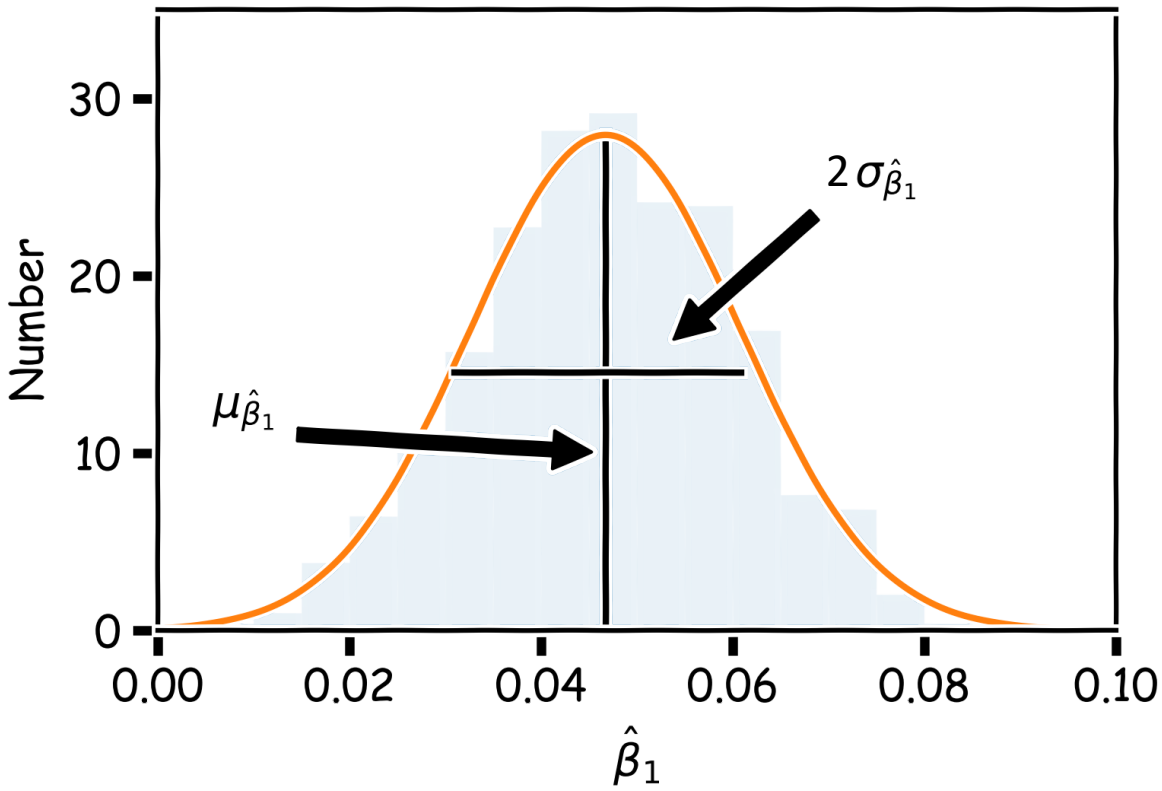
Repeat this for 100 times



Confidence intervals for the predictors estimates (cont)

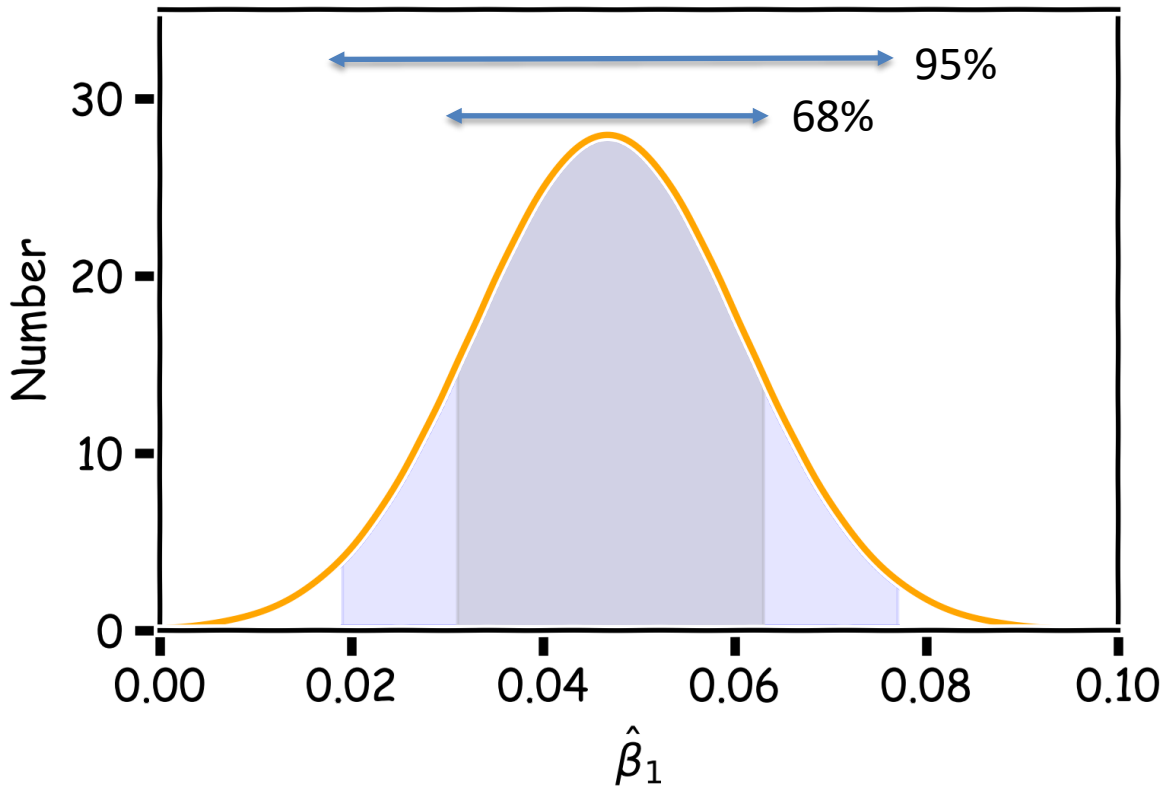
We can now estimate the mean and standard deviation of all the estimates $\hat{\beta}_1$.

The variance of $\hat{\beta}_0$ and $\hat{\beta}_1$ are also called their ***standard errors***, $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$.



Confidence intervals for the predictors estimates (cont)

Finally, we can calculate the confidence intervals, which are the ranges of values such that the **true** value of β_1 is contained in this interval with *n* percent probability.



Confidence intervals for the predictors estimates: Standard Errors

- We can empirically estimate the **standard errors**, $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$ of β_0 and β_1 through bootstrapping.

- If for each bootstrapped sample the estimated betas are: $\hat{\beta}_{0,i}$, $\hat{\beta}_{1,i}$, then

- $SE(\hat{\beta}_0) = \sqrt{\text{var}(\widehat{\beta_0})}$

- $SE(\hat{\beta}_1) = \sqrt{\text{var}(\widehat{\beta_1})}$

Confidence intervals for the predictors estimates: Standard Errors

- **Alternatively:**
- If we know the variance σ_ϵ^2 of the noise ϵ, we can compute $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$ analytically using the formulae below (no need to bootstrap):

$$SE\left(\widehat{\beta}_0\right) = \sigma_\epsilon \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_i (x_i - \overline{x})^2}}$$

$$SE\left(\widehat{\beta}_1\right) = \frac{\sigma_\epsilon}{\sqrt{\sum_i (x_i - \overline{x})^2}}$$

*** If data set is small this approximation does not work**

Standard Errors

- However, if we make the following assumptions,
  - the errors $\epsilon_i = y_i - \hat{y}_i$ and $\epsilon_j = y_j - \hat{y}_j$ are uncorrelated, for $i \neq j$,
  - each ϵ_i has a mean 0 and variance σ_ϵ^2,
- then, we can empirically estimate σ^2, from the data and our regression line:

$$\sigma_\epsilon \approx \sqrt{\frac{n \cdot \text{MSE}}{n-2}} = \sqrt{\frac{\sum_i (y_i - \widehat{y}_i)^2}{n-2}}$$

Remember:

$$y_i = f(x_i) + \epsilon_i \Longrightarrow \epsilon_i = y_i - f(x_i)$$

Hypothesis Testing

Hypothesis testing is a formal process through which we evaluate the validity of a statistical hypothesis by considering evidence for or against the hypothesis gathered by **random sampling** of the data.

1. State the hypotheses, typically a **null hypothesis**, H_0 and an **alternative hypothesis**, H_1, that is the negation of the former.
2. Choose a type of analysis, i.e. how to use sample data to evaluate the null hypothesis. Typically this involves choosing a single test statistic.
3. **Sample** data and compute the test statistic.
4. Use the value of the test statistic to either reject or not reject the null hypothesis.

Hypothesis testing

State Hypothesis:

Null hypothesis:

H_0: There is no relation between X and Y

The alternative:

H_a: There is some relation between X and Y

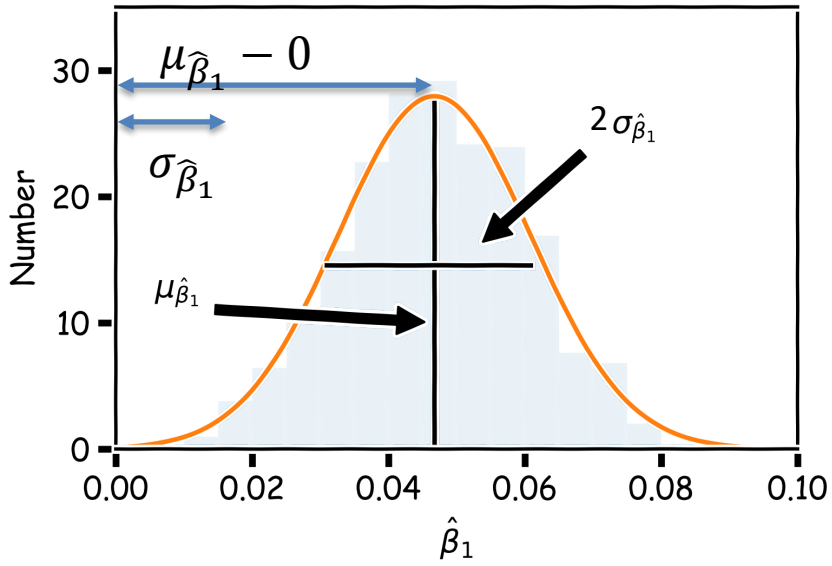
Mathematically, this corresponds to testing $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 \neq 0$. Since if $\beta_1 = 0$ then the model $Y = \beta_0 + \beta_1 X + \boldsymbol{\varepsilon}$ reduces to $Y = \beta_0 + \boldsymbol{\varepsilon}$, and X is not associated with Y .

To test the null hypothesis, we need to determine whether, our estimate for $\hat{\beta}_1$, is sufficiently far from zero that we can be confident that $\hat{\beta}_1$ is non-zero. We use the following t-statistic, which measures the number of standard deviations that $\hat{\beta}_1$ is away from 0

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$$

Hypothesis testing

The t-test statistic measures the distance from zero in units of standard deviation.



We evaluate how often a particular value of *t* can occur by accident. We expect that *t* will have a *t-distribution with n-2 degrees* of freedom.

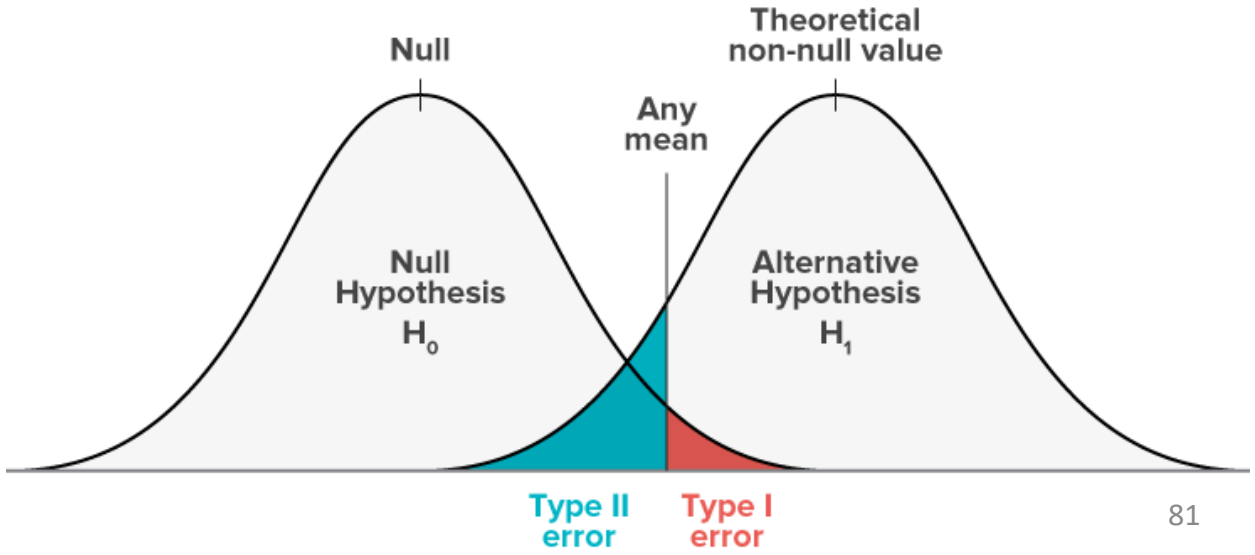
To compute the probability of observing any value equal to $|t|$ or larger, assuming $\hat{\beta}_1 = 0$ is easy. We call this probability the **p-value**.

a small p-value (<0.05) indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance.

Error Types

- A type I error (or error of the first kind) is the incorrect rejection of a true null hypothesis.
  - Usually a type I error leads one to conclude that a supposed effect or relationship exists when in fact it doesn't.
  - Examples of type I errors include a test that shows a patient to have a disease when in fact the patient does not have the disease, a fire alarm going on indicating a fire when in fact there is no fire, or an experiment indicating that a medical treatment should cure a disease when in fact it does not.
- A type II error (or error of the second kind) is the failure to reject a false null hypothesis.
  - Examples of type II errors would be a blood test failing to detect the disease it was designed to detect, in a patient who really has the disease; a fire breaking out and the fire alarm does not ring; or a clinical trial of a medical treatment failing to show that the treatment works when really it does

		Decision	
		Retain the null	**Reject the null**
Truth in the population	True	CORRECT $1 - \alpha$	TYPE I ERROR α
	False	TYPE II ERROR β	CORRECT $1 - \beta$ POWER



Important tests

- **T-test:** compare two groups, or two interventions on one group.
- **CHI-squared and Fisher's test.** Compare the counts in a "contingency table".
- **ANOVA:** compare outcomes under several discrete interventions.

T-test

- **Single-sample:** Compute the test statistic:

  $t = \frac{\bar{x}}{\bar{\sigma}}$ $\qquad t = \frac{Z}{s} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{s}$

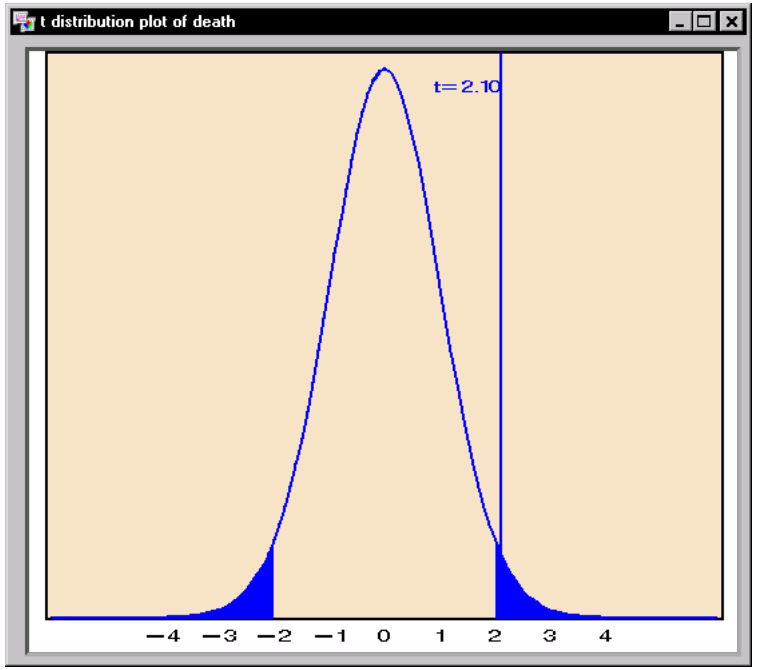
- where $\bar{x}$ is the sample mean and $\bar{\sigma}$ is the sample standard deviation, which is the square root of the sample variance Var(x).
- If X is normally distributed, t is **almost** normally distributed, but not quite because of the presence of $\bar{\sigma}$. It has a **t-distribution**.
- You use the single-sample test for **one group** of individuals in **two conditions**. Just subtract the two measurements for each person, and use the difference for the single sample t-test.
- This is called a **within-subjects** design.

T-statistic and T-distribution

- We use the t-statistic from the last slide to test whether the mean of our sample could be zero.
- If the underlying population has mean zero, the t-distribution should be distributed like this:

- The area of the tail beyond our measurement tells us how likely it is under the null hypothesis.

- If that probability is low (say < 0.05) we reject the null hypothesis.



Two sample T-test

- In this test, there are two samples x_1 and x_2 of sizes n_1 and n_2. A t-statistic is constructed from their sample means and sample standard deviations:
- $t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma_{\bar{x}_1 - \bar{x}_2}}$
- where: $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}$ and σ_1 and σ_2 are sample stdevs,
- You should try to understand the formula, but you shouldn't need to use it. most stats. software exposes a function that takes the samples x_1 and x_2 as inputs directly.
- This design is called a between-subjects test.

Chi-squared test

- Often you will be faced with discrete (count) data. Given a table like this:

	Prob(X)	Count(X)
X=0	0.3	10
X=1	0.7	50

- Where Prob(X) is part of a null hypothesis about the data (e.g. that a coin is fair).
- The CHI-squared statistic lets you test whether an observation is consistent with the data:

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

- Oi is an observed count, and Ei is the expected value of that count. It has a chi-squared distribution, whose p-values you compute to do the test.

Fisher's exact test

- In case we only have counts under different conditions

	Count1(X)	Count2(X)
X=0	a	b
X=1	c	d

- We can use Fisher's exact test (n = a+b+c+d):

$$p = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{n}{a+c}} = \frac{(a+b)!\ (c+d)!\ (a+c)!\ (b+d)!}{a!\ \ b!\ \ c!\ \ d!\ \ n!}$$

- Which gives the probability directly (its not a statistic).

One-Way ANOVA

- ANOVA (ANalysis Of VAriance) allows testing of multiple differences in a single test. Suppose our experiment design has an independent variable Y with four levels:

Y

Primary School	High School	College	Grad degree
4.1	4.5	4.2	3.8

- The table shows the mean values of a response variable (e.g. avg number of Facebook posts per day) in each group.
- We would like to know in a single test whether the response variable depends on Y, at some particular significance such as 0.05.

ANOVA

- In ANOVA we compute a single statistic (an F-statistic) that compares variance between groups with variance within each group.

$$F = \frac{VAR_{between}}{VAR_{within}}$$

- The higher the F-value is, the less probable is the null hypothesis that the samples all come from the same population.
- We can look up the F-statistic value in a cumulative F-distribution (similar to the other statistics) to get the p-value.
- ANOVA tests can be much more complicated, with multiple dependent variables, hierarchies of variables, correlated measurements etc.

Non-Parametric Tests

- All the tests so far are parametric tests that assume the data are normally distributed, and that the samples are independent of each other and all have the same distribution (IID).

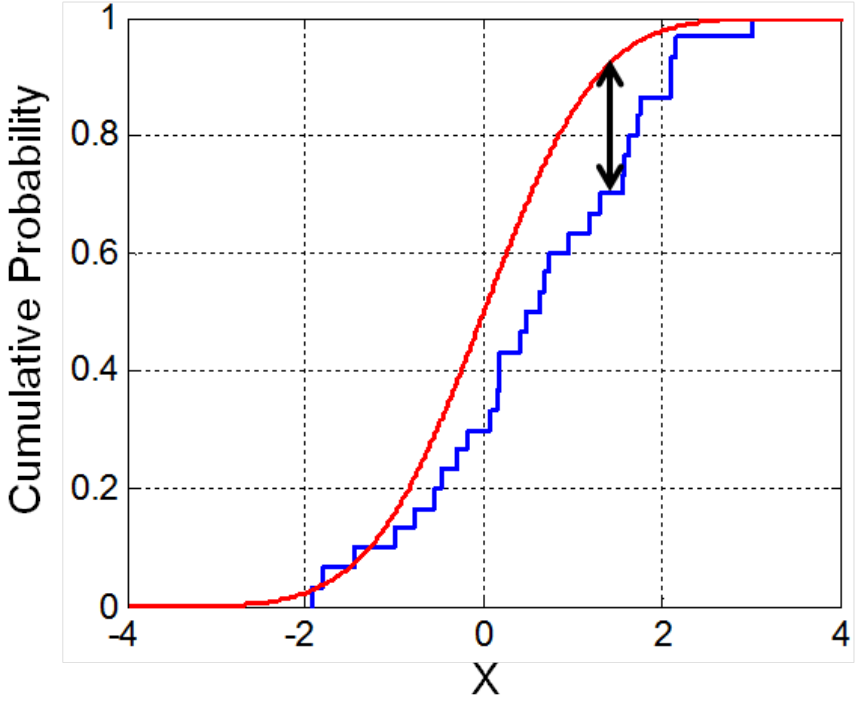
- They may be arbitrarily inaccurate if those assumptions are not met. Always make sure your data satisfies the assumptions of the test you're using. e.g. watch out for:
  - Outliers – will corrupt many tests that use variance estimates.
  - Correlated values as samples, e.g. if you repeated measurements on the same subject.
  - Skewed distributions – give invalid results.

Non-Parametric Tests

- These tests make no assumption about the distribution of the input data, and can be used on very general datasets:
  - K-S test
  - Permutation tests
  - Bootstrap confidence intervals

K-S test

- The K-S (Kolmogorov-Smirnov) test is a very useful test for checking whether two (continuous or discrete) distributions are the same.
- In the one-sided test, an observed distribution (e.g. some observed values or a histogram) is compared against a reference distribution.
- In the two-sided test, two observed distributions are compared.
- The K-S statistic is just the max distance between the CDFs of the two distributions.
- While the statistic is simple, its distribution is not!
- But it is available in most stat packages.



K-S test

- The K-S test can be used to test **whether a data sample has a normal distribution** or not.
- Thus it can be used as a sanity check for any common parametric test (which assumes normally-distributed data).
- It can also be used to compare distributions of data values in a large data pipeline: **Most errors will distort the distribution of a data parameter and a K-S test can detect this**.

Classification

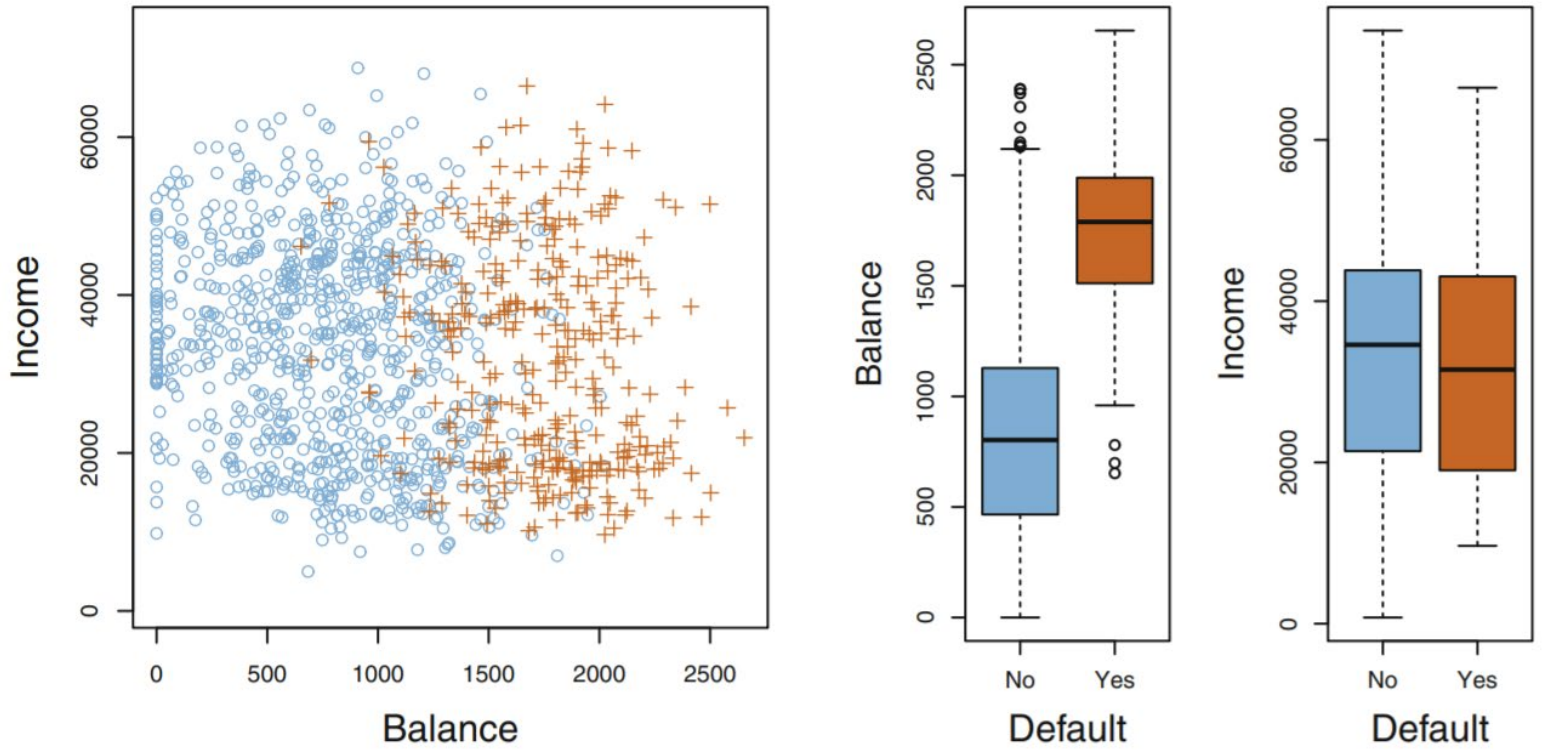
- Up to this point, the methods we have seen have centered around modeling and predicting a **quantitative** response variable (ex, number of taxi pickups, number of bike rentals, etc.). Linear **regression** performs well under these situations

- When the response variable is **categorical**, then the problem is no longer called a regression problem but is instead labeled as a **classification problem**.

- The goal is to attempt to classify each observation into a category (aka, class or cluster) defined by *Y*, based on a set of predictor variables *X*.

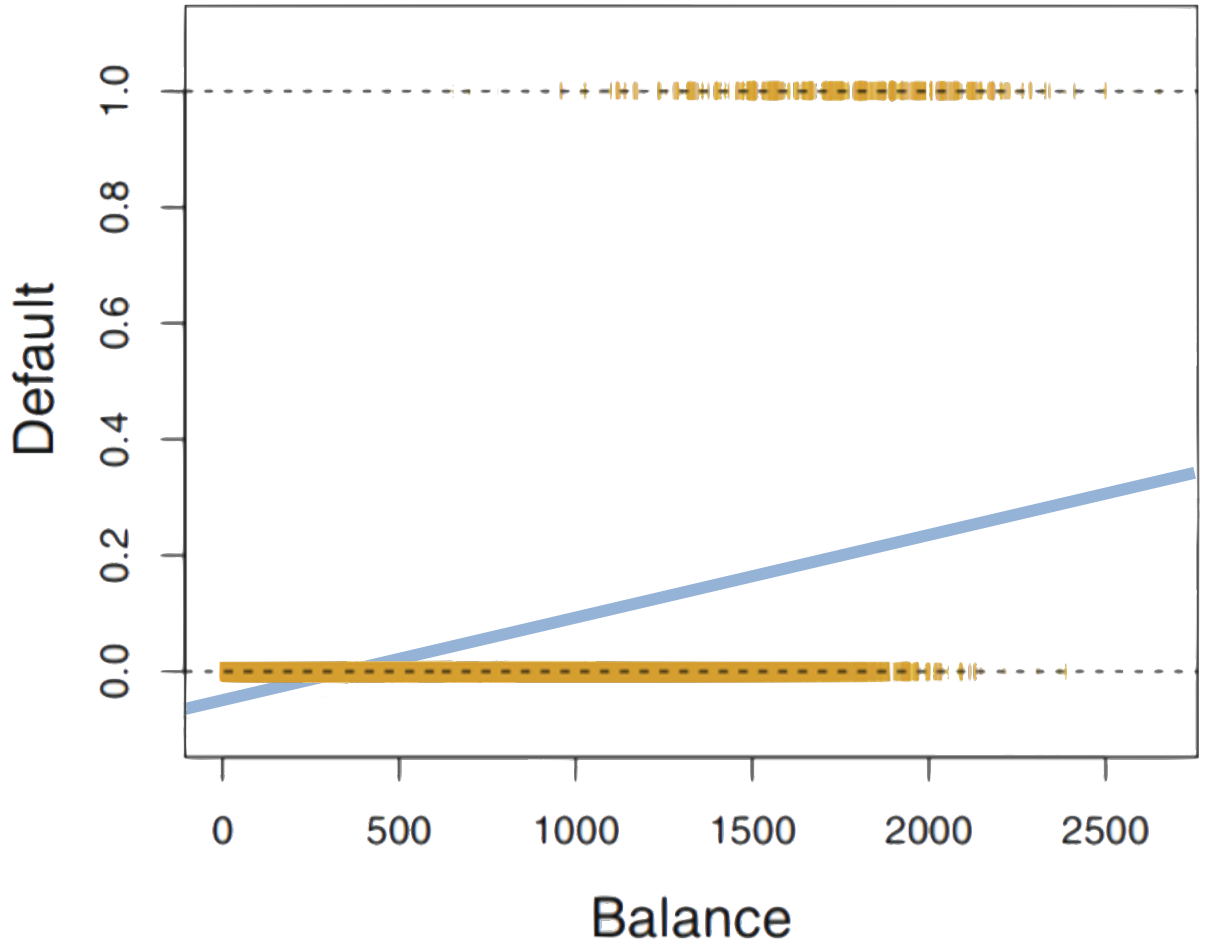
Example

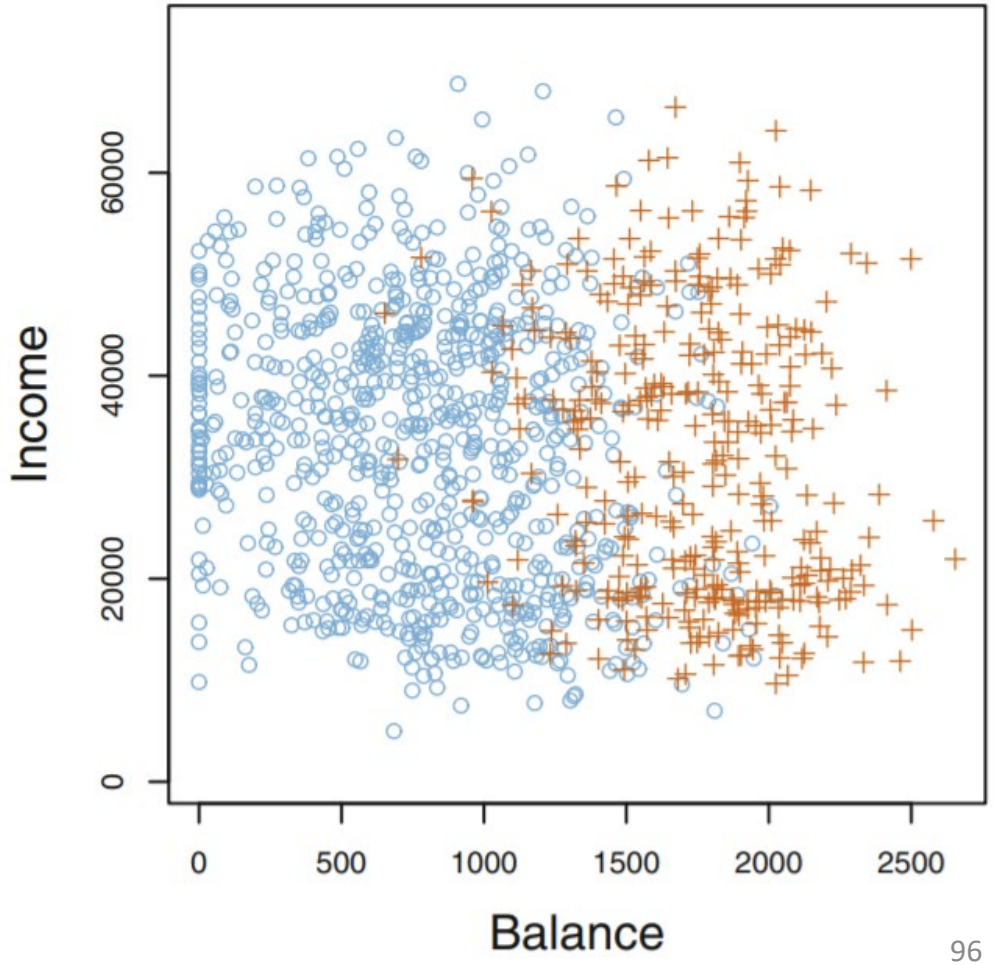
- We are interested in predicting whether an individual will default on their credit card payment based on annual income and monthly credit card balance.
- The individuals who defaulted on their credit card payments are shown in orange and those who did not are shown in blue.



Why not Linear Regression?

- We model default =1 and not default = 0
- Applying Linear Regression, we will get:





Logistic Regression

- Rather than modelling this response Y directly, logistic regression models the **probability** that Y belongs to a particular category
- For the Default data, logistic regression models the probability of default.
  - The probability of default given balance can be written as **P(default = Yes | balance),** which we abbreviate as **p(balance)**
  - The values **p(balance)** will range between 0 and 1.
  - Then, for any given value of balance, a prediction can be made for default.
  - For example, one might predict default = Yes for any individual for whom p(balance) > 0.5.
  - Alternatively, if a company wishes to be conservative in predicting individuals who are at risk for default, then they may choose to use a lower threshold, such as p(balance) > 0.1.

The Logistic Model

- We need to model the relationship between $p(X) = P(Y = 1|X)$ and X

  – We can try to apply linear regression model $p(X) = \beta_0 + \beta_1 X$, but we will get the model shown two slides before

- To avoid this problem, we must model $p(X)$ using a function that gives outputs between 0 and 1 for all values of X.
- Many functions meet this description, but in logistic regression, we use the **logistic function**,

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

The Logistic Model

- After a bit of manipulation of the original formula, we find that

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$

- The quantity $p(X)/[1-p(X)]$ is called the **odds** and can take on any value between 0 and ∞.
- Values of the odds close to 0 and ∞ indicate very low and very high probabilities, respectively
- For example,
  - If on average, 1 in 5 people will default, we have p(X)=0.2 implies an odds of 0.2/(1–0.2) = 0.25.
  - Likewise, on average, 9 out of every 10 people will default since p(X)=0.9 implies an odds of 0.9/(1–0.9) = 9

The Logistic Model

- By taking the logarithm of both sides, we arrive at

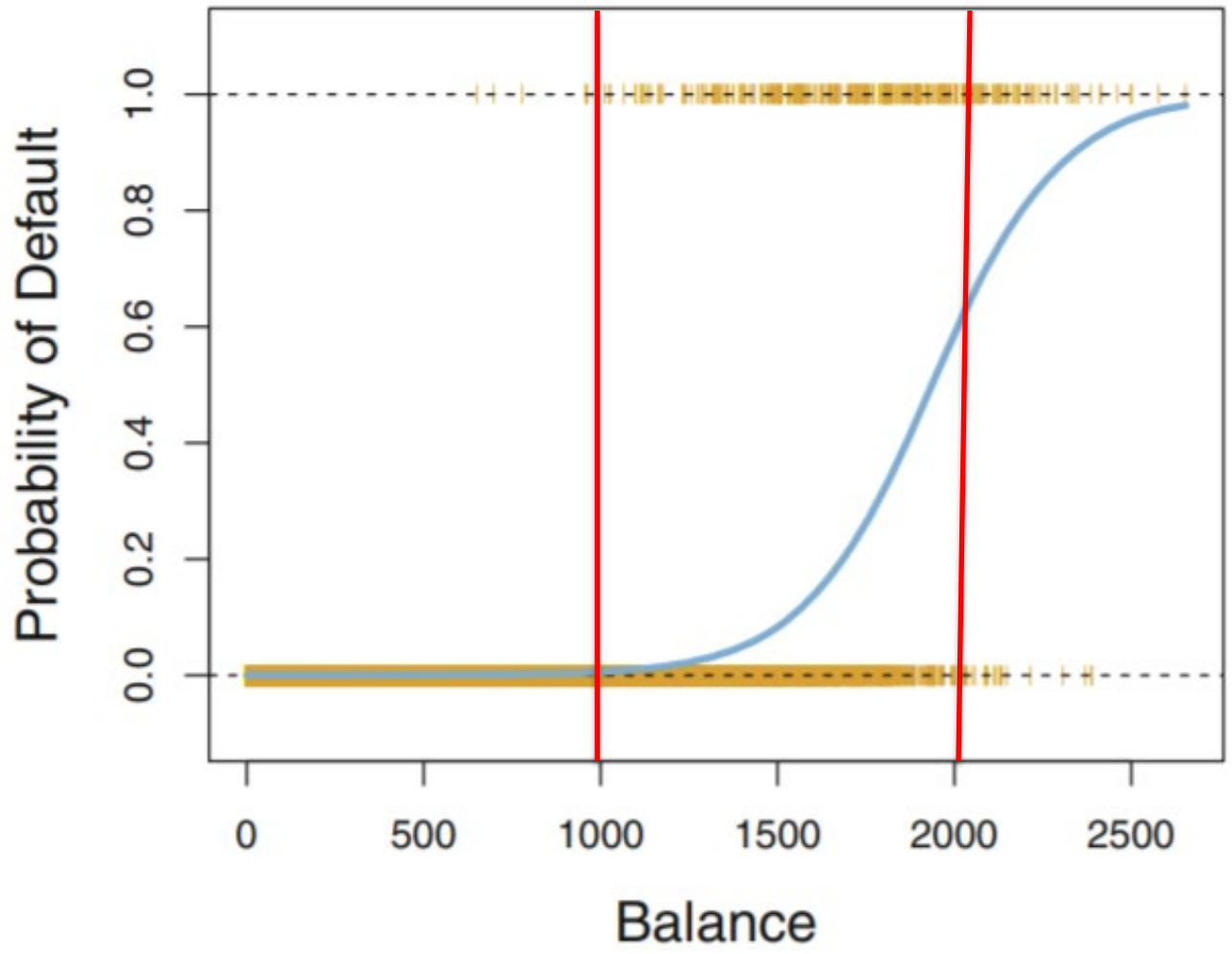
$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

- The left-hand side is called the **log odds** or **logit**
- The coefficients β_0 and β_1 need to be estimated based on the available training data
- The **maximum likelihood method** is used to fit a logistic regression model and can be interpreted as follows:
  - we seek estimates for β_0 and β_1 such that the predicted probability $\hat{p}(x_i)$ of default for each individual corresponds as closely as possible to the individual's observed default status.
- Mathematical equation called a likelihood function is used to estimate β_0 and β_1 :

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'}))$$

Making Predictions

- Solving the model, a we get $\beta_0 = -10.6513$ and $\beta_1 = 0.0055$



$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} =$$

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1,000}}{1+e^{-10.6513+0.0055\times 1,000}} =$$

$$= 0.586$$

$$= 0.00576$$

Multiple Logistic Regression

- We now consider the problem of predicting a binary response using multiple predictors

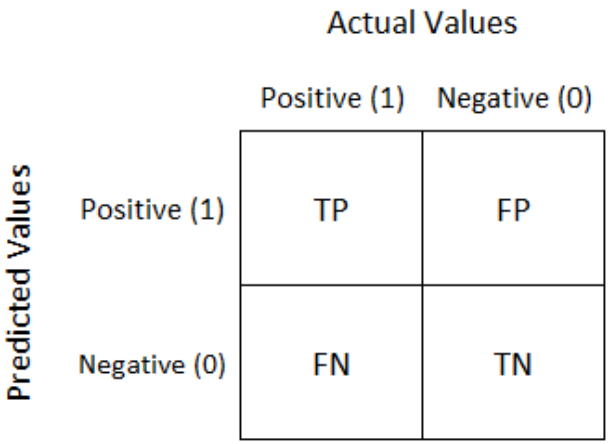
$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

Important measures for classification and diagnostic testing

- Accuracy
- Precision
- Recall
- Sensitivity
- Specificity

Why not to use “accuracy” directly



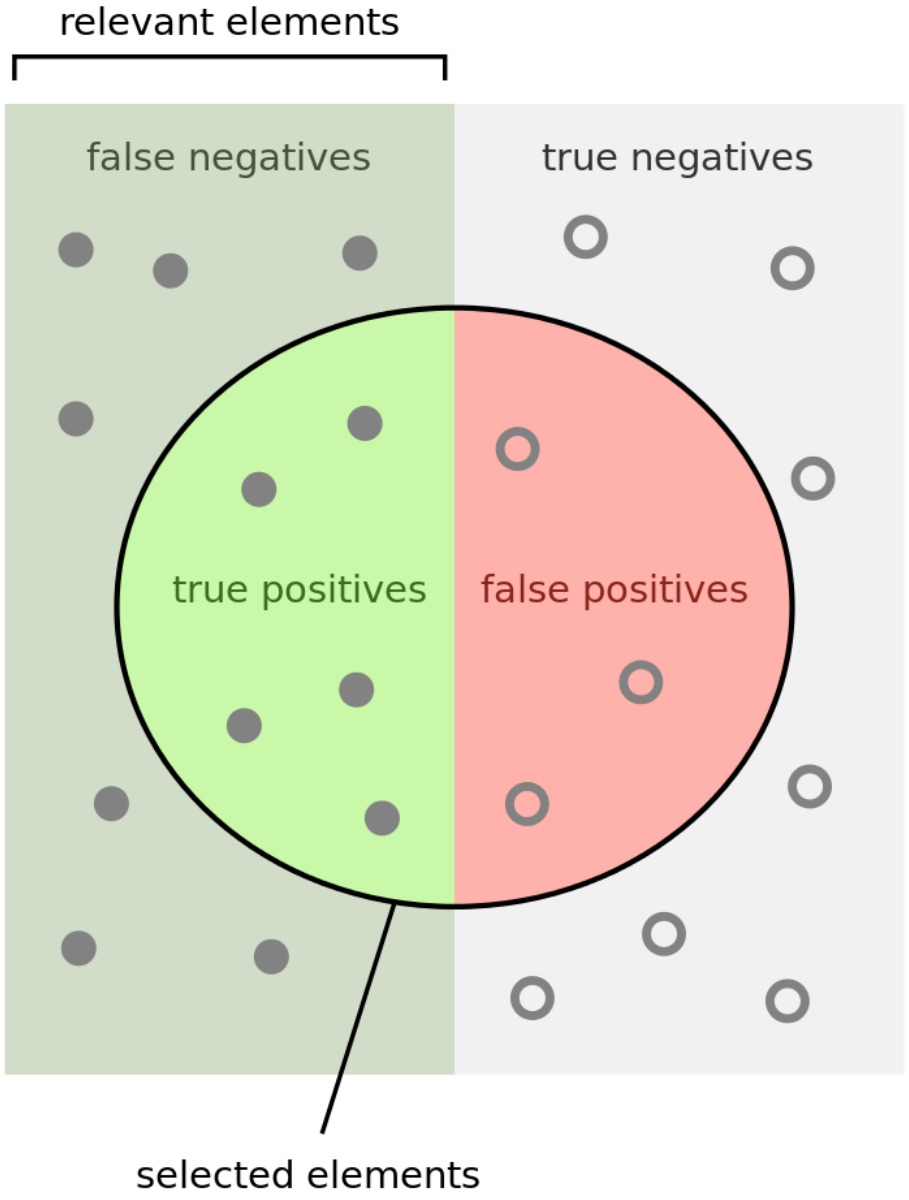
- The simplest measure of performance would be the fraction of items that are correctly classified, or the “**accuracy**” which is:

$$\frac{tp + tn}{tp + tn + fp + fn}$$

- (tp = true positive, fn = false negative etc.).

- But this measure is dominated by the larger set (of positives or negatives) and favors trivial classifiers.
  - e.g. if 5% of items are truly positive, then a classifier that always says “negative” is 95% accurate.

Precision and Recall



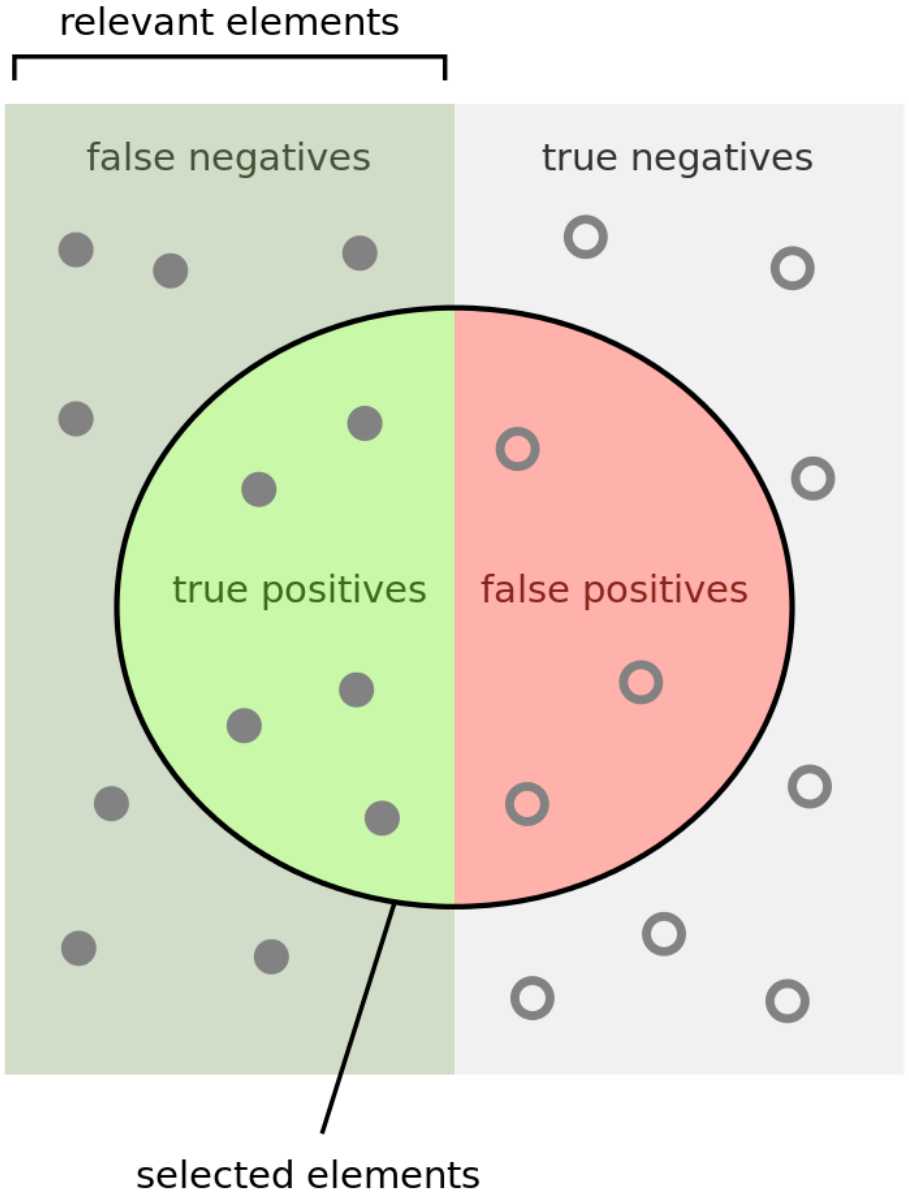
How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{selected elements}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{relevant elements}}$$

Sensitivity and Specificity



How many relevant items are selected? e.g. How many sick people are correctly identified as having the condition.

$$\text{Sensitivity} = \frac{\text{true positives}}{\text{relevant elements}}$$

How many negative selected elements are truly negative? e.g. How many healthy peple are identified as not having the condition.

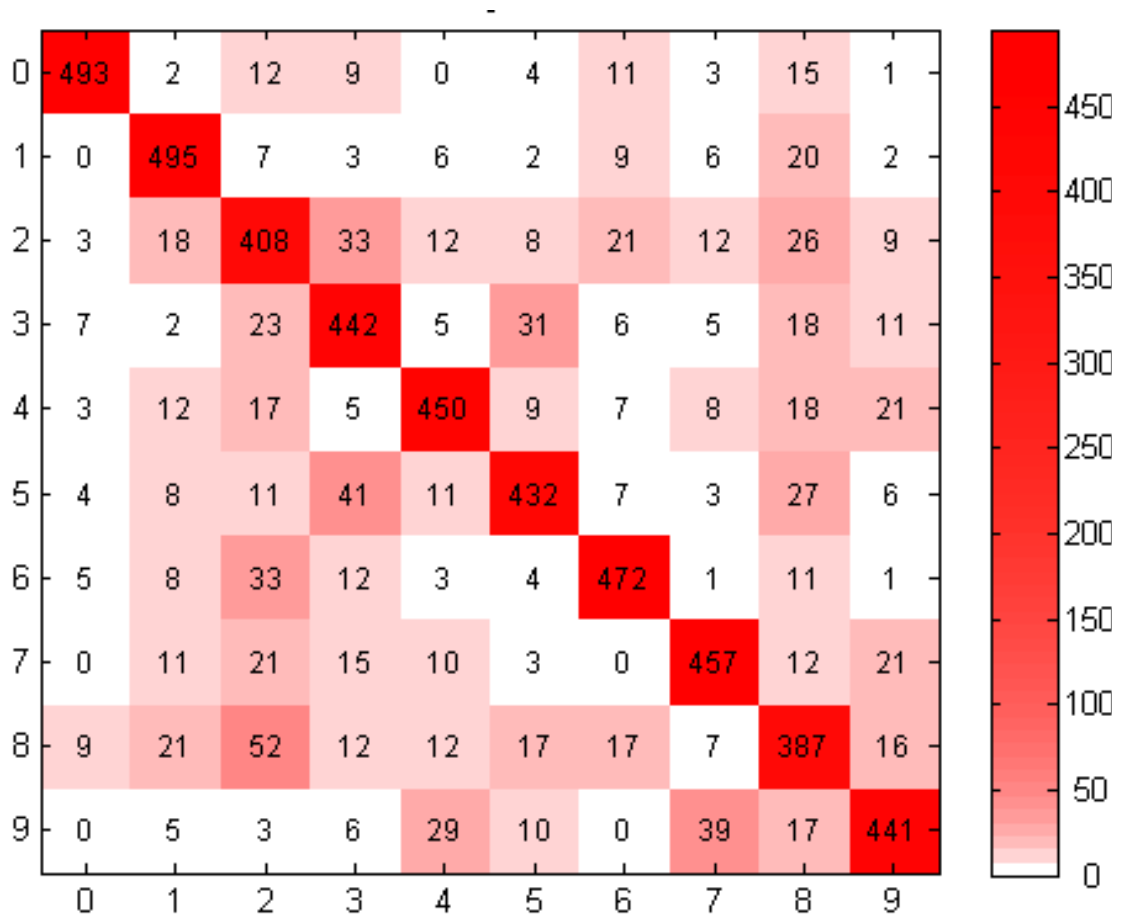
$$\text{Specificity} = \frac{\text{true negatives}}{\text{false positives}}$$

Important measures for classification and diagnostic testing

		True condition				
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$	F_1 score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$		

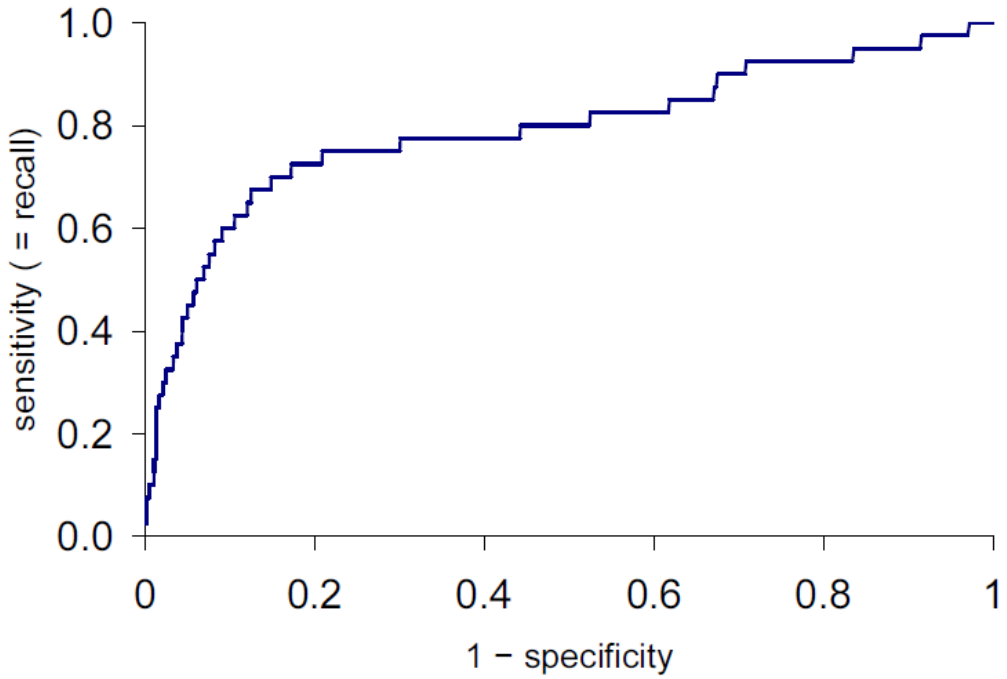
Confusion matrix

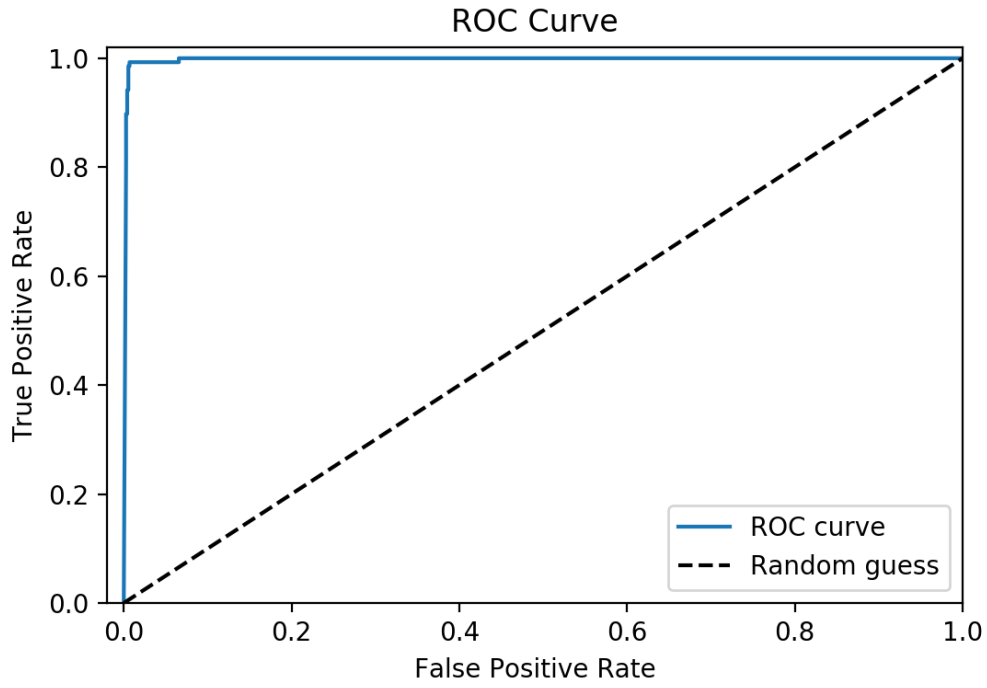
- Is an error matrix that allows visualization of the performance of an algorithm.
- Each row of the matrix represents the instances in a predicted class, while each column represents the instances in an actual class (or vice versa).



ROC plots

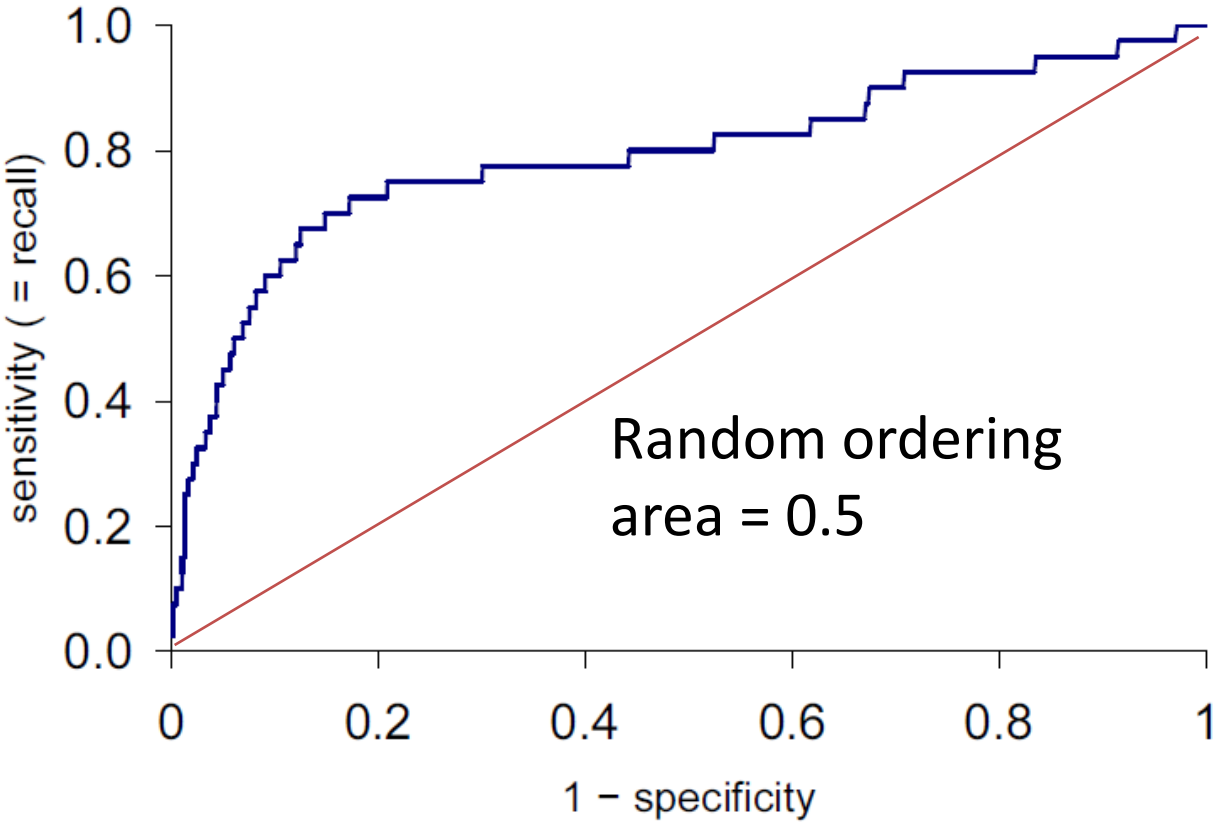
- ROC is Receiver-Operating Characteristic
  - Y-axis: true positive rate = tp/(tp + fn), same as recall (or sensitivity)
  - X-axis: false positive rate = fp/(fp + tn) = 1 - specificity





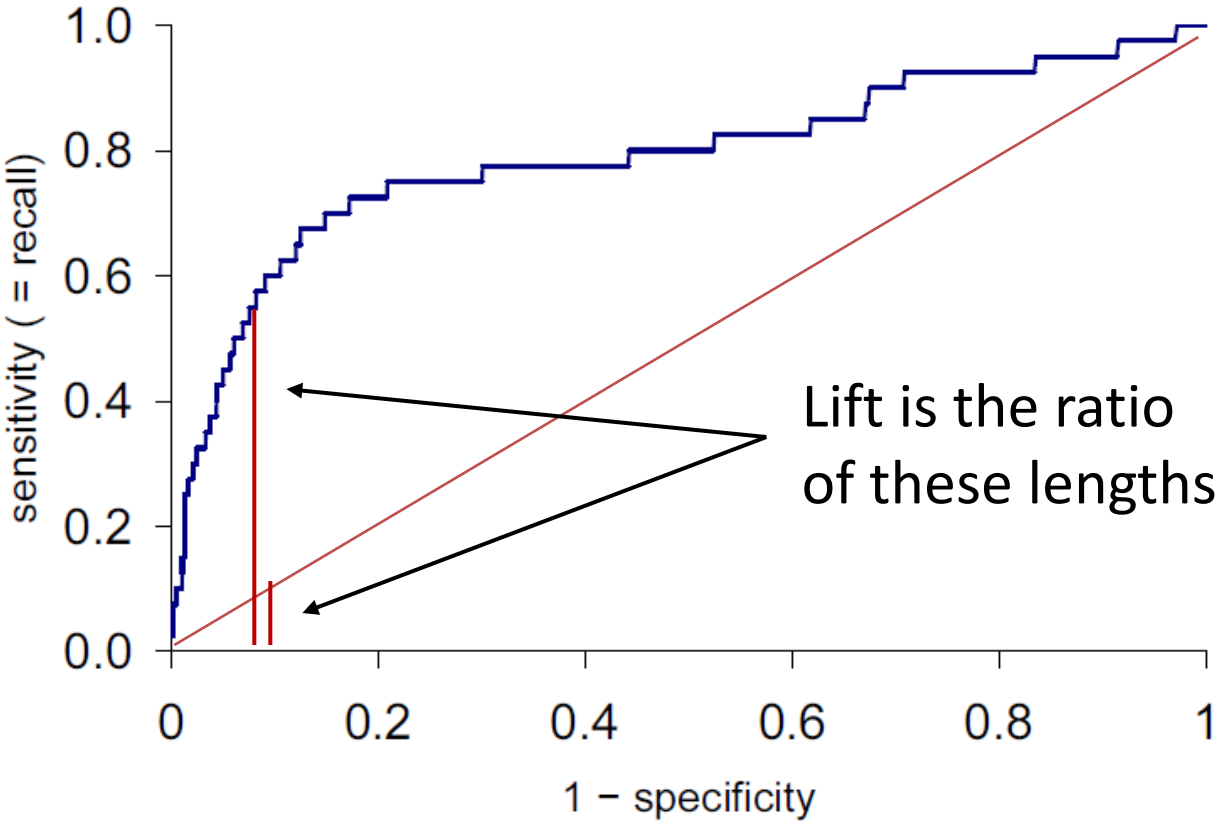
ROC AUC

- ROC AUC is the “Area Under the Curve” – a single number that captures the overall quality of the classifier.
- It should be between 0.5 (random classifier) and 1.0 (perfect).



Lift Plot

- A variation of the ROC plot is the lift plot, which compares the performance of the actual classifier against random ordering or sometimes against another classifier.



Lift Plot

- Lift plots emphasize initial precision (typically what you care about) and performance in a problem-independent way.

- Note: The lift plot points should be computed at regular spacing, e.g., 1/00 or 1/1000. Otherwise, the initial lift value can be excessively high and unstable.

